

INTRODUCTION TO **VLSI** DESIGN

Er. Sailesh Mishra

B.Tech (ECE | Silicon University)

Specialization in Microelectronics & VLSI

RC CIRCUIT

Er. Sailesh Mishra

B.Tech (ECE | Silicon University)

Specialization in Microelectronics & VLSI

BASICS ...

capacitance = $10 \mu\text{F}$
 $= 10 \times 10^{-6} \text{ F}$
 $V = 5 \text{ V}$
 $Q = C \cdot V$
 $= 10 \mu\text{F} \times 5$
 $= 50 \mu\text{C}$

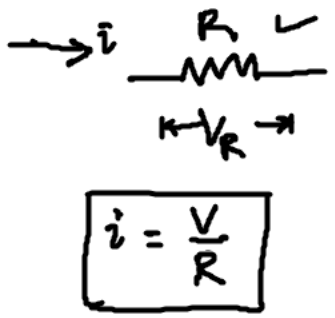
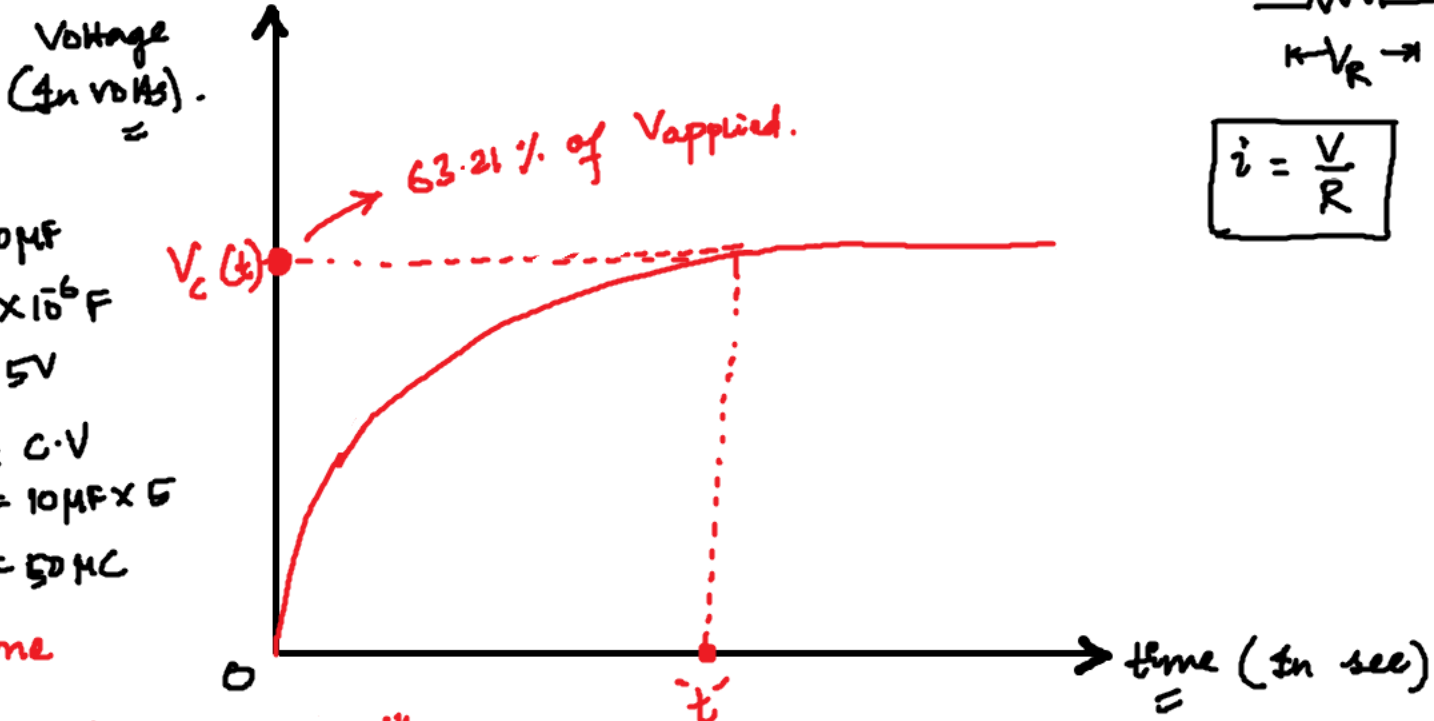
$V = \frac{Q}{C}$

$Q = ne$

$$\frac{50 \mu\text{C}}{1.6 \times 10^{-19}} = n_e = 3.125 \times 10^{14}$$

$$= 300 \text{ no. of } e^-$$

1SD +ve $\rightarrow$ +ve METAL PLATE.
 1SD -ve $\rightarrow$ -ve METAL PLATE.



Similarly;

ex. SiO_2 : $V_L = 1$
 $K = 3.97 \approx 4$

$\text{Al}_2\text{O}_3 = 9.0$ (v. high)
 $\text{Air} = 1.0$ (K).
 $\text{H}_2\text{O} = 20-25$.

$Q = CV$

V or Battery is always const.

V
(mV)
volts

$V_C(t)$: CHARGING $\rightarrow V_C(t) = V_{app} (1 - e^{-t/\tau})$
 $\downarrow$
 $1.25V = 1.8V (1 - e^{-t/\tau})$



V_{app}
 $1.8V$

$V' \rightarrow 1.8V$

$\tau = R \cdot C$
 $= 1k\Omega \times 10\mu F$
 $= 1 \times 10^3 \times 10 \times 10^{-6} F$

$1.2V \rightarrow 1.25V$

0.9

0.45

$V' \rightarrow 0V$

t_n

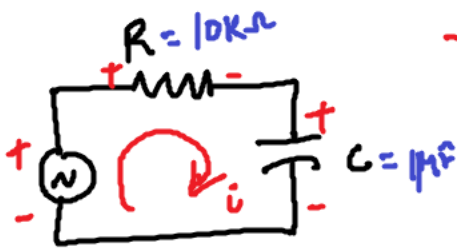
$V_C(t)$: DISCHARGING $\rightarrow V_C(t) = V_{app} \times (e^{-t/\tau})$

t (in ms)

Example-1 :-



$V_{\text{pulse}} = 1.8\text{V}$



$$\rightarrow V_{\text{pulse}} - V_R - V_C = 0$$

$$\Rightarrow 1.8 - V_R - V_C = 0$$

$\Rightarrow 1.8 - V_C = V_R$: voltage drop across $R = 10\text{k}\Omega = 10 \times 10^3 \Omega$.
voltage drop across $C = 1\mu\text{F} = 1 \times 10^{-6}\text{F}$.

$$P = \frac{V^2}{R} + \frac{1}{2} CV^2$$

Time period = 8ms
 $T_{\text{ON}} = 4\text{ms}$
 $T_{\text{rise}} = T_{\text{fall}} = 0\text{s}$

Q1 Find $i \rightarrow R$

Q2 Find $i \rightarrow C$

$$\tau = R \cdot C = 10\text{k}\Omega \times 1\mu\text{F} = 10 \times 10^3 \Omega \times 10^{-6}\text{F} = 10^{-2}\text{s}$$

$$\rightarrow i_R(t) = \frac{V}{R} = \frac{V_N}{R} = \frac{V_{\text{pulse}} - V_C}{R} = \frac{1.8\text{V} - V_{\text{app}} \times (1 - e^{-t/\tau})}{R}$$

$$\rightarrow i_C(t) = C \cdot \frac{dV}{dt} = \frac{1\mu\text{F} \times V_{\text{app}} \times (1 - e^{-t/\tau})}{t}$$

Example-1 :



Time period = 8ms

$T_{ON} = 4ms$

$T_{rise} = T_{fall} = 0s$

Q1: Find $V_C(t)$ at $2ms$; when CAPACITOR is charging...

Q2: Find $V_C(t)$ at $5ms$; when CAPACITOR is discharging...

$$\rightarrow V_{pulse} - V_R - V_C = 0$$

$$\Rightarrow 1.8 - V_R - V_C = 0$$

$$\Rightarrow 1.8 - V_C = V_R$$

$\rightarrow$ voltage drop across $R = 10k\Omega = 10 \times 10^3 \Omega$.

$\rightarrow$ voltage drop across $C = 1\mu F = 1 \times 10^{-6} F$.

$$\rightarrow \tau = R \cdot C = 10k\Omega \times 1\mu F = 10 \times 10^3 \Omega \times 10^{-6} F = 10^{-2} = 10^{-2}$$

$$= \frac{1}{100} = 10ms$$

$$\rightarrow V_C(t) = V_{app} \times (1 - e^{-t/\tau})$$

$$\rightarrow V_C(t) = V_{app} \times e^{-t/\tau}$$

$t: @ 0ms \rightarrow V_C(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V \times (1 - e^{-0/10 \times 10^{-3}}) = 1.8 \times (1 - 1) = 0V$ ✓

$t: @ 1ms \rightarrow V_C(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V \times (1 - e^{-1ms/10ms}) = 1.8 \times (1 - e^{-0.1}) = 0.171V$ ✓

$t: @ 2ms \rightarrow V_C(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V \times (1 - e^{-2ms/10ms}) = 1.8 \times (1 - e^{-0.2}) = 0.326V$ ✓

$$t: @ 0ms \rightarrow V_c(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V \times (1 - e^{-0}) = 1.8 \times (1 - 1) = \boxed{0V}$$

$$t: @ 1ms \rightarrow V_c(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V \times (1 - e^{-\frac{1ms}{10ms}}) = 1.8 (1 - e^{-0.1}) = \boxed{0.171V} \checkmark$$

$$t: @ 2ms \rightarrow V_c(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V \times (1 - e^{-\frac{2ms}{10ms}}) = 1.8 (1 - e^{-0.2}) = \boxed{0.326V} \checkmark$$

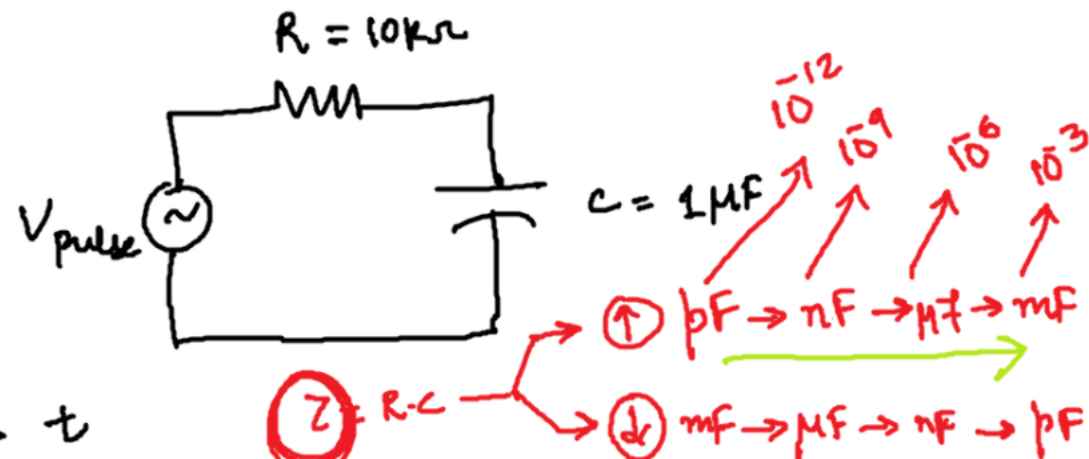
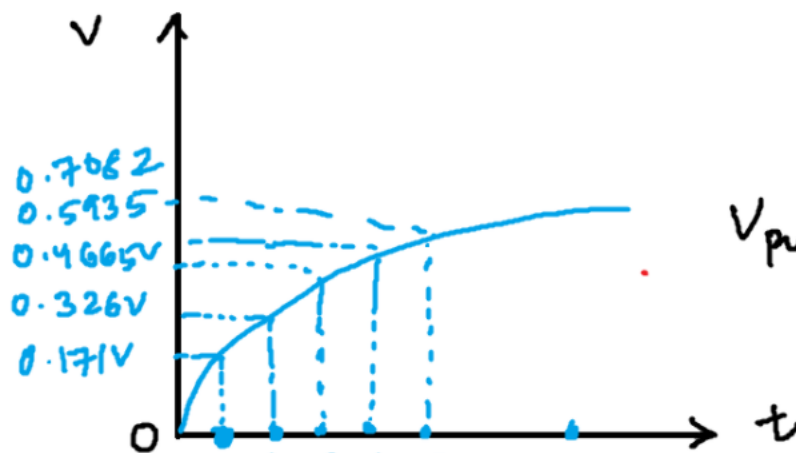
$$t: @ 3ms \rightarrow V_c(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8 (1 - e^{-0.3}) = \boxed{0.4665V} \checkmark$$

$$t: @ 4ms \rightarrow V_c(t) = 1.8 (1 - e^{-0.4}) = \boxed{0.5934V}$$

$$t: @ 5ms \rightarrow V_c(t) = 1.8 (1 - e^{-0.5}) = \boxed{0.7082V} \rightarrow 6ms = \boxed{0.8121V} \rightarrow 7ms = \boxed{0.9041V} \rightarrow 8ms = \boxed{0.9912V}$$

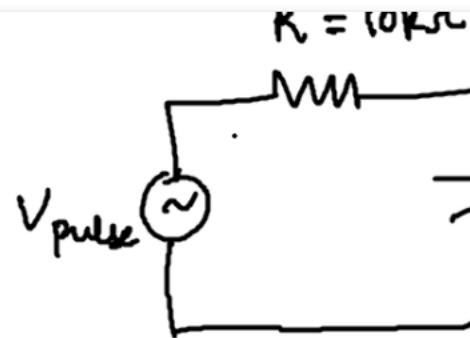
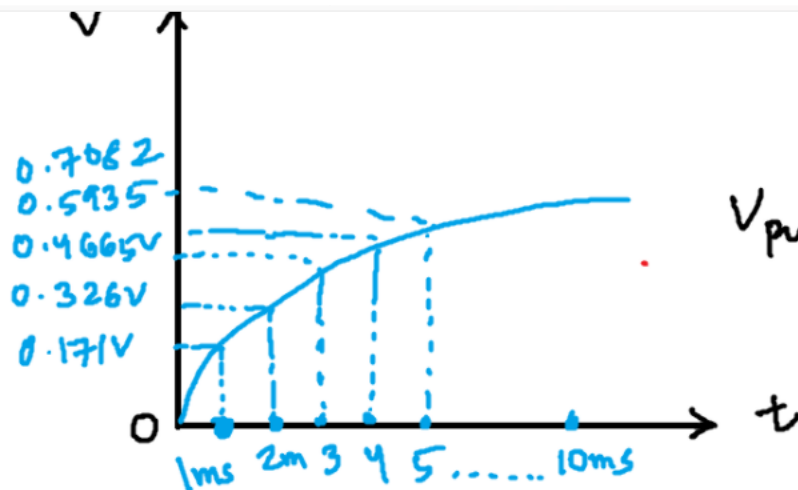
$\uparrow$ → FASTER charging
discharging

$\downarrow$ → SLOWER charging
discharging



$Z \uparrow \rightarrow$ FASTER charging discharging

$Z \downarrow \rightarrow$ SLOWER charging discharging



$C = 1\mu F$

$\uparrow$ pF $\rightarrow$ nF $\rightarrow$ μF $\rightarrow$ mF

$\downarrow$ mF $\rightarrow$ μF $\rightarrow$ nF $\rightarrow$ pF

(10^3) (10^{12})

$$V_C(t) = V_{\text{pulse}} \times (1 - e^{-t/\tau})$$

$$= 1.8V \left(1 - e^{-\frac{1\mu s}{10\mu s}}\right) = 1.8 \times (1 - e^{-0.1})$$

$$= \boxed{0.171V}$$

$V \rightarrow V_{\text{applied}}$



@ 2ms (t)

$$V(t) = 1.8 \times (1 - e^{-0.2})$$

discharging.

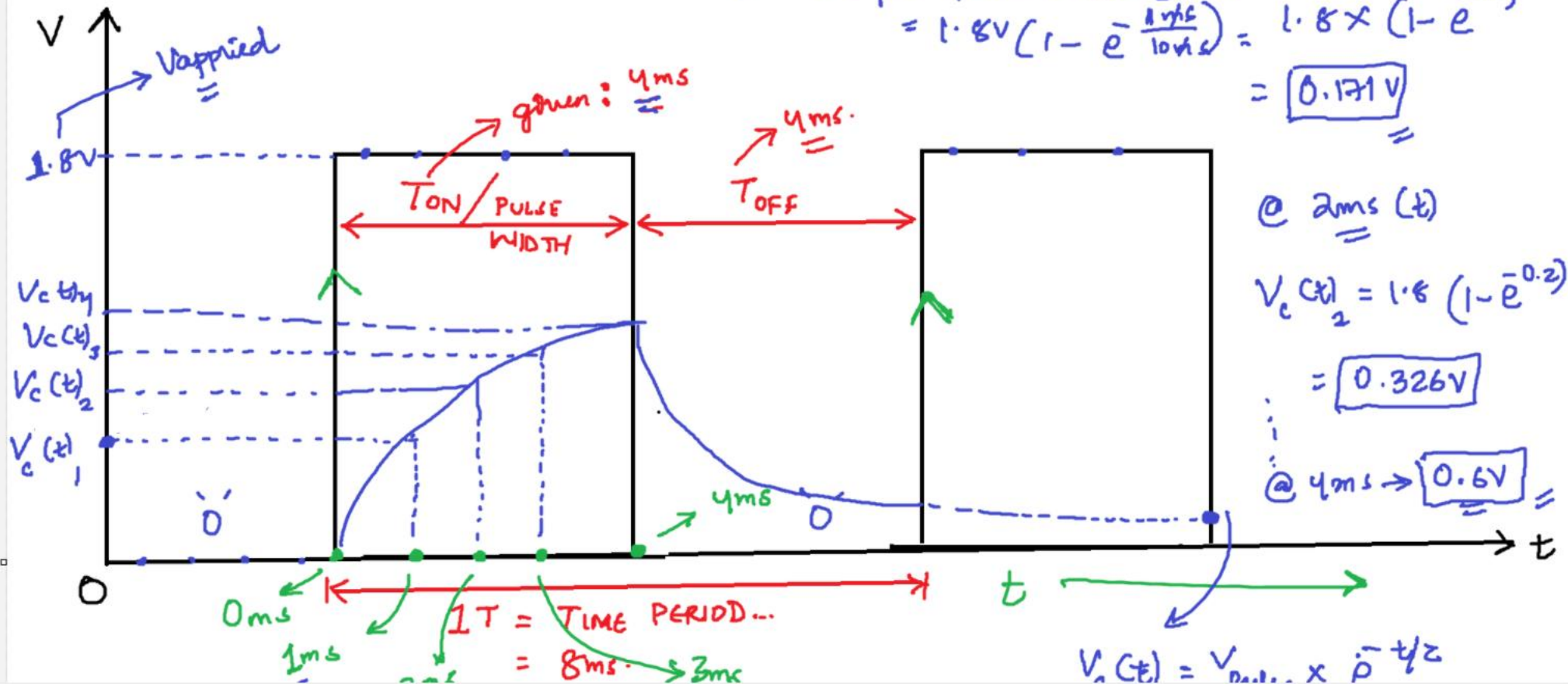
1ms 2ms 3 4 5..... 10ms

(d) $mt \rightarrow \mu t \rightarrow nt \rightarrow pt$
(10^3) (10^{12})

$$V_c(t)_1 = V_{\text{pulse}} \times (1 - e^{-t/\tau})$$

$$= 1.8V \left(1 - e^{-\frac{1\text{ms}}{10\text{ms}}}\right) = 1.8 \times (1 - e^{-0.1})$$

$$= \boxed{0.171V}$$



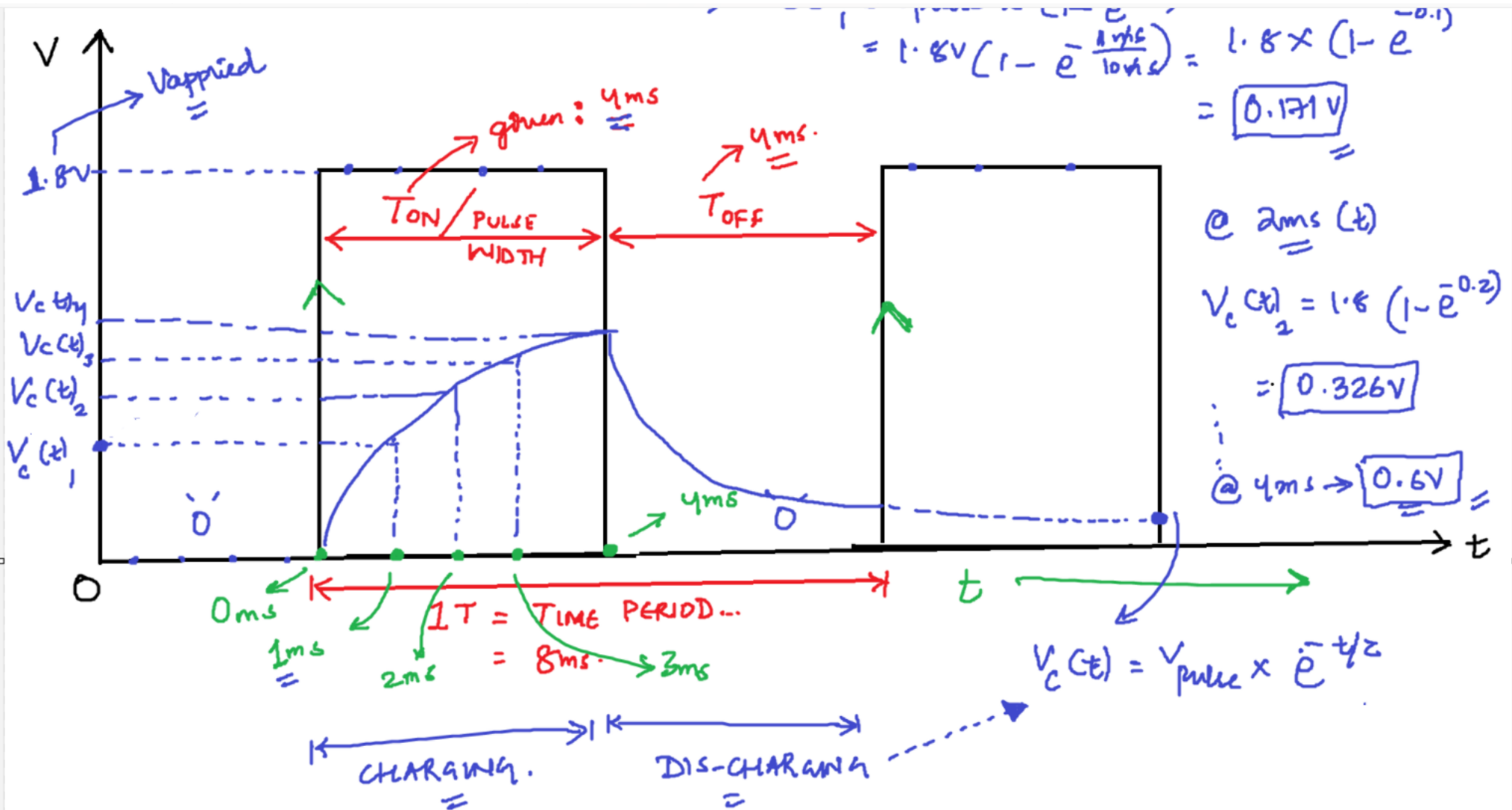
@ 2ms (t)

$$V_c(t)_2 = 1.8 (1 - e^{-0.2})$$

$$= \boxed{0.326V}$$

@ $4\text{ms} \rightarrow \boxed{0.6V}$

$$V_c(t) = V_{\text{pulse}} \times e^{-t/\tau}$$



CONCLUSION $\rightarrow \therefore T_{ON} = 4ms$

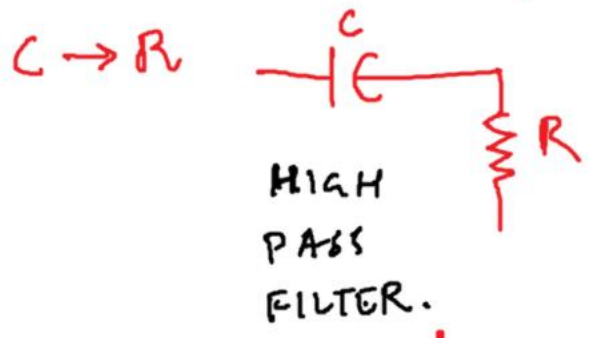
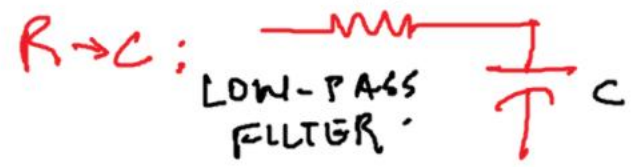
But @ 4ms ; Cap $\rightarrow 0.6V$ i.e. $\frac{1}{3}^{rd}$ of the Applied Voltage. (1.8V)

$\rightarrow$ Hence; it Needs Enough time to get CHARGED to its FULL POTENTIAL.

$\hookrightarrow$ This is NOTHING BUT DELAY !!

S_o; R-C ckt produces delays to the ckt.

$\boxed{*}$ ALSO ACTS AS A "FILTER"
 $\downarrow$
SS, DSP...

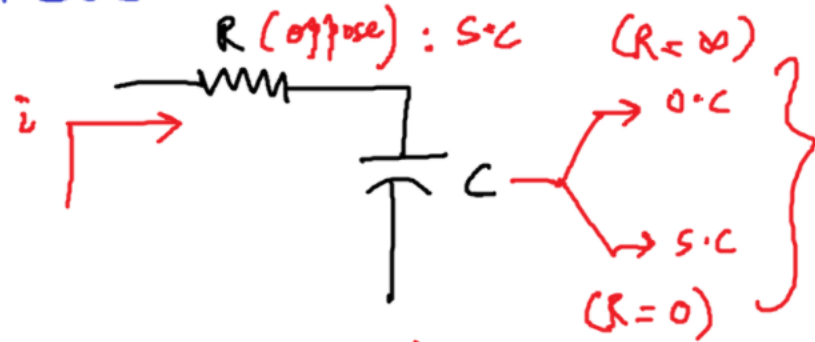


CONCEPT OF LOW-PASS FILTER :-

R (oppose) : $S \propto C$ ($R < \infty$)

=
i.e. ATTENUATE
1.79V
=

CONCEPT OF LOW-PASS FILTER :-



$t \rightarrow$
 $f = \frac{1}{t}$ Ⓢ

$\bar{R} \rightarrow \infty \rightarrow O.C$
 $\bar{R}' \rightarrow 0 \rightarrow S.C$

PASS
 FILTER.

$\rightarrow$ impedance.
 $R_c = X_c = \frac{1}{\omega C} = \frac{1}{2\pi f C}$

- D.C $\rightarrow f = 0 \text{ Hz} \rightarrow 0 \text{ Hz}$
- A.C $\rightarrow 50-60 \text{ Hz} = \infty \text{ Hz}$

$f = \infty$

$R_c = X_c = \frac{1}{2\pi f C} = \frac{1}{\infty} = 0$

$R = 0 \rightarrow S.C$

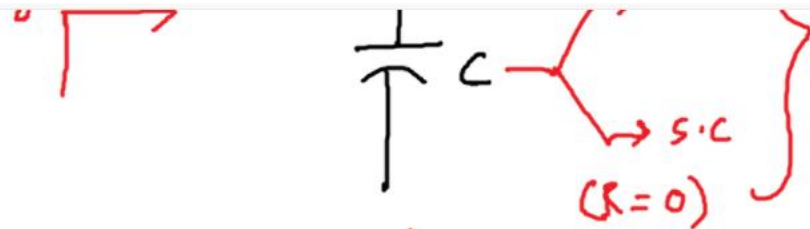
$f = 0$

$R_c = X_c = \frac{1}{2\pi f C} = \frac{1}{0} = \infty$

$R = \infty \rightarrow O.C$



$\rightarrow$ HIGH PASS } A-C Analysis.



$$\left. \begin{array}{l} R \rightarrow \infty \rightarrow \text{O.C} \\ R' \rightarrow 0 \rightarrow \text{S.C} \end{array} \right\} R_c = X_c = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

- D.C $\rightarrow f = 0 \text{ Hz} \rightarrow 0 \text{ Hz}$
- A.C $\rightarrow 50-60 \text{ Hz} = \infty \text{ Hz}$

$t \rightarrow$
 $f = \frac{1}{t}$ ①

$$f = \infty$$

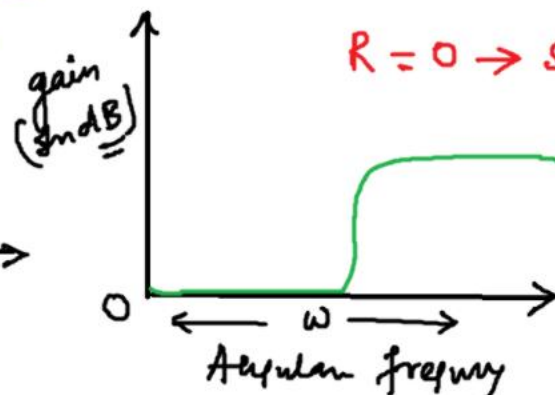
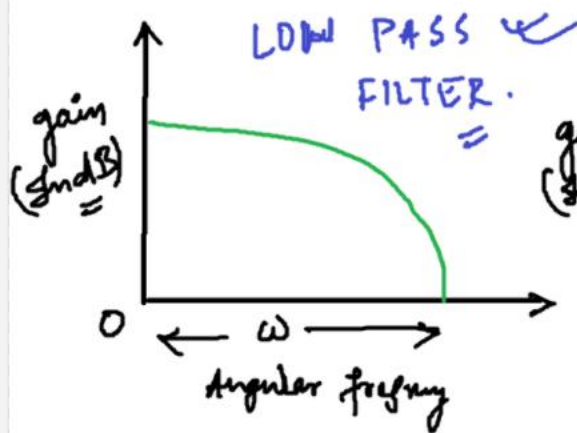
$$f = 0$$

$$R_c = X_c = \frac{1}{2\pi f C} = \frac{1}{\infty} = 0$$

$$R_c = X_c = \frac{1}{2\pi f C} = \frac{1}{0} = \infty$$

$$R = 0 \rightarrow \text{S.C}$$

$$R = \infty \rightarrow \text{O.C}$$



HIGH PASS FILTER

A-C Analysis
AC amplitude ✓
Phase = 0° ✓



Navigator

Schematic
rc_ckt_dc

OBJECTS

All
Instances 5
Nets 4
Pins 2
Nets and Pins

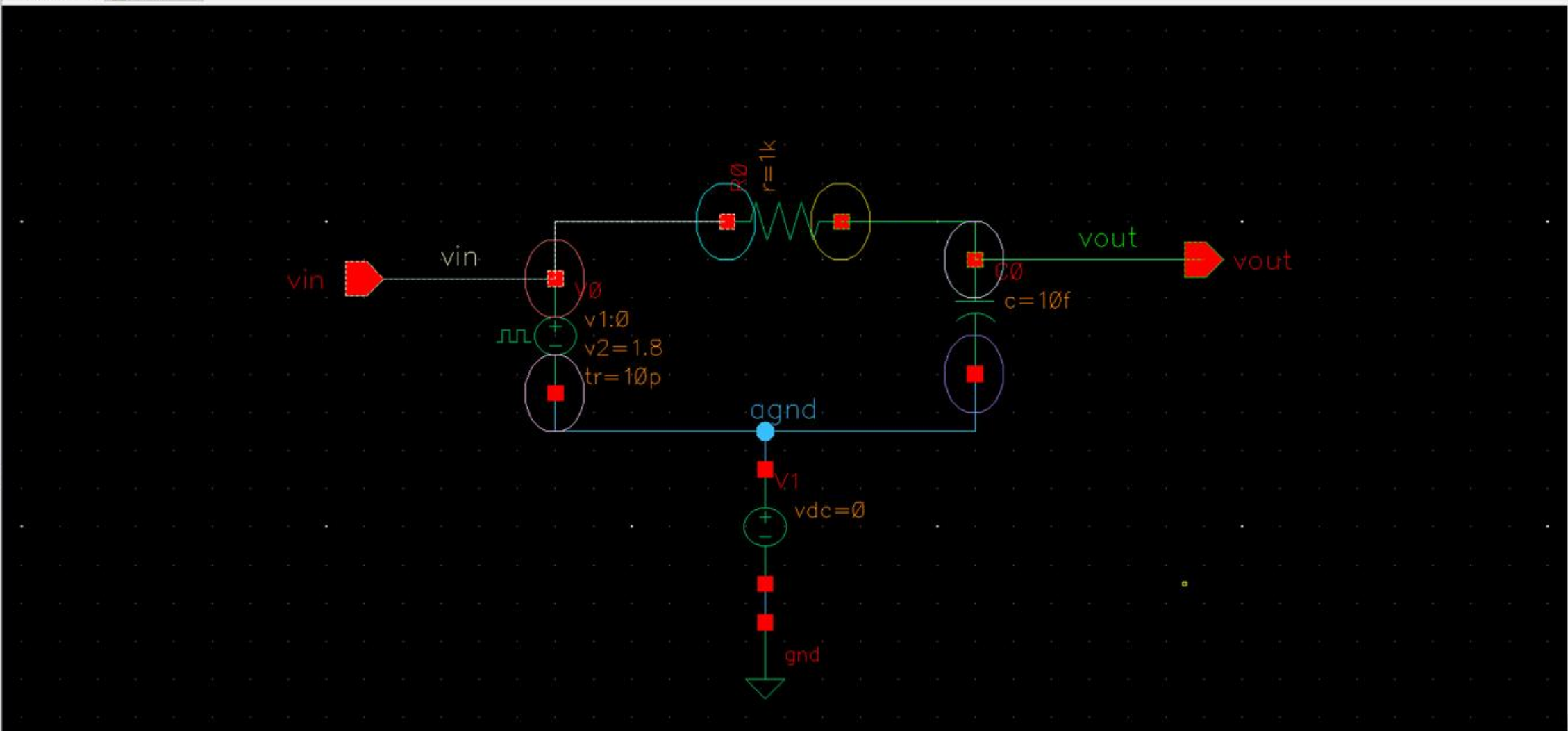
GROUPS

Cells
Types

Property Editor

Schematic Attributes

Library	sailesh
Cell	rc_ckt_dc
View	schematic
Mode	editable
Last Saved	Sat Feb 8 ...
Locked By	vlab14@td...
Units	inch

 $R = 1k \Omega ; C = 10f F$ 

mouse L: schSingleSelectPt()

M: hiZoomAbsoluteScale(getCurrentWindow() 0.9)

R: schHiMousePopUp()

1(2)

Cmd: Set: 0 Status: Ready Results: ExplorerRun.0 sailesh rc_ckt_dc schematic Simulator: spectre aps Batch

Filter Filter

sailesh_rc_ckt_dc_1

Simulator spectre

Analyses

- tran 0 800p moderate
- dc t 0 1.8 Automatic Start-Stop /V0

Click to add analysis

Design Variables

Click to add variable

Parameters

CornerRadius

Reliability Analyses

Monte Carlo Sampling

Checks/Asserts

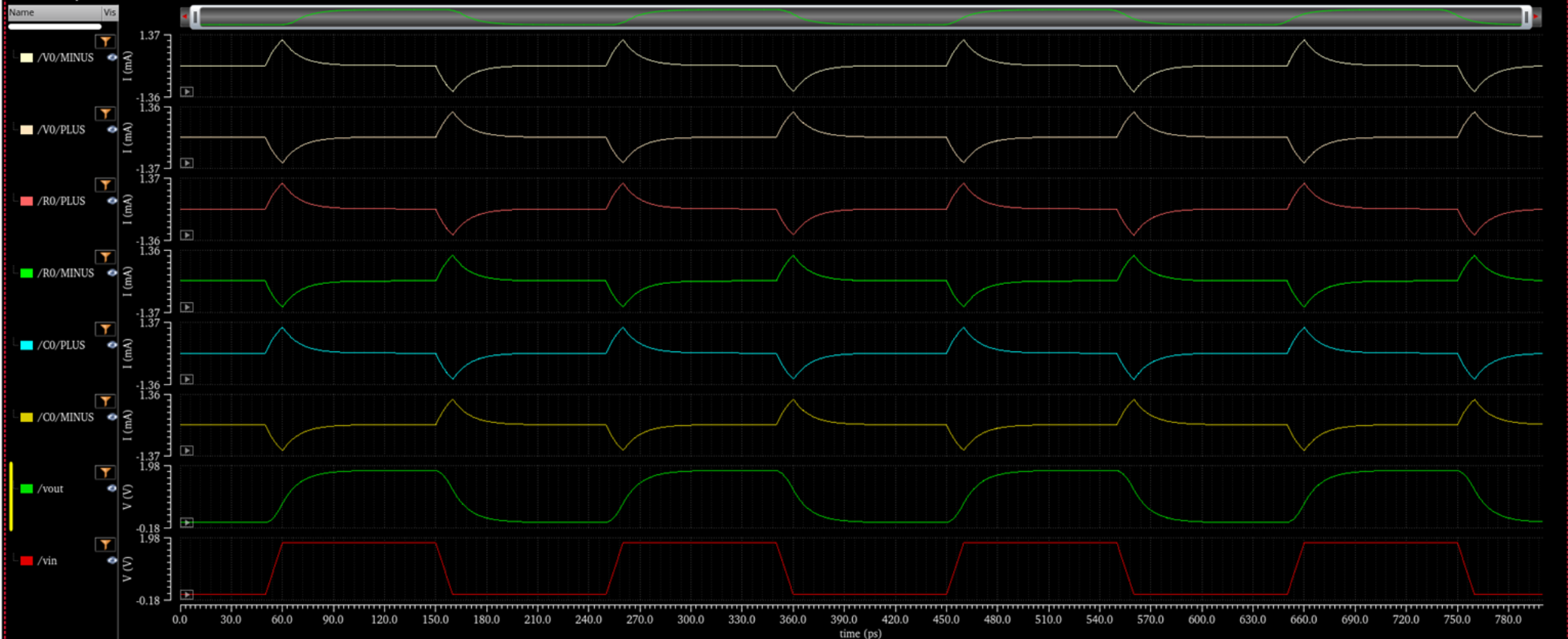
Name	Type	Details	Value	Plot	Save	Spec
signal (I)	/V0/MINUS			☒	☒	
signal (I)	/V0/PLUS			☒	☒	
signal	/vout			☒	☒	
signal (I)	/R0/PLUS			☒	☒	
signal (I)	/R0/MINUS			☒	☒	
signal (I)	/C0/PLUS			☒	☒	
signal (I)	/C0/MINUS			☒	☒	
signal	/vin			☒	☒	



sailesh_rc_ckt_dc_1 x

Transient Response

Sat Feb 8 10:57:13 2025 1



mouse L:

M:

R:

6(20) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom



sailesh_rc_ckt_dc_1

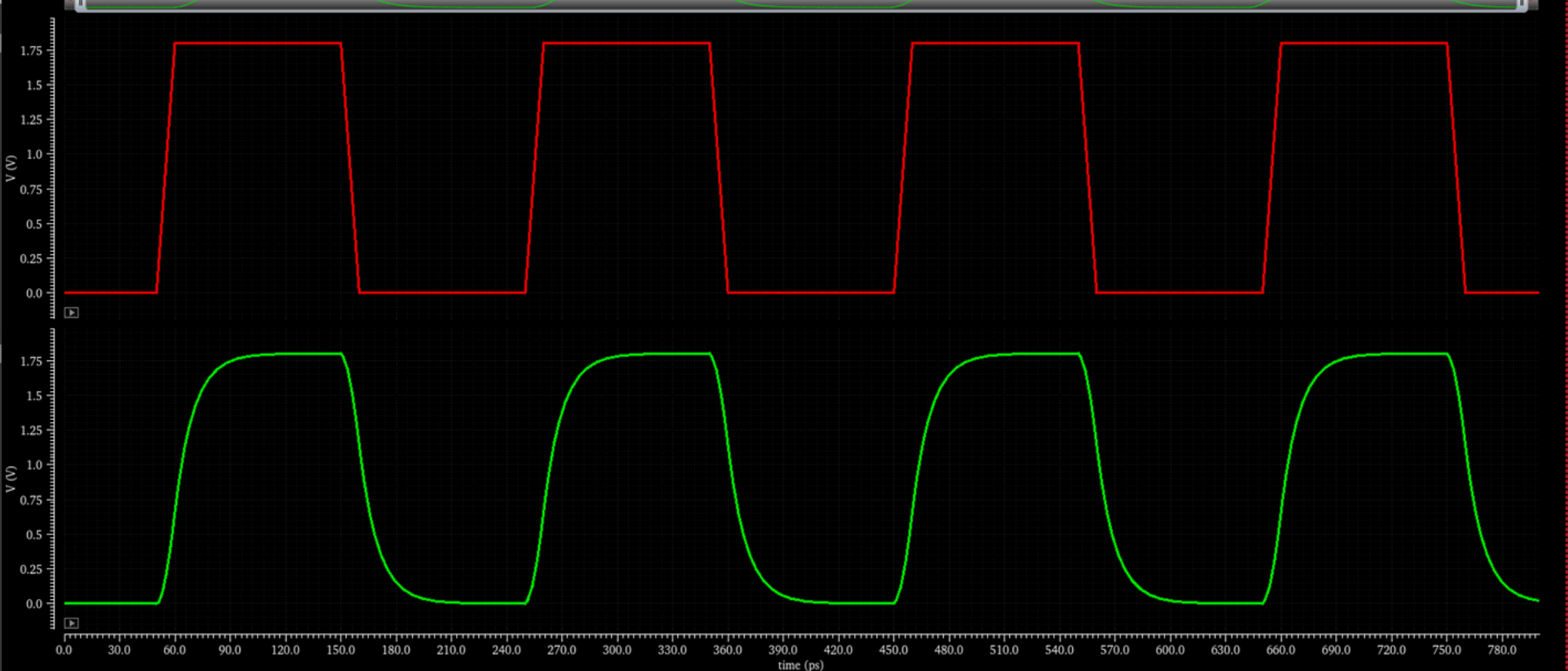
Transient Response

Sat Feb 8 10:57:13 2025

Name Vis

/vin

/vout



mouse L:

M:

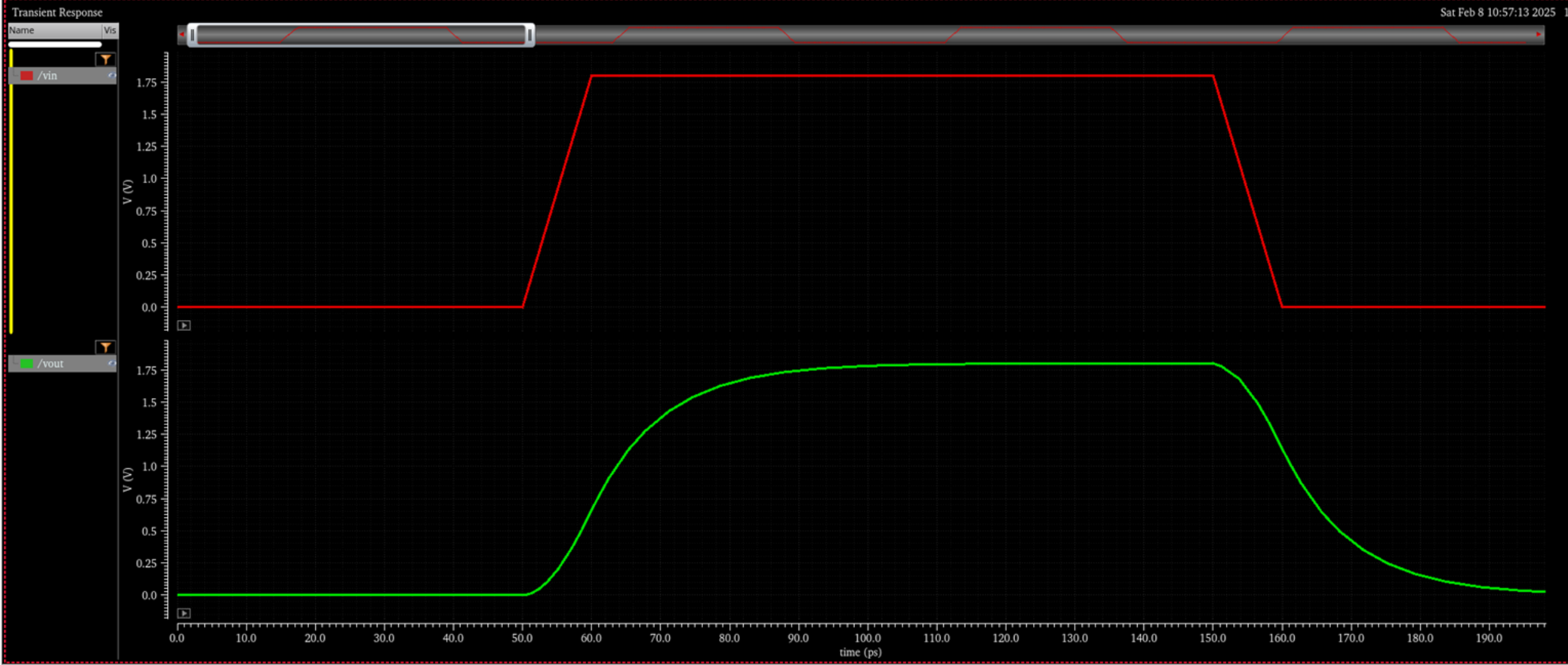
R:

6(20) no error

File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help

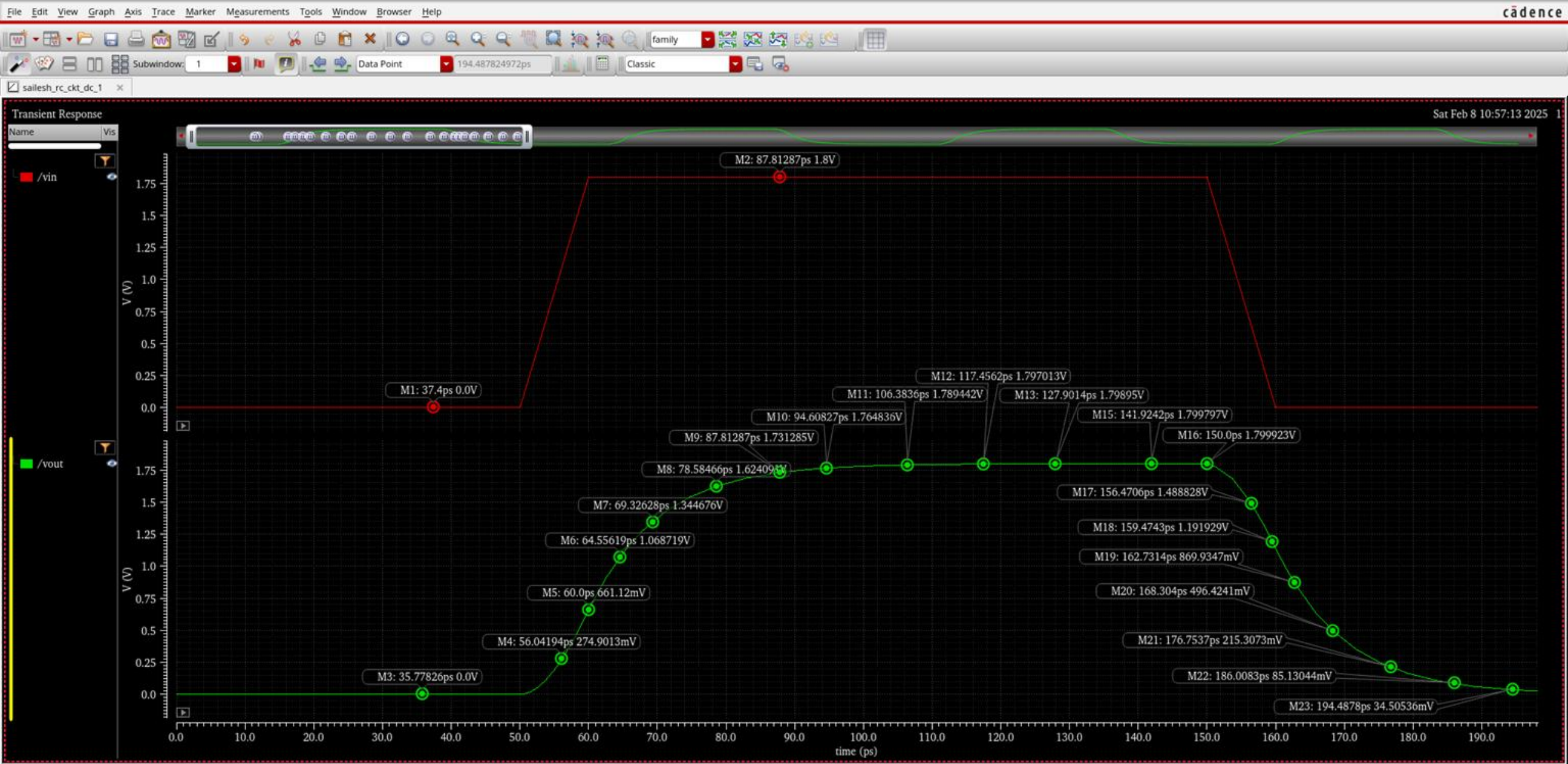


sailesh_rc_ckt_dc_1



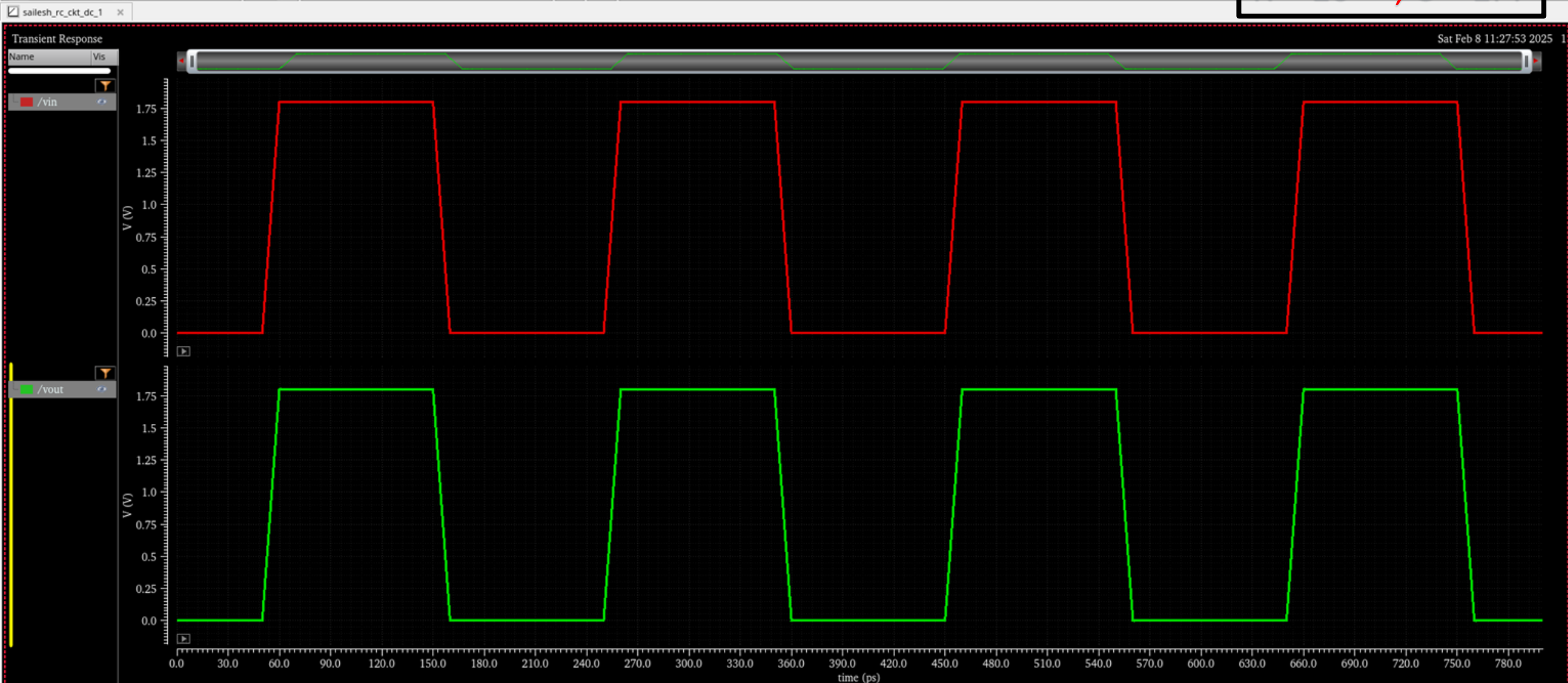
mouse L: M: R:

6(20) Trace: Avin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom





$R = 10 \Omega$; $C = 1 \text{fF}$





Subwindow: 1 Data Point: 737.085445548ps Classic

sailesh_rc_ckt_dc_1

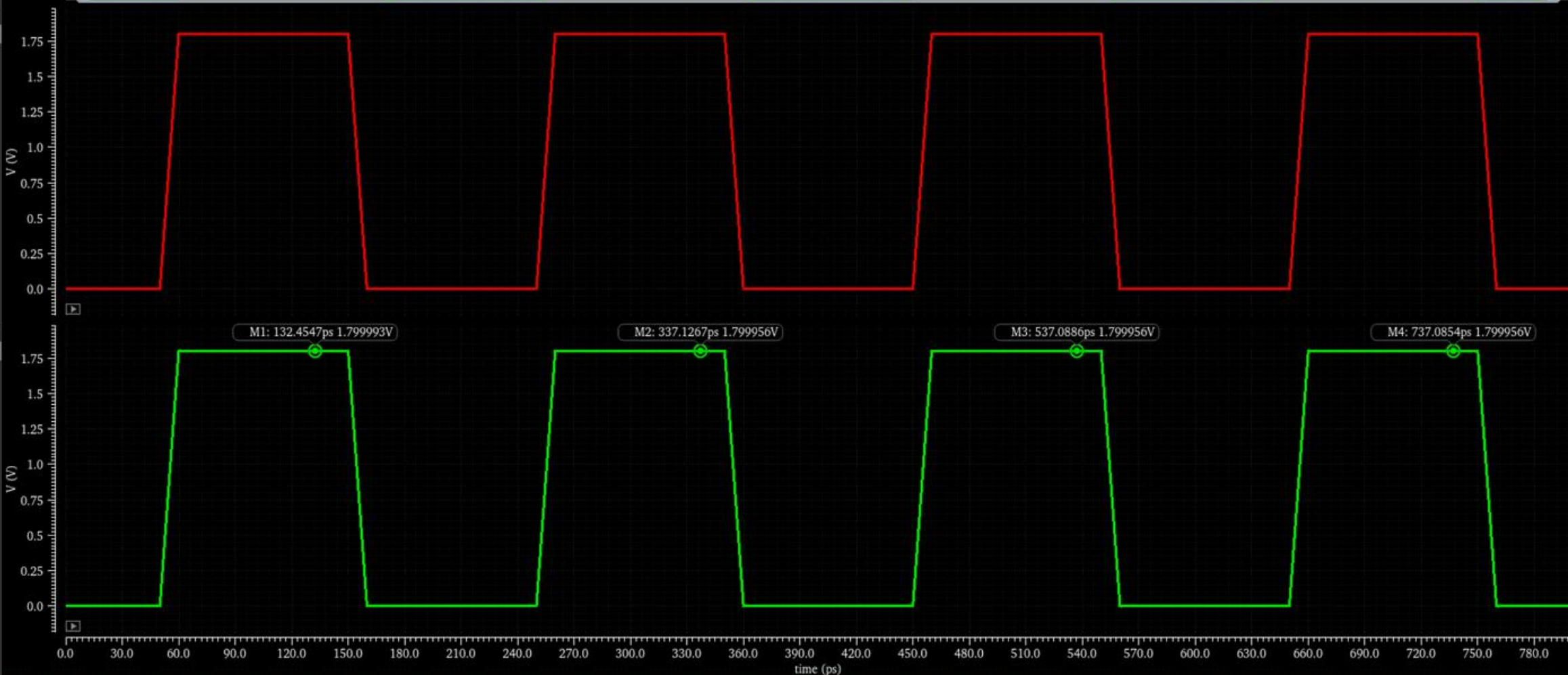
Transient Response

Sat Feb 8 11:27:53 2025 1

Name Vis

/vin

/vout

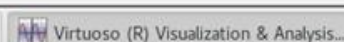


mouse L:

M:

R:

4(11) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom





$R = 10 \Omega$; $C = 10 \text{fF}$

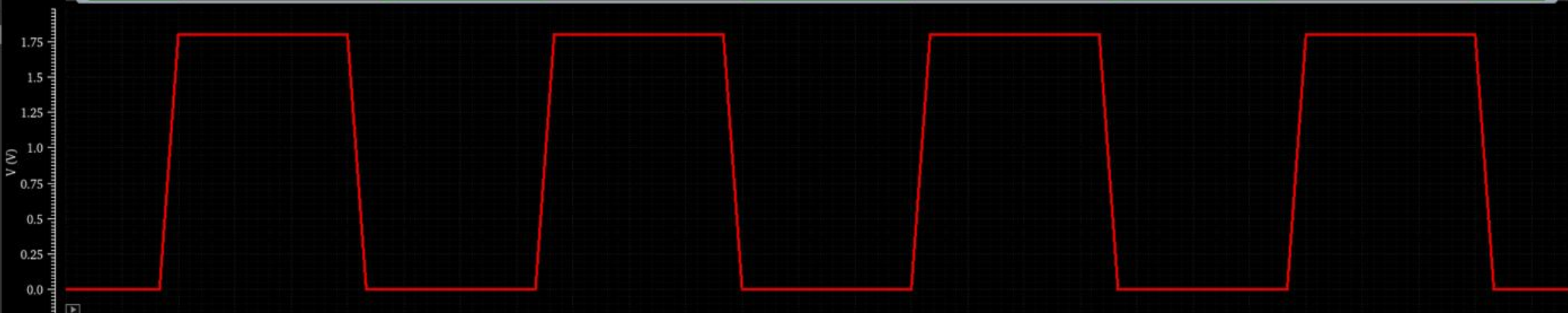
sailesh_rc_ckt_dc_1

Transient Response

Sat Feb 8 11:30:27 2025 1

Name Vis

/vin



/vout



mouse L:

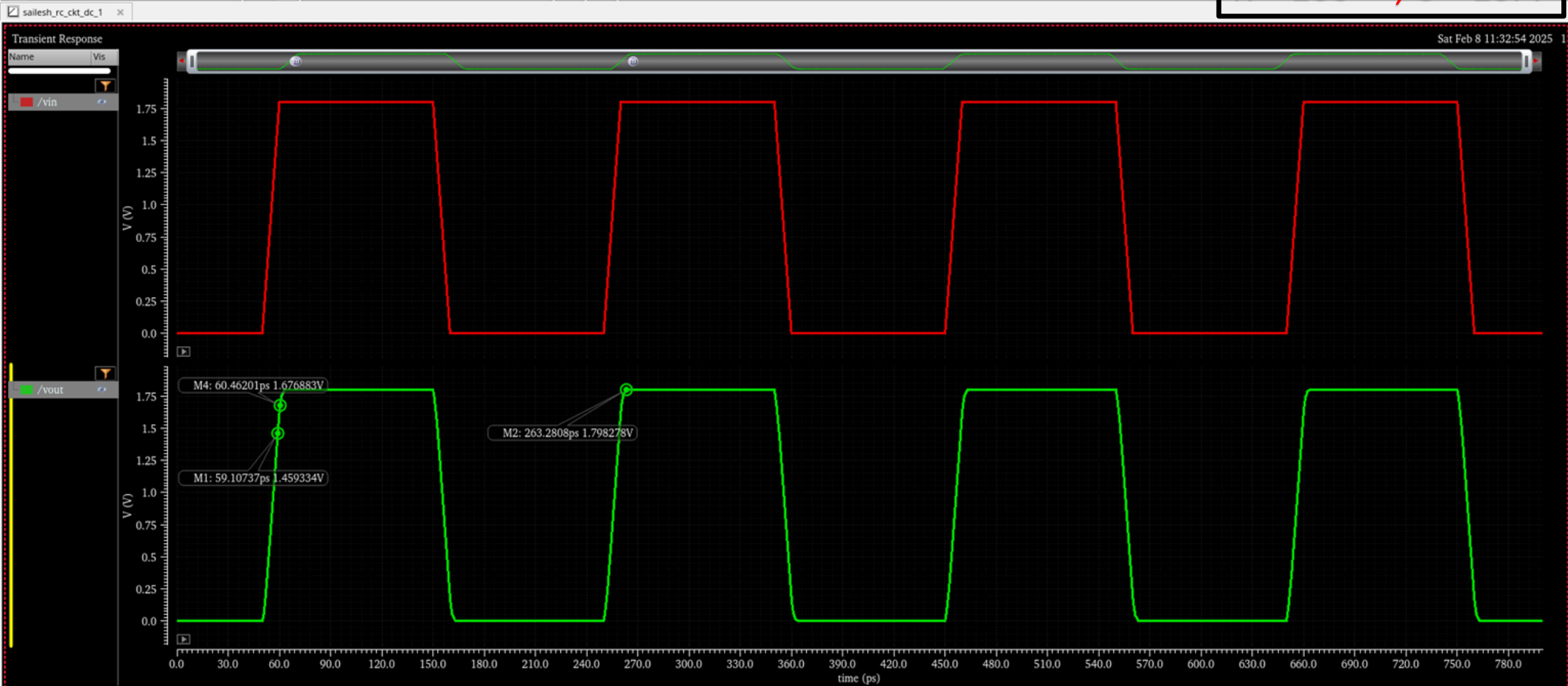
M:

R:

5(14) Trace: Avout; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom



$R = 100 \Omega$; $C = 10f F$





$R = 100 \Omega$; $C = 100f F$

sailesh_rc_ckt_dc_1 x

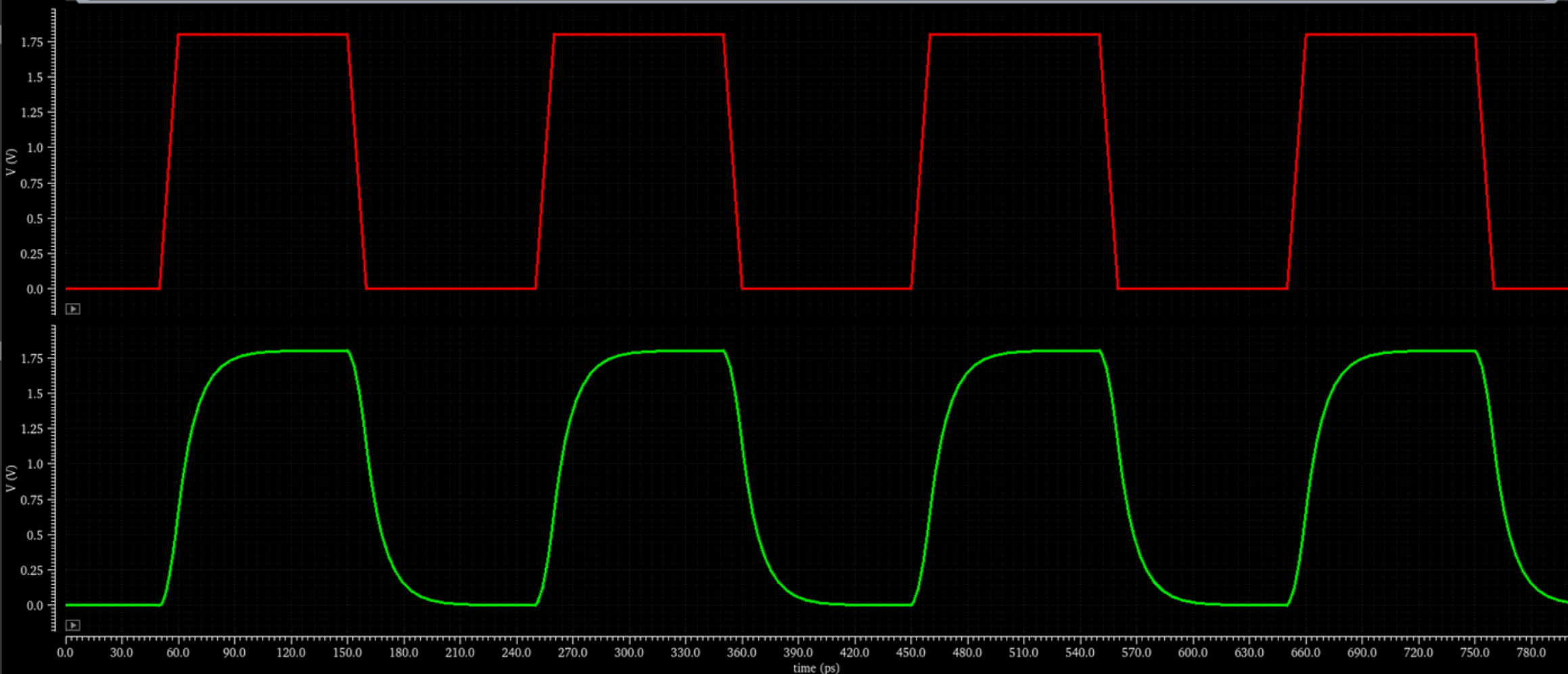
Transient Response

Sat Feb 8 11:35:01 2025 1

Name Vis

/vin

/vout

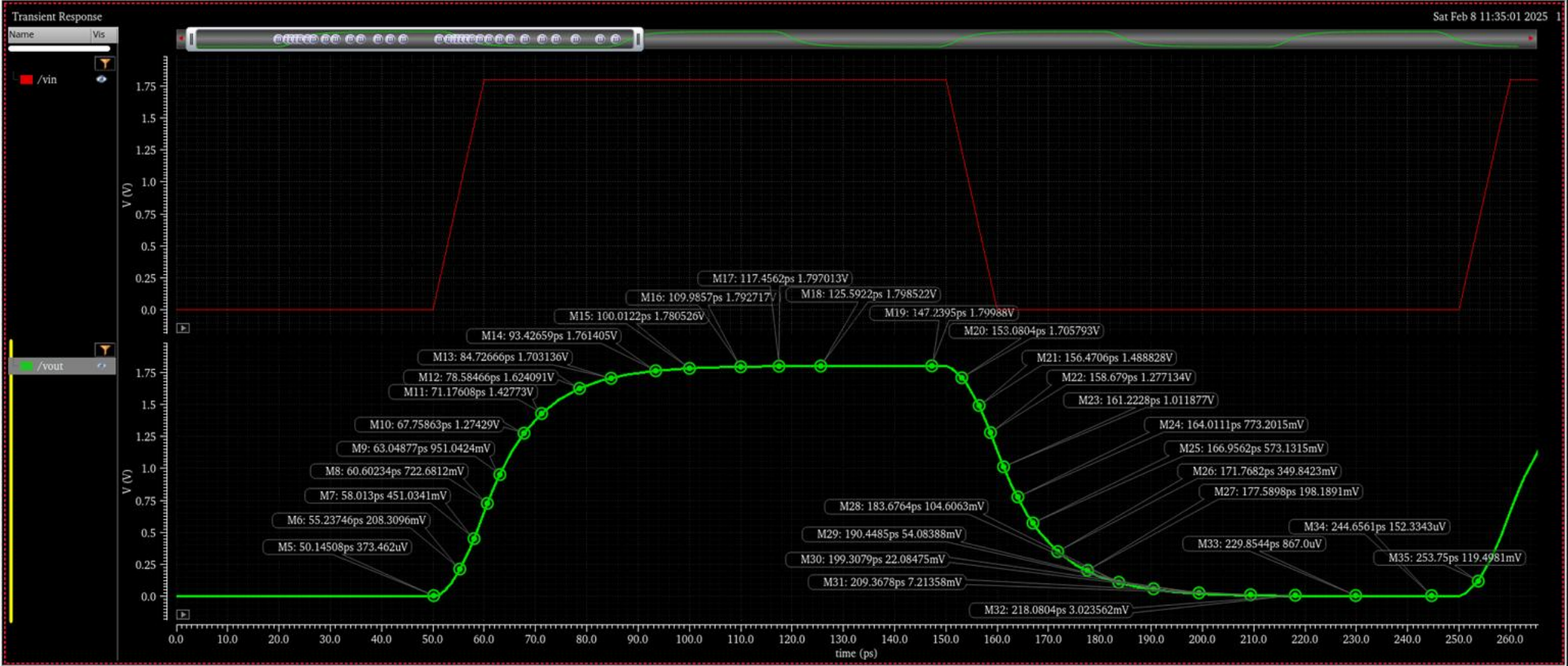


mouse L:

M:

R:

7(21) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom





$R = 1k\ \Omega ; C = 10f\ F$

sailesh_rc_ckt_dc_1 x

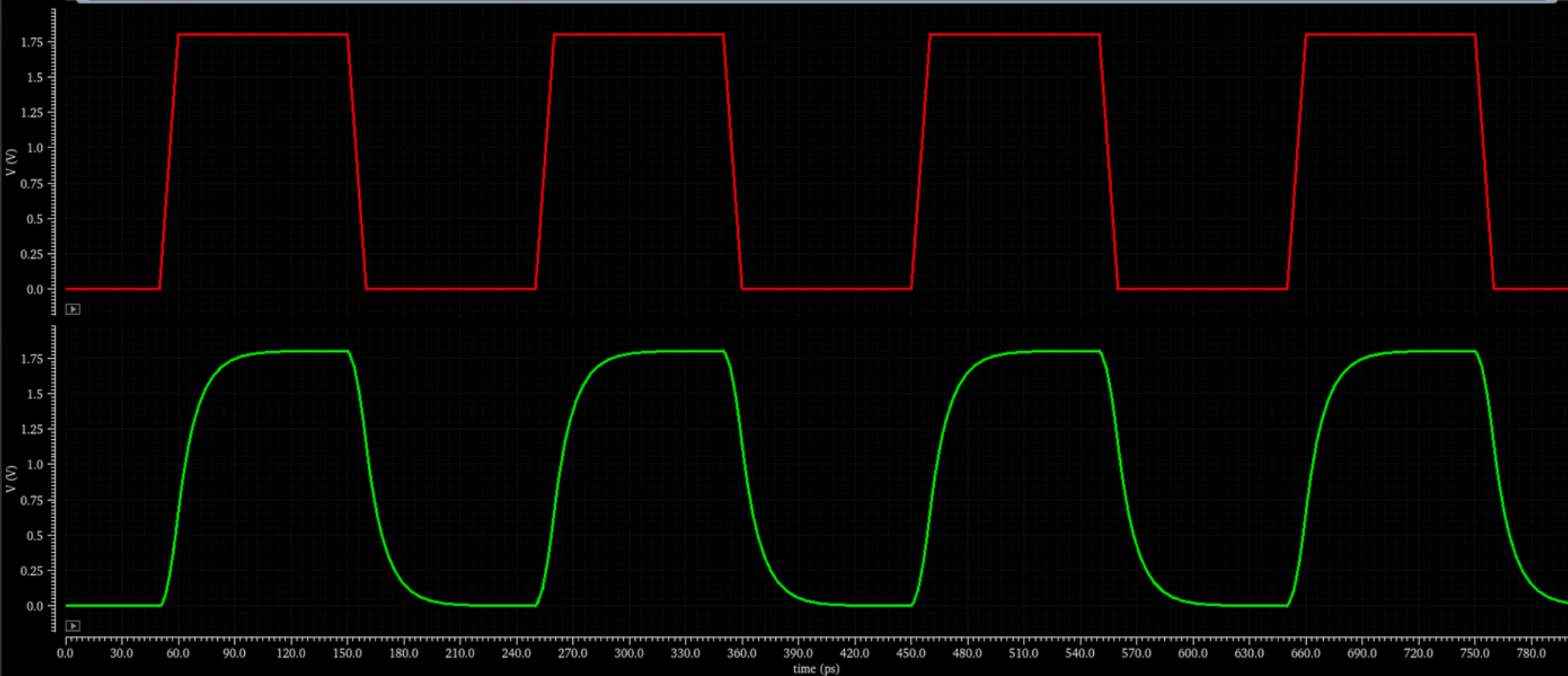
Transient Response

Sat Feb 8 11:44:06 2025 1

Name Vis

/vin

/vout



mouse L:

M:

R:

8(24) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom

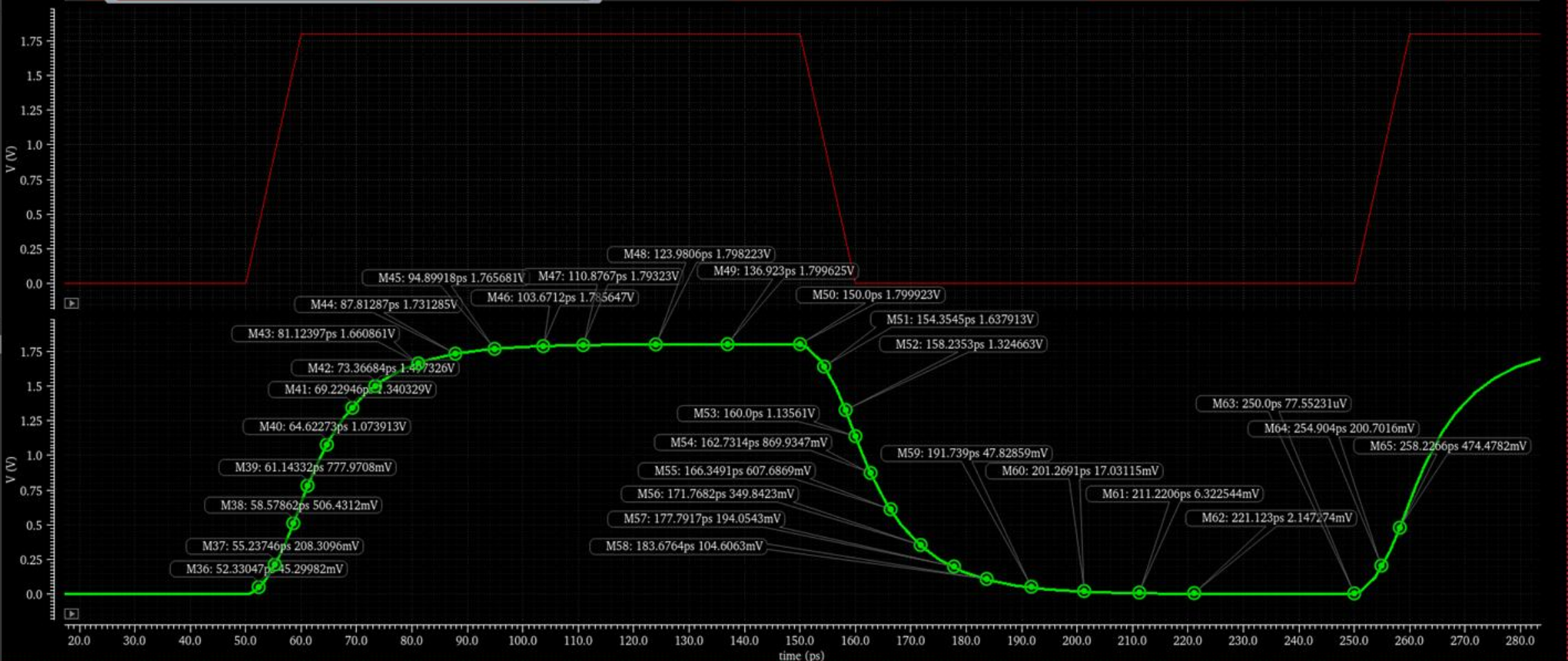
Transient Response

Sat Feb 8 11:44:06 2025

Name Vis

/vin

/vout



mouse L:

M:

R:

8(24) Trace: Avout; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom



$R = 1k\ \Omega ; C = 100f\ F$

sailesh_rc_ckt_dc_1 ×

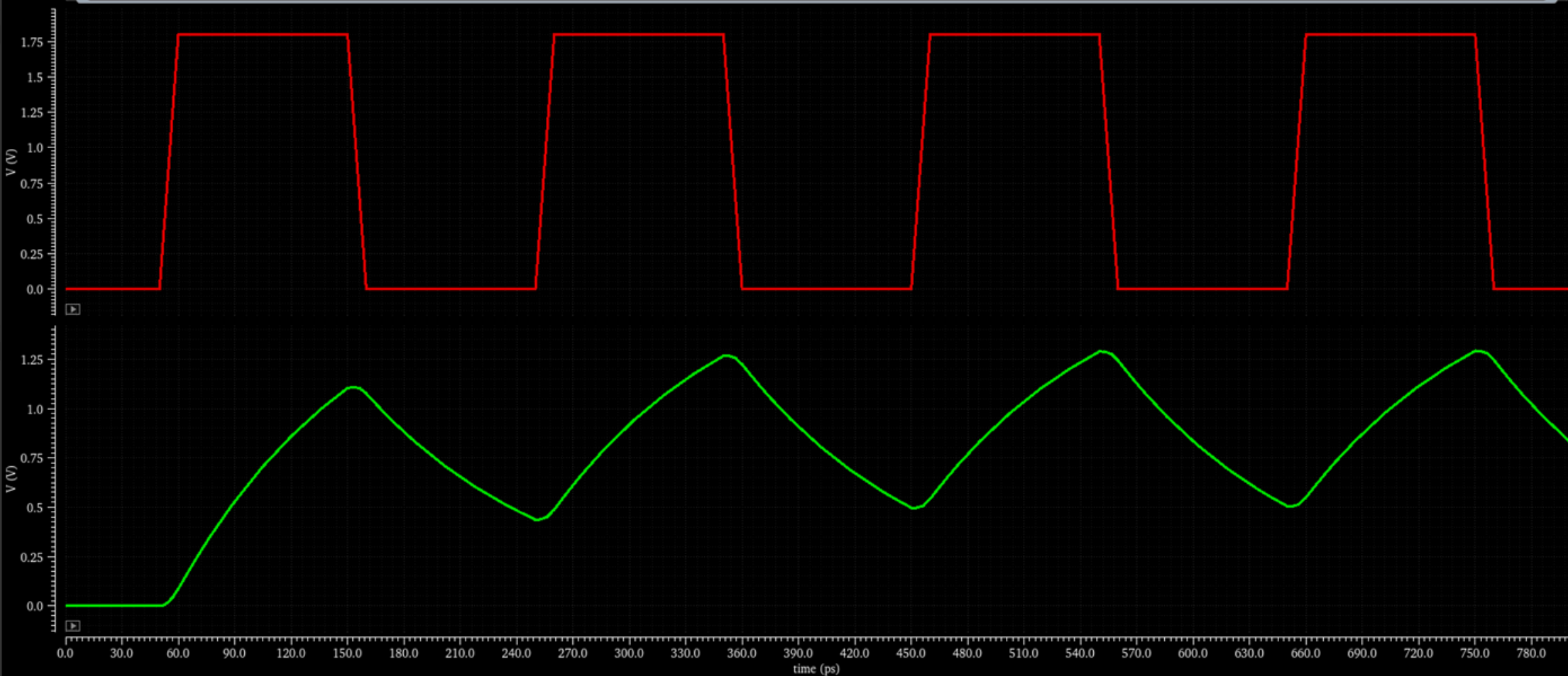
Transient Response

Sat Feb 8 11:47:27 2025 1

Name Vis

/vin

/vout



mouse L:

M:

R:

9(27) no error

Virtuoso (R) Visualization & Analysis XL: sailesh rc_ckt_dc maestro

File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help

cadence



Subwindow: 1 Data Point: 750.0ps Classic

sailesh_rc_ckt_dc_1

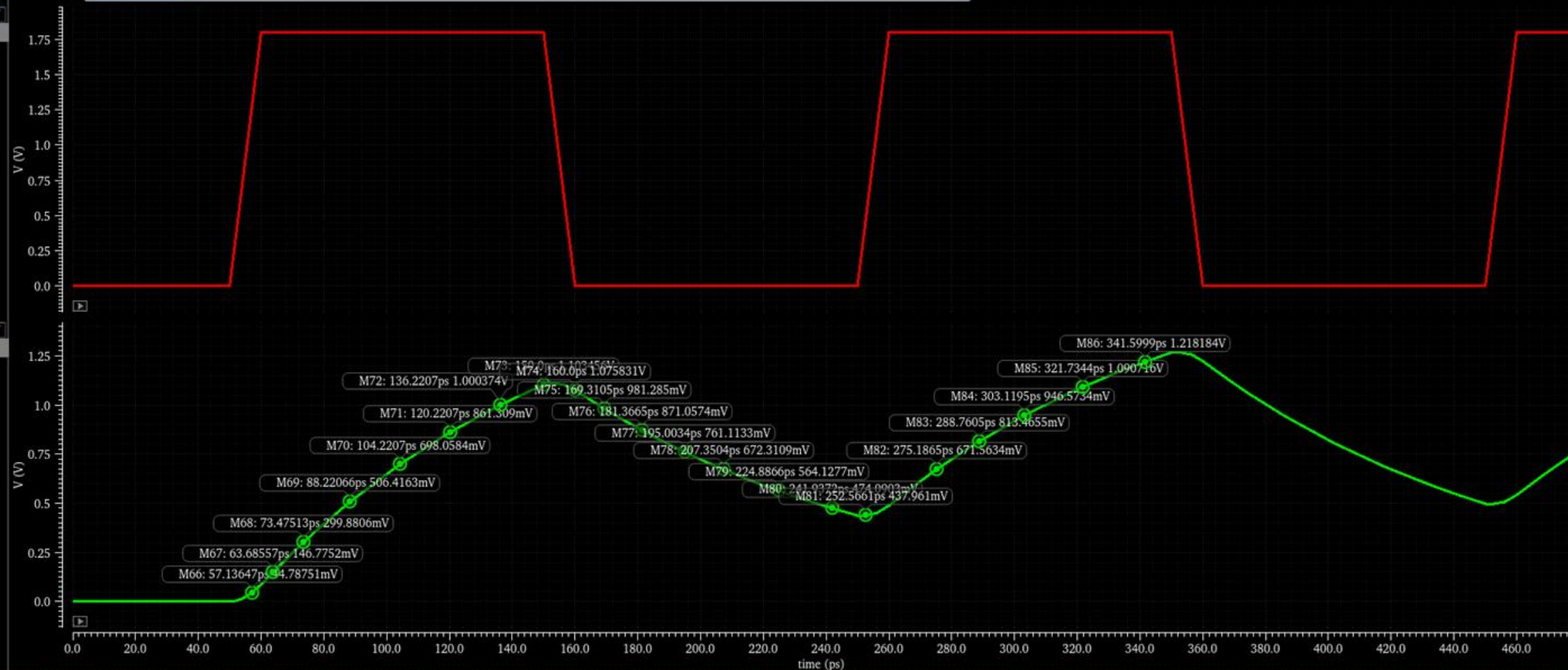
Transient Response

Sat Feb 8 11:47:27 2025 1

Name Vis

/vin

/vout

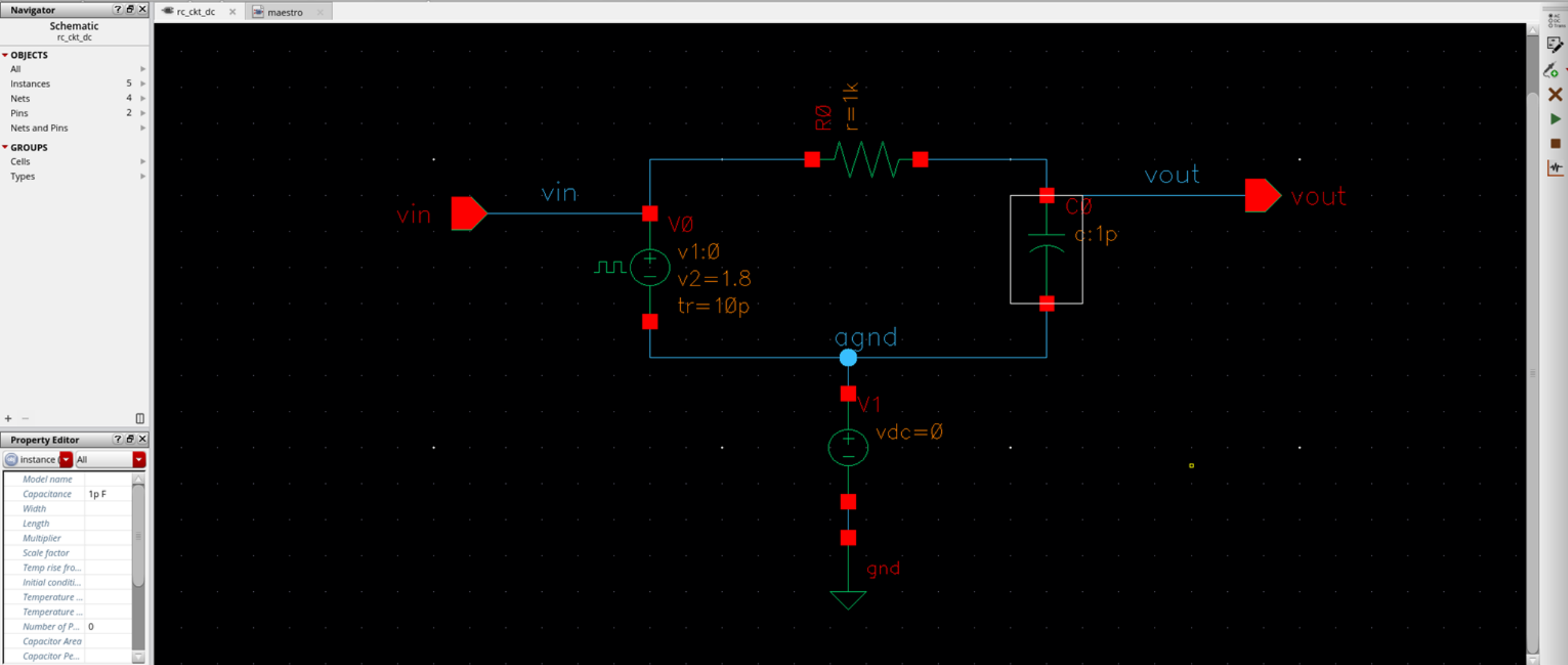


mouse L:

M:

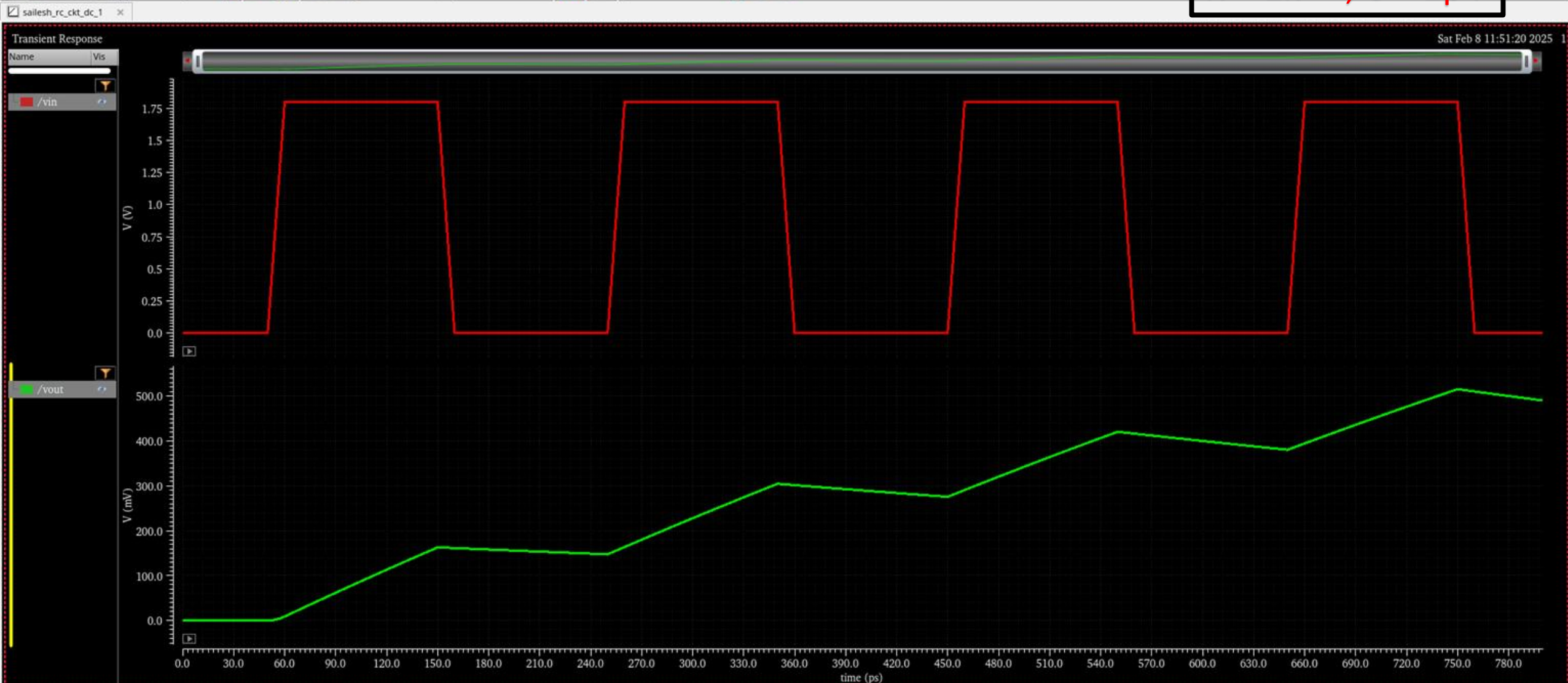
R:

9(27) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom





$R = 1k\ \Omega$; $C = 1p\ F$





$R = 1k\ \Omega$; $C = 1p\ F$

sailesh_rc_ckt_dc_1

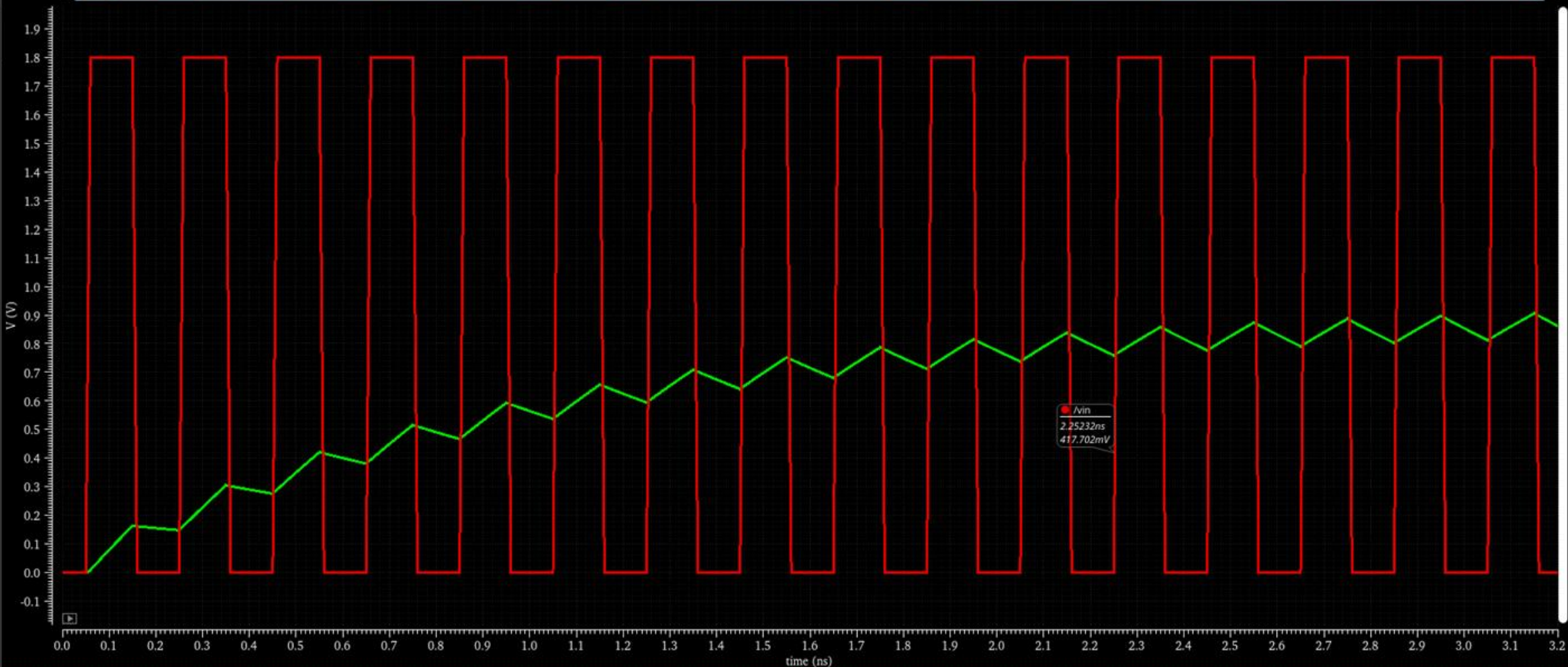
Transient Response

Sat Feb 8 13:03:12 2025 1

Name Vis

/vout

/vin



mouse L:

M:

R:

23(70) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom



- All
- Instances 5
- Nets 4
- Pins 2
- Nets and Pins

- Cells
- Types

Model name	1p F
Capacitance	
Width	
Length	
Multiplier	
Scale factor	
Temp rise fro...	
Initial conditi...	
Temperature ...	
Temperature ...	
Number of P...	0
Capacitor Area	
Capacitor Pe...	
Name	C0

Choosing Analyses -- ADE Explorer ✕

Analysis	☒ tran	☐ dc	☐ ac	☐ noise
	☐ xf	☐ sens	☐ dcmatch	☐ acmatch
	☐ stb	☐ pz	☐ lf	☐ sp
	☐ envlp	☐ pss	☐ pac	☐ pstb
	☐ pnoise	☐ pxf	☐ psp	☐ qpss
	☐ qpac	☐ qpnoise	☐ qpxf	☐ qpss
	☐ hb	☐ hbac	☐ hbstb	☐ hbnoise
	☐ hbsp	☐ hbxp		

Transient Analysis

Stop Time

Accuracy Defaults (errpreset)

☐ conservative
 ☒ moderate
 ☐ liberal

☐ Transient Noise

Dynamic Parameter ☐

Enabled ☒

Options...

OK
Cancel
Defaults
Apply
Help



$R = 10k \Omega ; C = 1p F$

sailesh_rc_ckt_dc_1

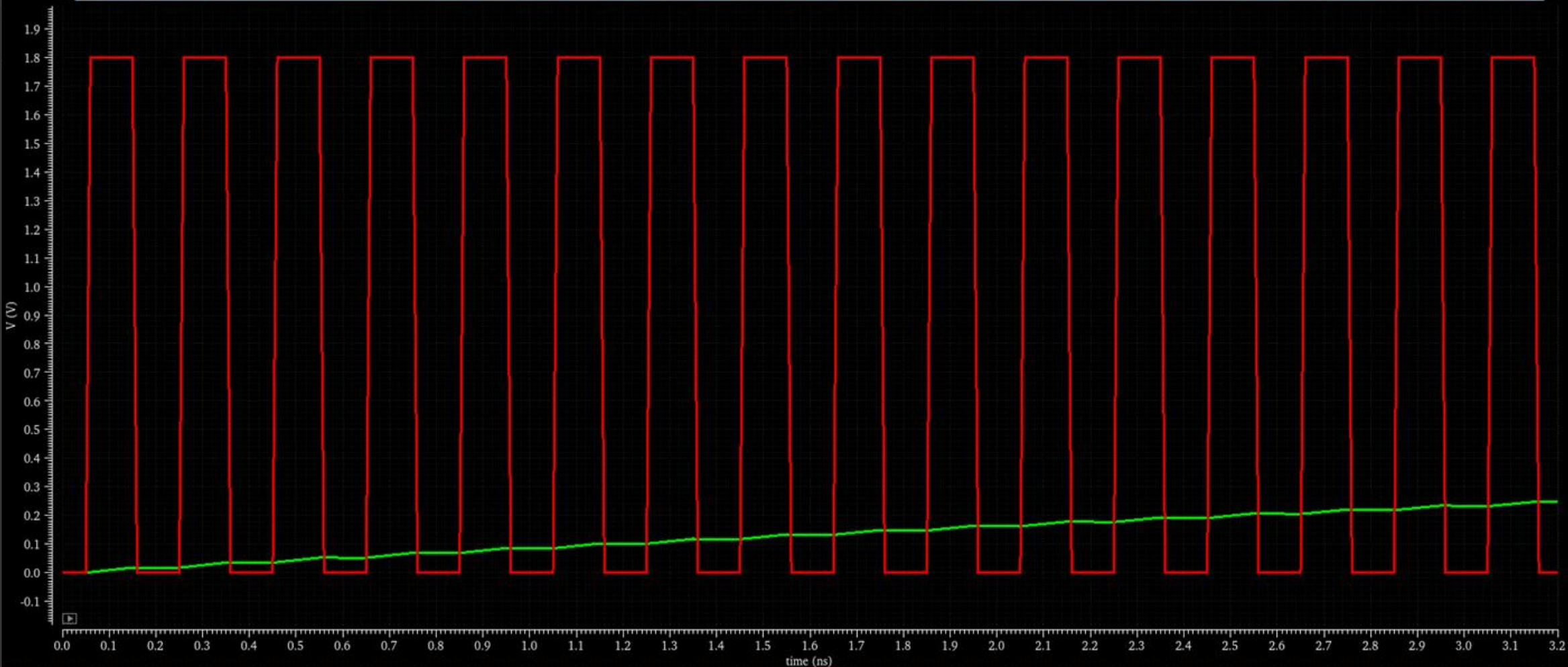
Transient Response

Sat Feb 8 13:05:23 2025 1

Name Vis

/vout

/vin



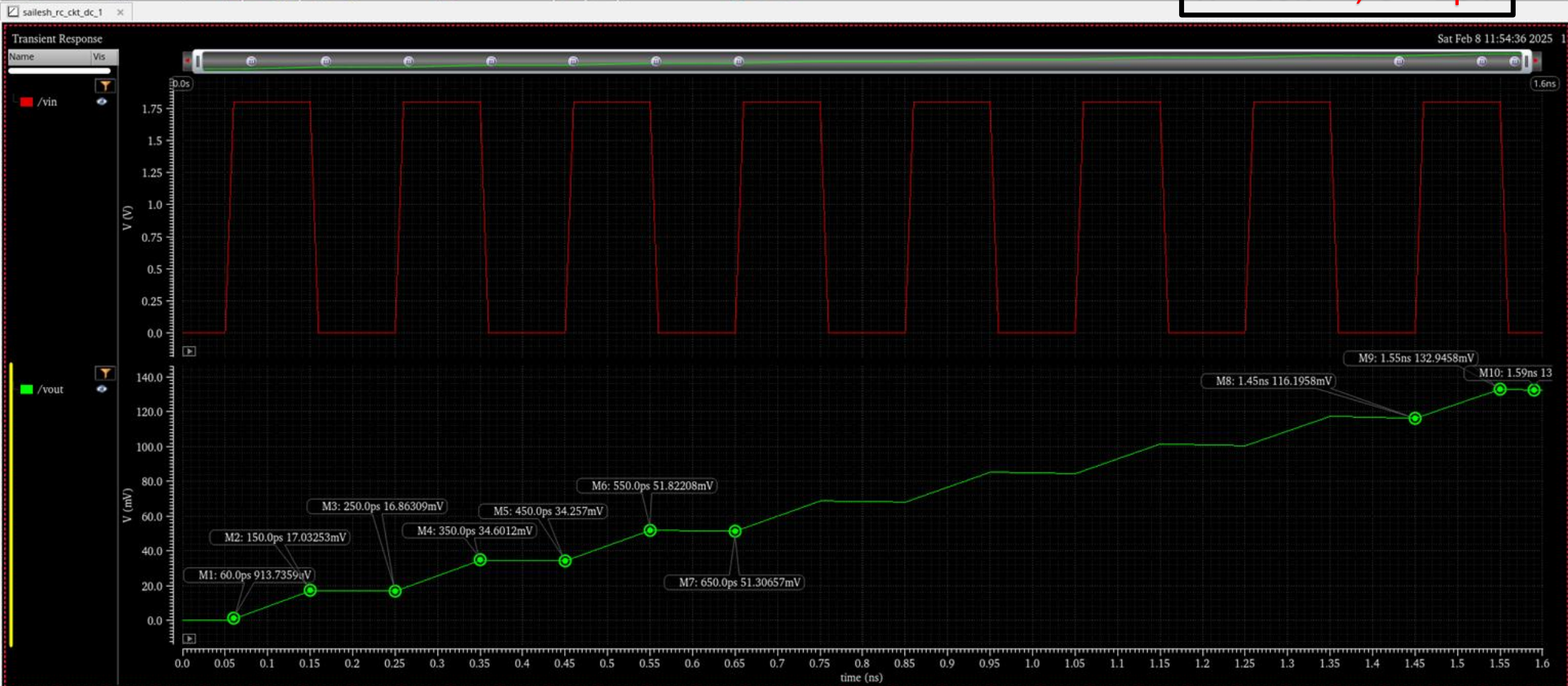
mouse L:

M:

R:

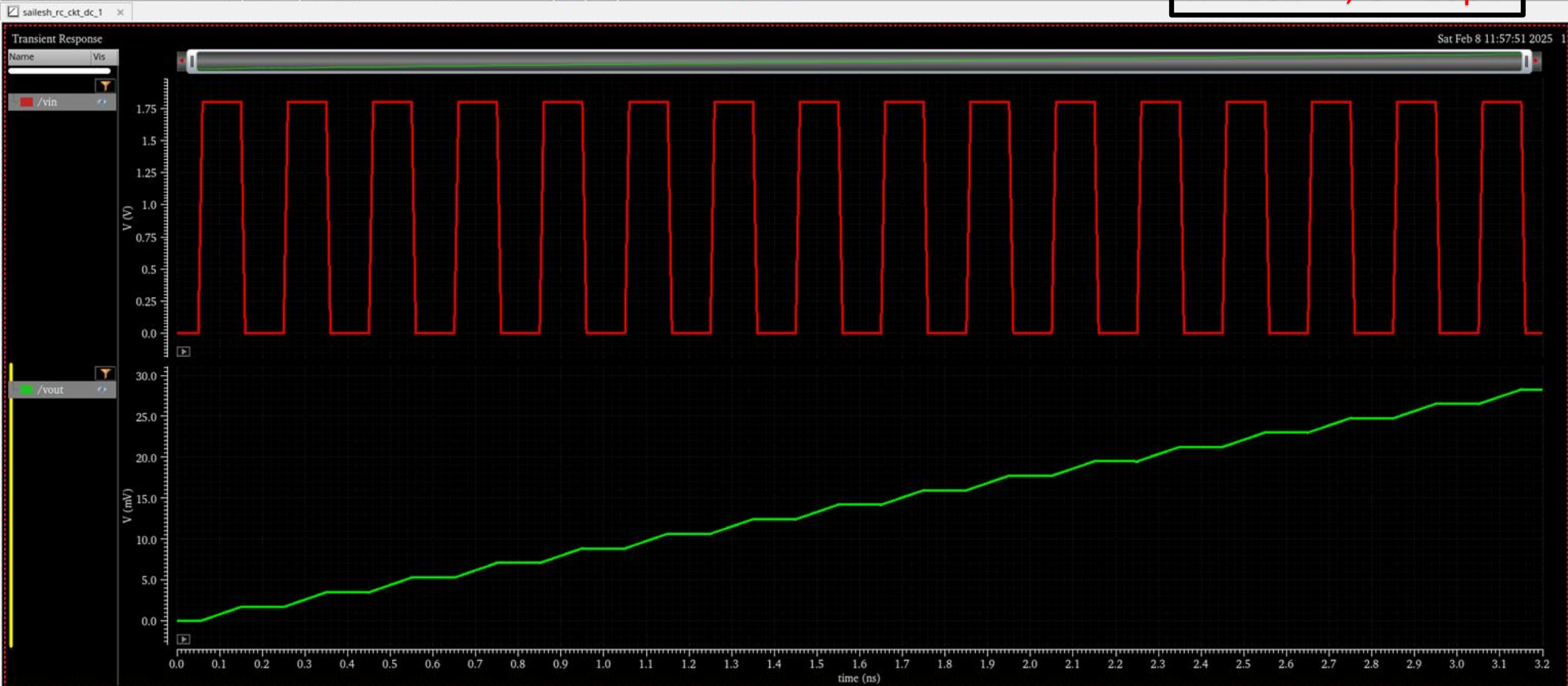
24(73) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom

$R = 10k\ \Omega$; $C = 1p\ F$





$R = 10k \Omega$; $C = 10p F$





sailesh_rc_ckt_dc_1

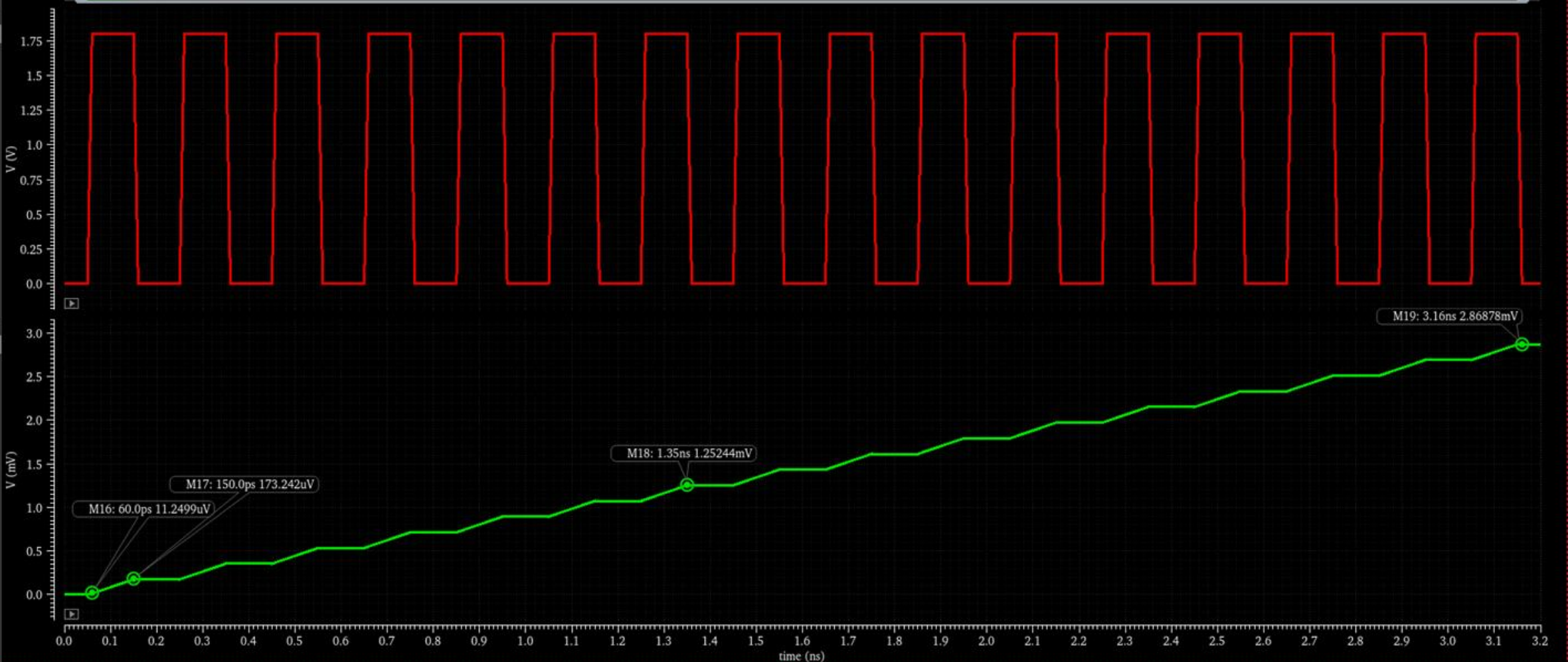
Transient Response

Sat Feb 8 12:00:55 2025 1

Name Vis

/vin

/vout



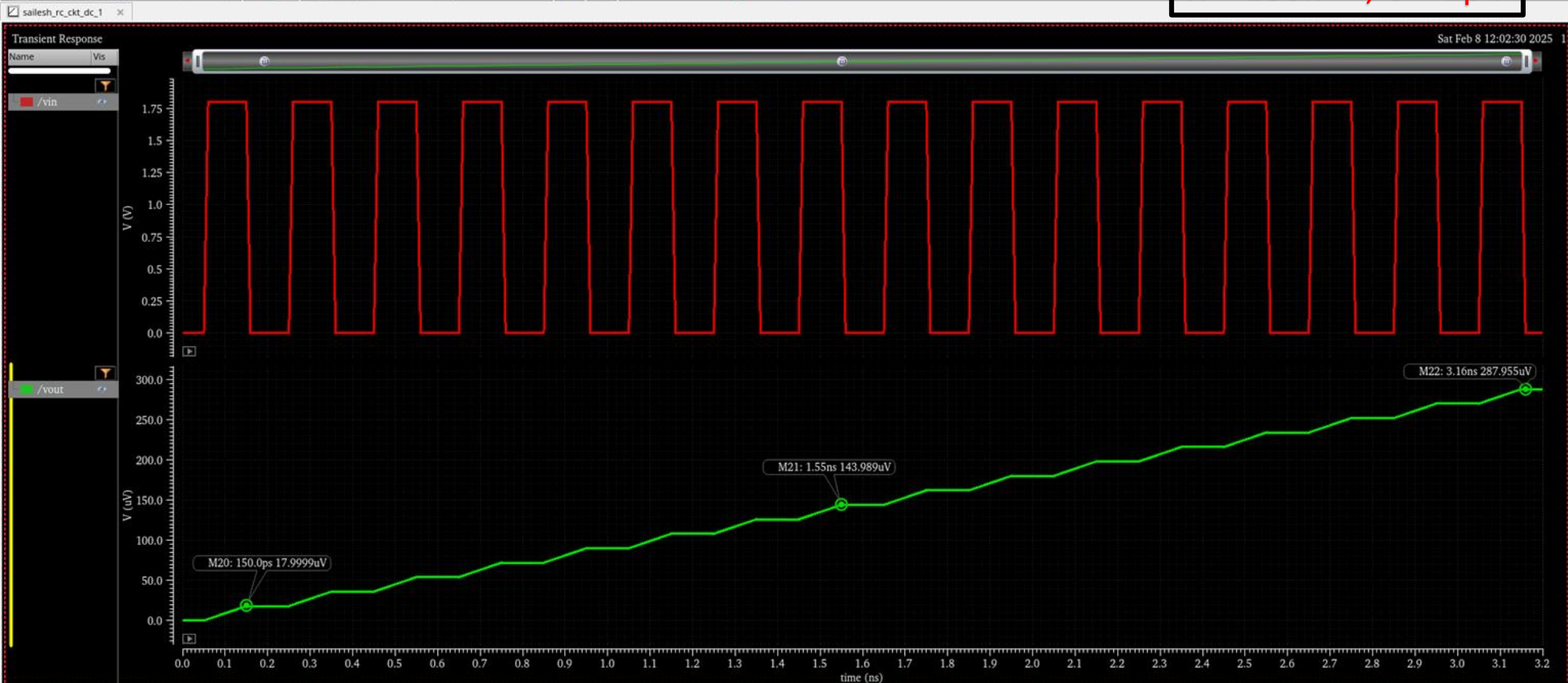
mouse L:

M:

16(49) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom

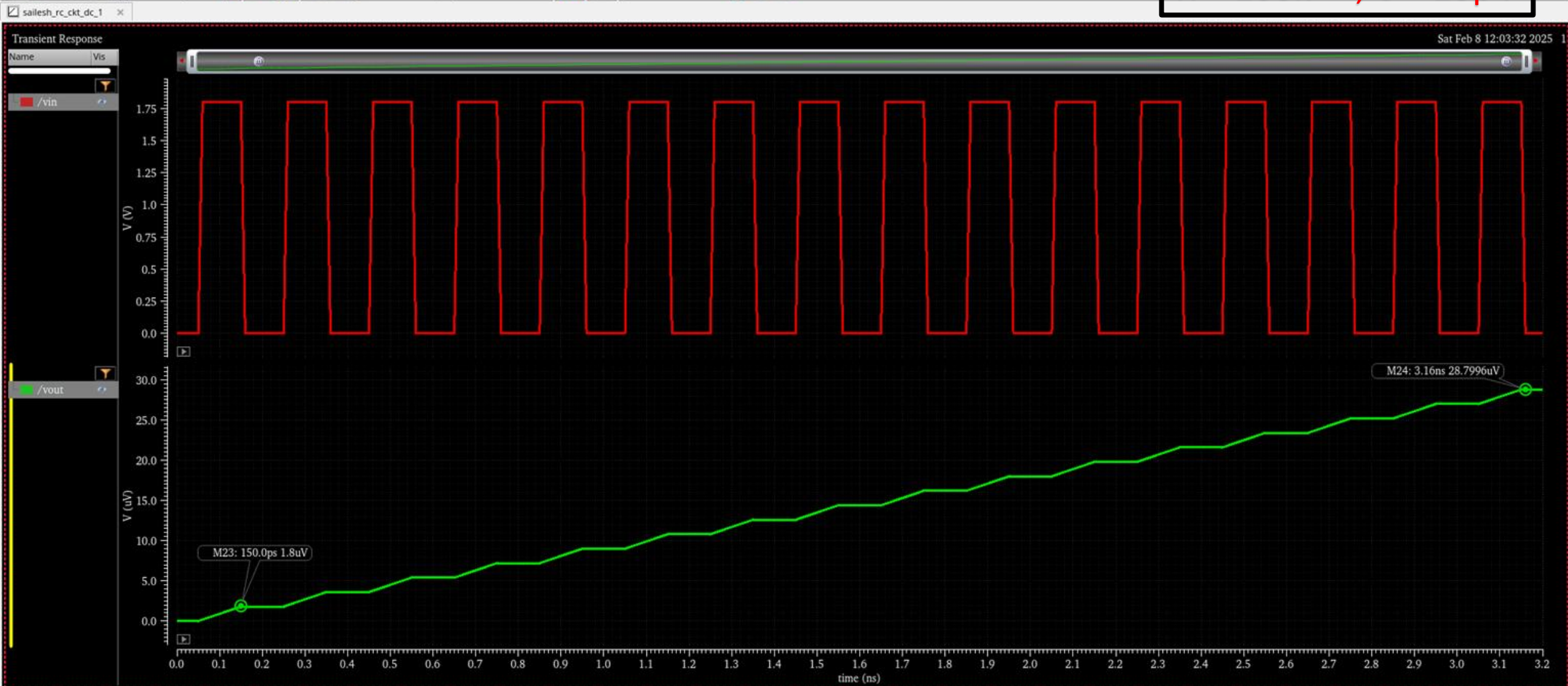


$R = 100k \Omega$; $C = 1p F$



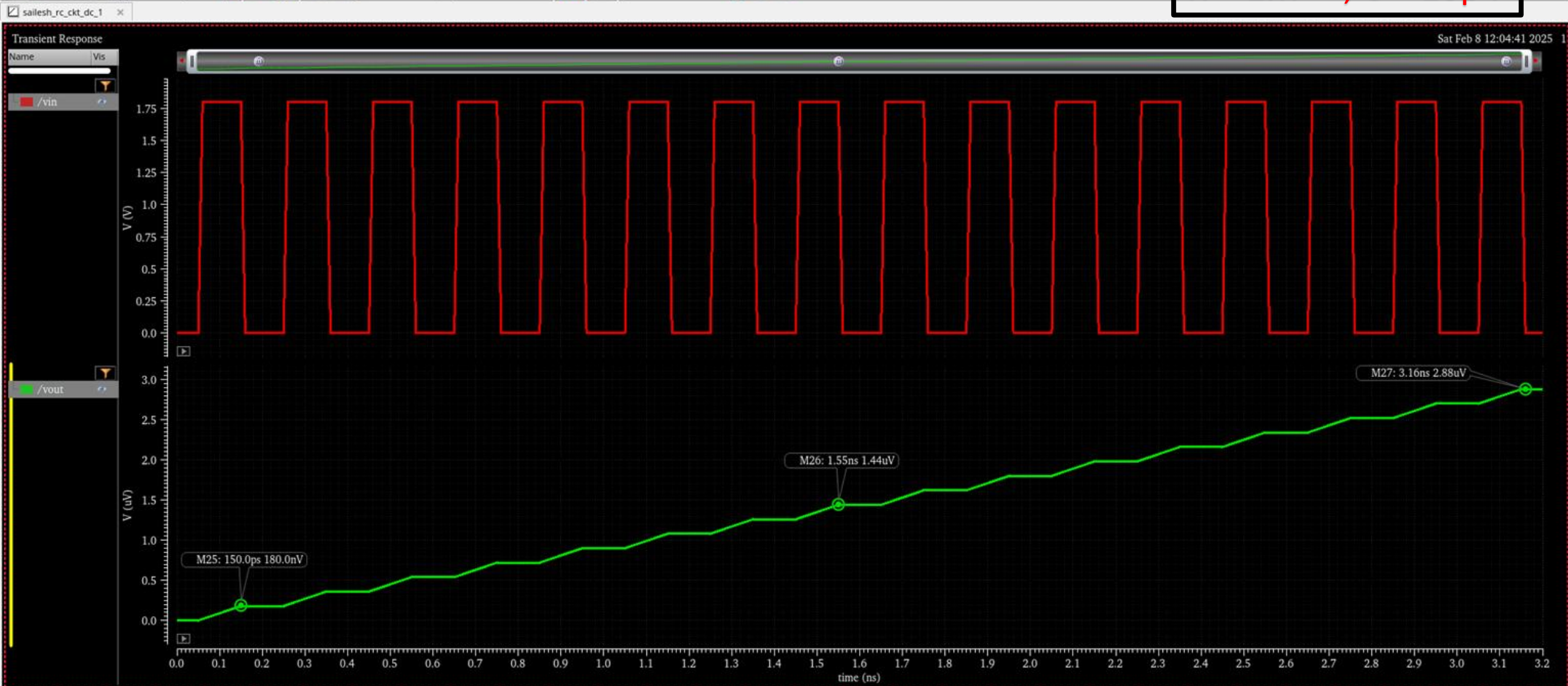


$R = 100k \Omega$; $C = 10p F$

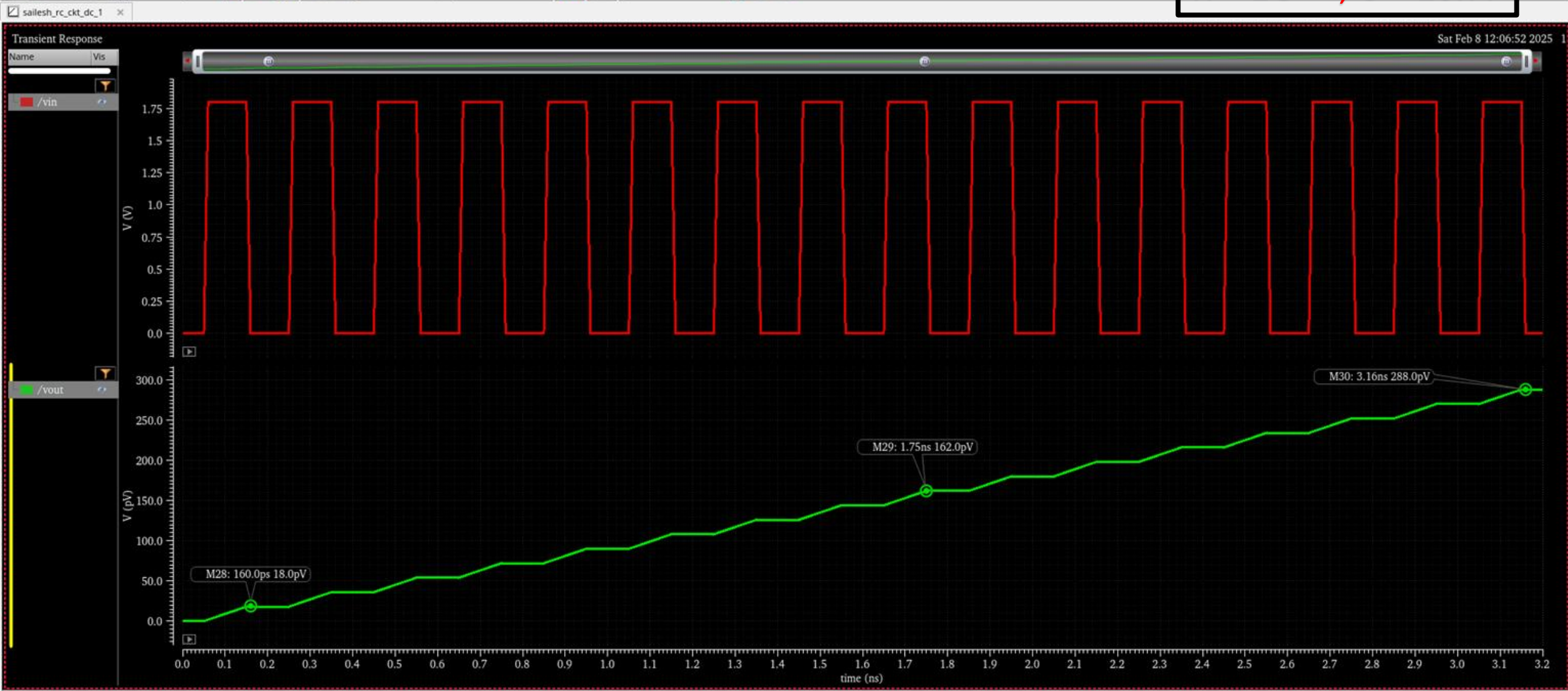




$R = 1\text{M } \Omega$; $C = 10\text{p F}$



$R = 1\text{G } \Omega ; C = 10\text{n F}$



Actual graphs . . .



$R = 100k \Omega ; C = 1p F$

sailesh_rc_ckt_dc_1

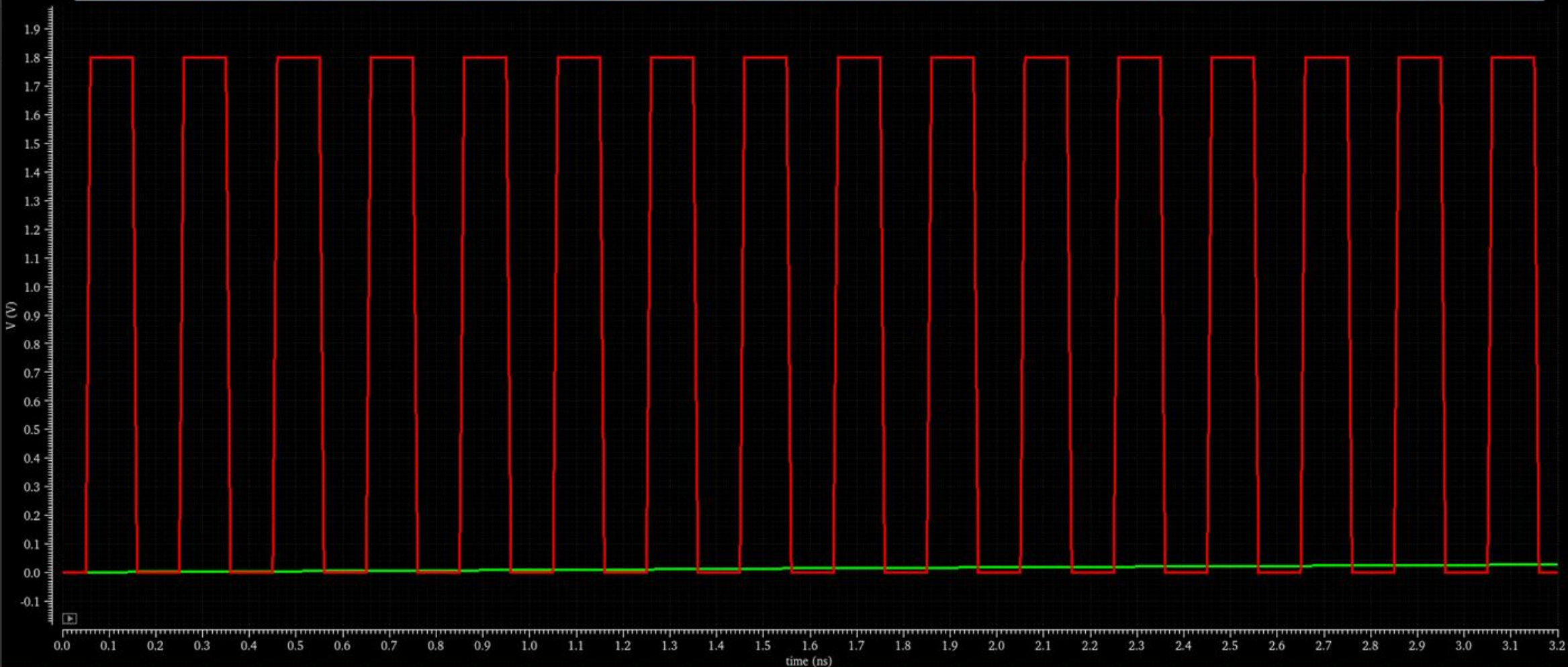
Transient Response

Sat Feb 8 13:06:08 2025 1

Name Vis

/vout

/vin



mouse L:

M:

R:

25(76) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom



$R = 100k \Omega$; $C = 10p F$

sailesh_rc_ckt_dc_1

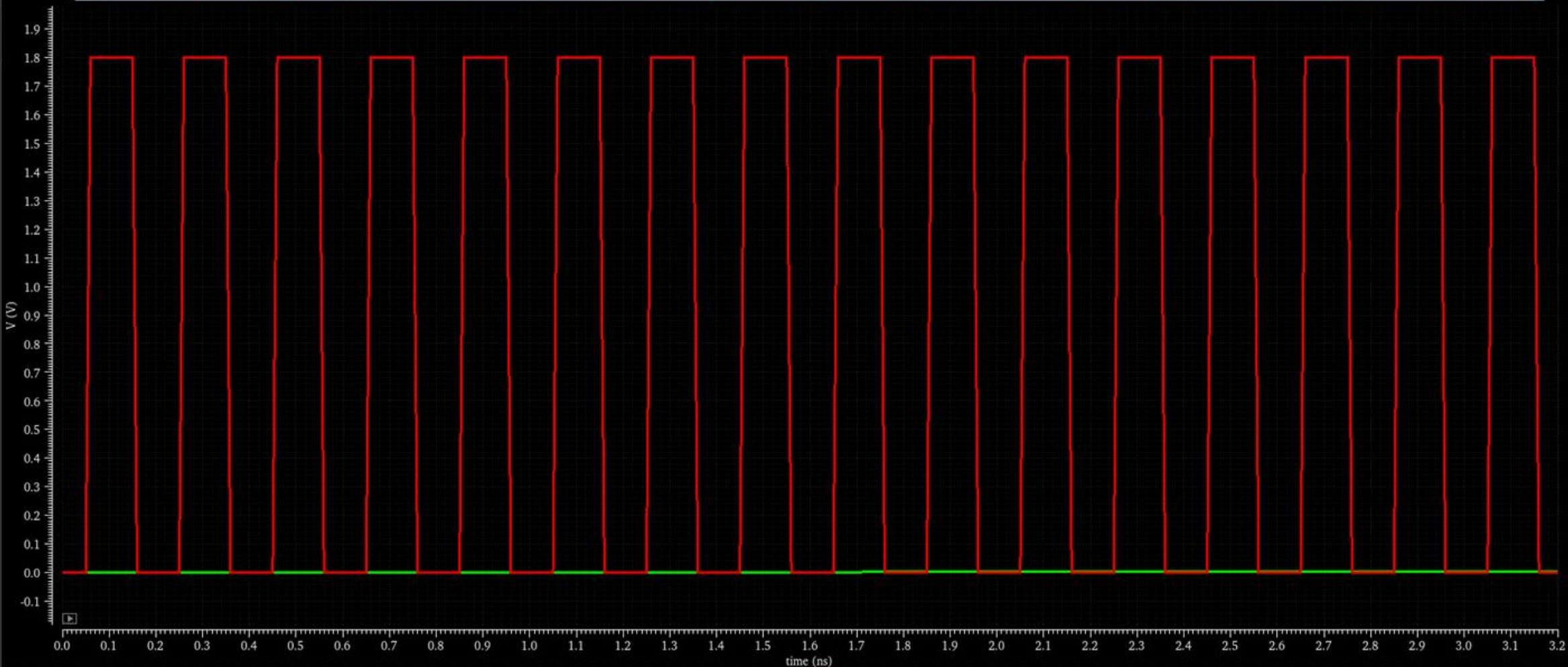
Transient Response

Sat Feb 8 13:07:20 2025 1

Name Vis

/vout

/vin



mouse L:

M:

R:

26(79) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom



$R = 100k \Omega$; $C = 1n F$

sailesh_rc_ckt_dc_1

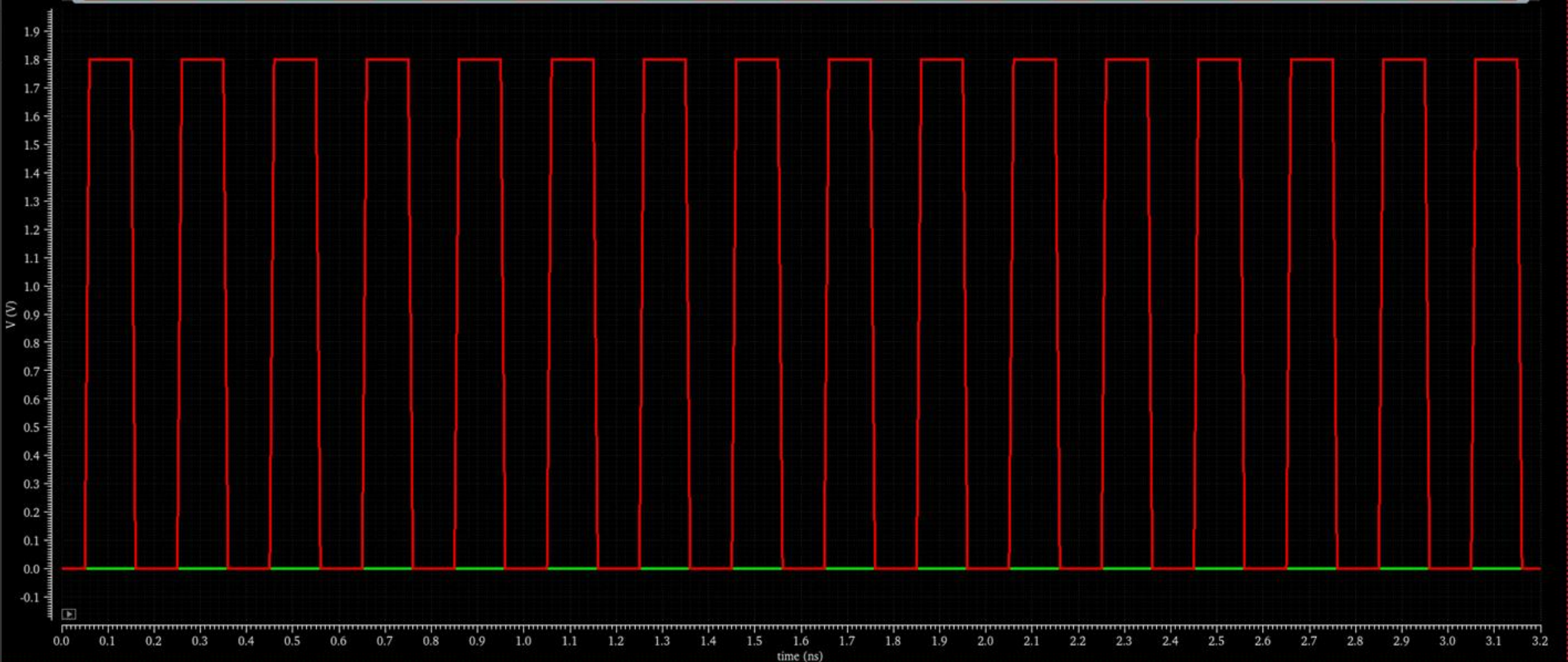
Transient Response

Sat Feb 8 13:08:13 2025 1

Name Vis

/vout

/vin



mouse L:

M:

R:

27(82) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom

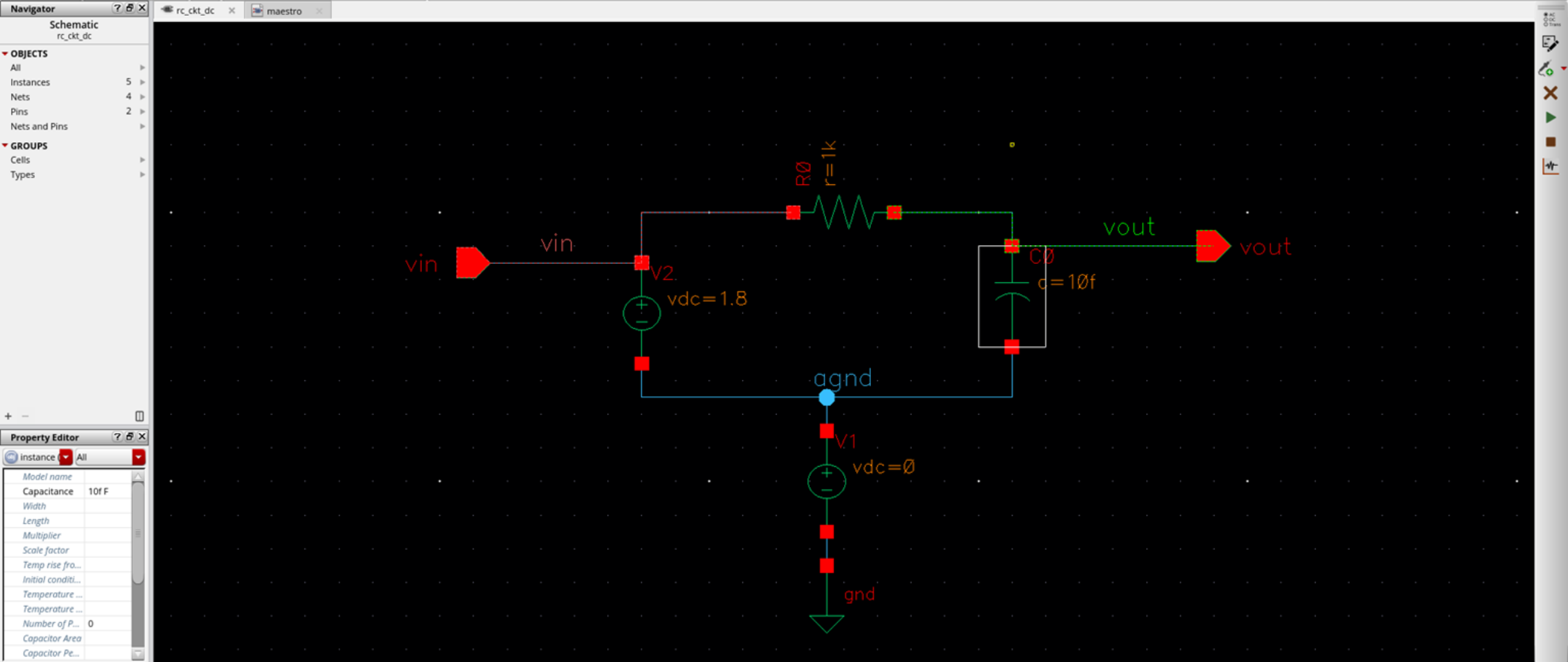
RC FILTER CIRCUITS

Er. Sailesh Mishra

B.Tech (ECE | Silicon University)

Specialization in Microelectronics & VLSI

Basic Search Replace (None) dc Sim Time 0



Choosing Analyses -- ADE Explorer

- Analysis
- | | | | |
|------------------------------|-------------------------------------|-------------------------------|-------------------------------|
| ☐ tran | ☒ dc | ☐ ac | ☐ noise |
| ☐ xf | ☐ sens | ☐ dcmatch | ☐ acmatch |
| ☐ stb | ☐ pz | ☐ lf | ☐ sp |
| ☐ envlp | ☐ pss | ☐ pac | ☐ pstb |
| ☐ pnoise | ☐ pxf | ☐ psp | ☐ qpss |
| ☐ qpac | ☐ qpnoise | ☐ qpxf | ☐ qpssp |
| ☐ hb | ☐ hbac | ☐ hbstb | ☐ hbnoise |
| ☐ hbisp | ☐ hbxf | | |

DC Analysis

Save DC Operating Point ☒
Hysteresis Sweep ☐

Sweep Variable

- ☐ Temperature
☐ Design Variable
☒ Component Parameter
☐ Model Parameter

Component Name

/V2

Select Component

Parameter Name

dc

Sweep Range

- ☒ Start-Stop
☐ Center-Span

Start

0

Stop

1.8

Sweep Type

Automatic

Add Specific Points ☐Add Points By File ☐Enabled ☒

Options...

OK

Cancel

Defaults

Apply

Help

	Value	Plot	Save	Spec
		☒	☒	
		☒	☒	

Virtuoso (R) Visualization & Analysis XL: sailesh rc_ckt_dc maestro



File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help

cadence



sailesh_rc_ckt_dc_1 x

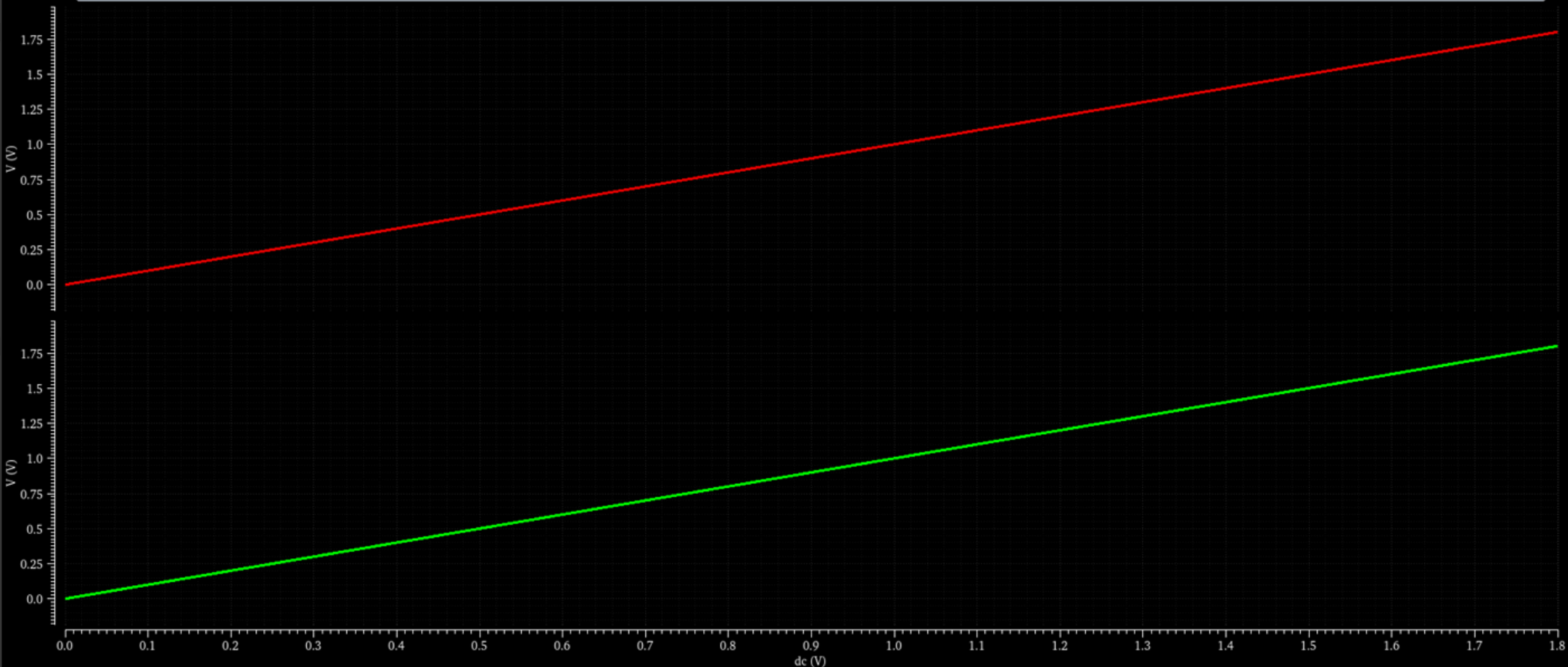
DC Response

Wed Feb 12 13:45:22

Name Vis

/vin

/vout



mouse L:

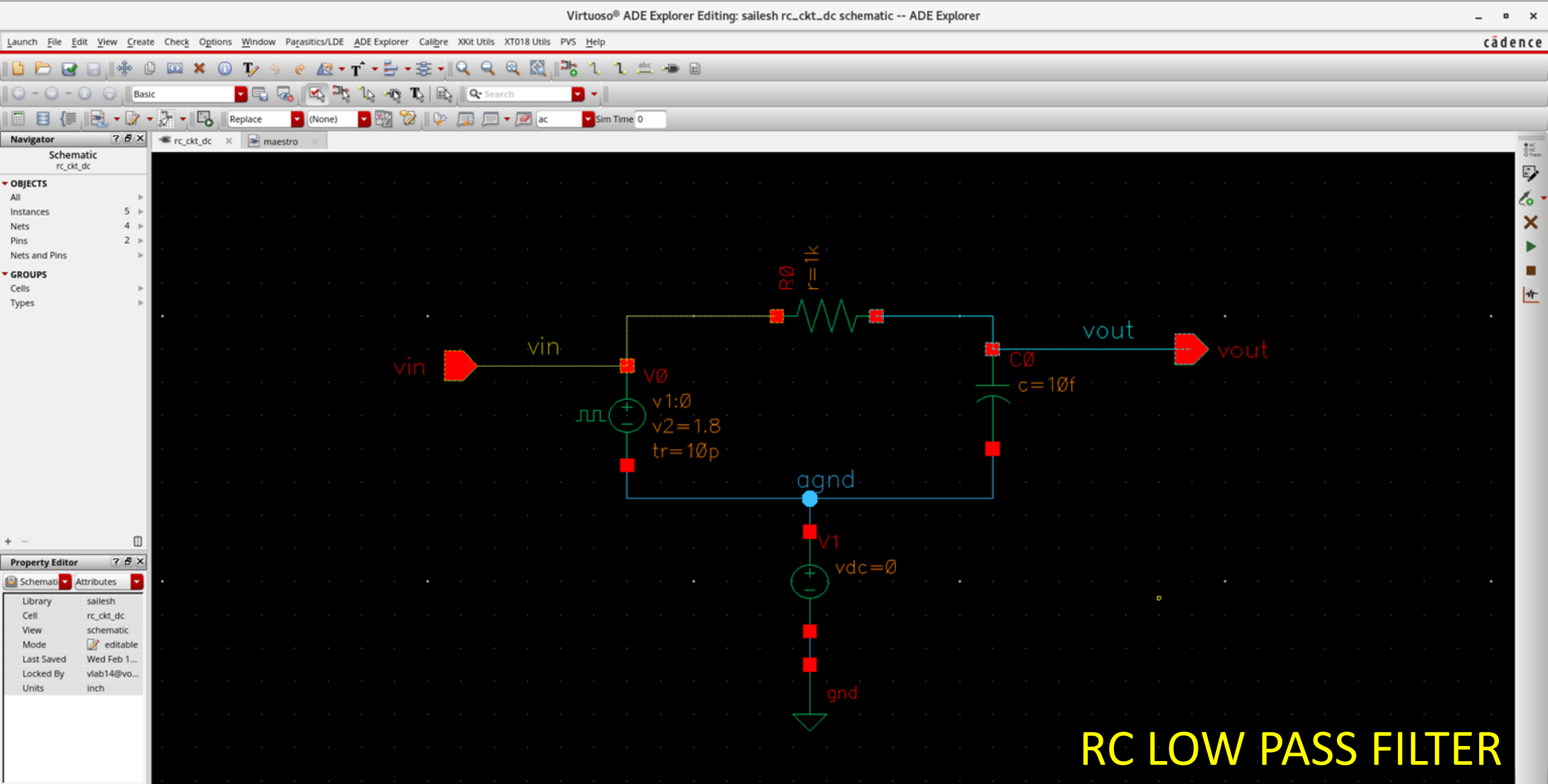
M:

R:

8(22) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom

AC ANALYSIS

FREQUENCY RESPONSE



Setup

Name	Value
Filter	Filter
sailesh_rc_ckt_dc_1	
Simulator	spectre
Analyses	
ac	1 1G 100 Logarithmic Points Per D...
Design Variables	
Parameters	
Corners	
Reliability Analyses	
Monte Carlo Sampling	
Checks/Asserts	

Name	Type	Details	Value	Plot	Save	Spec
	signal	/vout		☒	☒	
	signal	/vin		☒	☒	

Choosing Analyses -- ADE Explorer

Analysis

☐ tran ☐ dc ☒ ac ☐ noise

☐ xf ☐ sens ☐ dcmatch ☐ acmatch

☐ stb ☐ pz ☐ lf ☐ sp

☐ envlp ☐ pss ☐ pac ☐ pstb

☐ pnoise ☐ pxf ☐ psp ☐ qpss

☐ qpac ☐ qpnoise ☐ qpxf ☐ qpssp

☐ hb ☐ hbac ☐ hbstb ☐ hbnoise

☐ hbasp ☐ hbxf

AC Analysis

Sweep Variable

☒ Frequency

☐ Design Variable

☐ Temperature

☐ Component Parameter

☐ Model Parameter

☐ None

Sweep Range

☒ Start-Stop Start Stop

☐ Center-Span

Sweep Type

☐ Points Per Decade ☒ Logarithmic ☐ Number of Steps

Add Specific Points ☐

Add Points By File ☐

Specialized Analyses

Enabled ☒

Options...

OK Cancel Defaults Apply Help

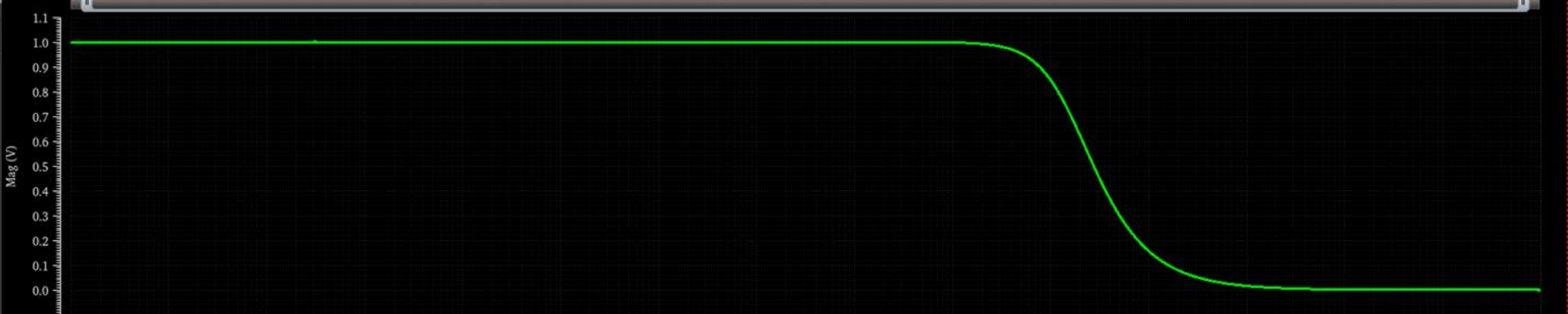


sailesh_rc_ckt_dc_1

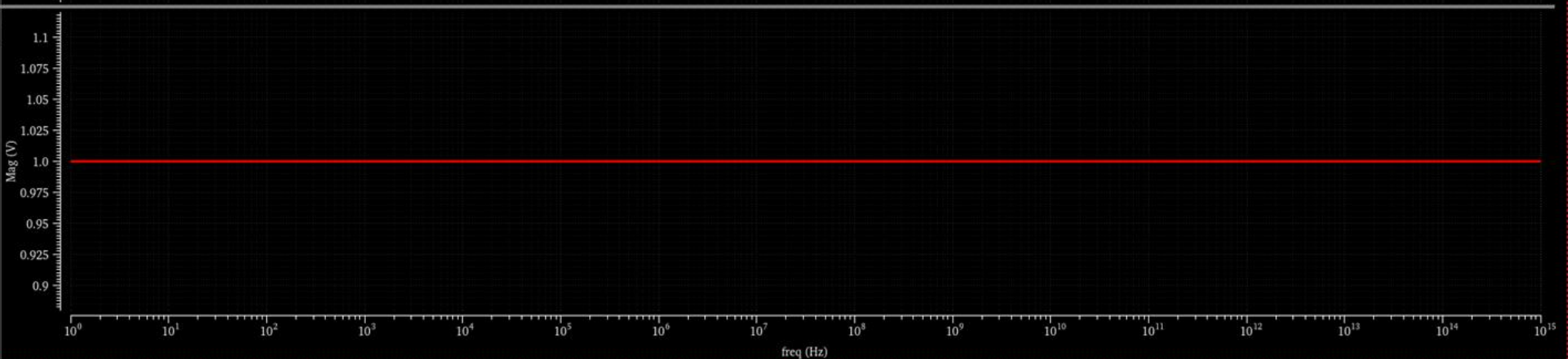
AC Response

Name Vis

/vout



/vin



mouse L:

M:

R:

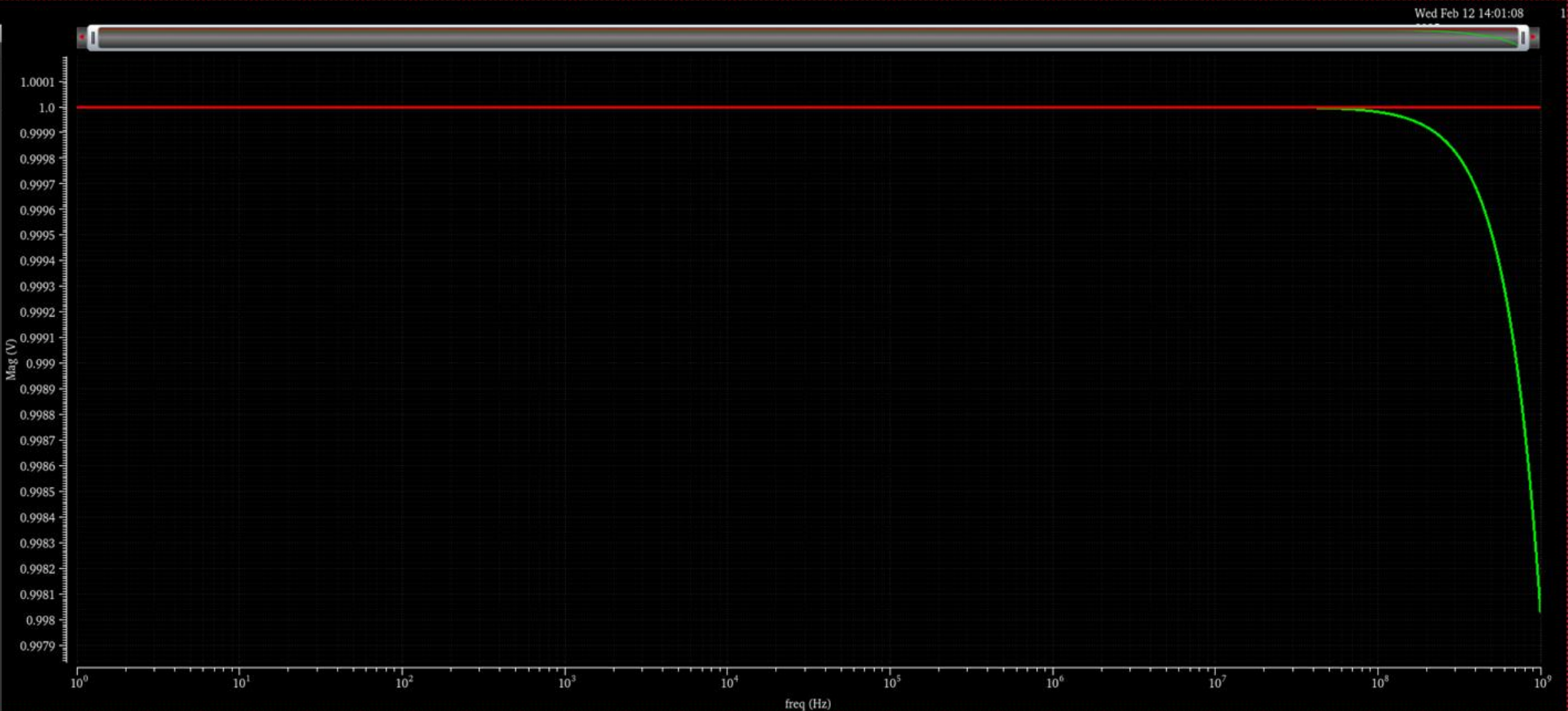
16(50) no error



sailesh_rc_ckt_dc_1

AC Response

Name	Vis
/vout	
/vin	



mouse L:

M:

R:

9(26) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom

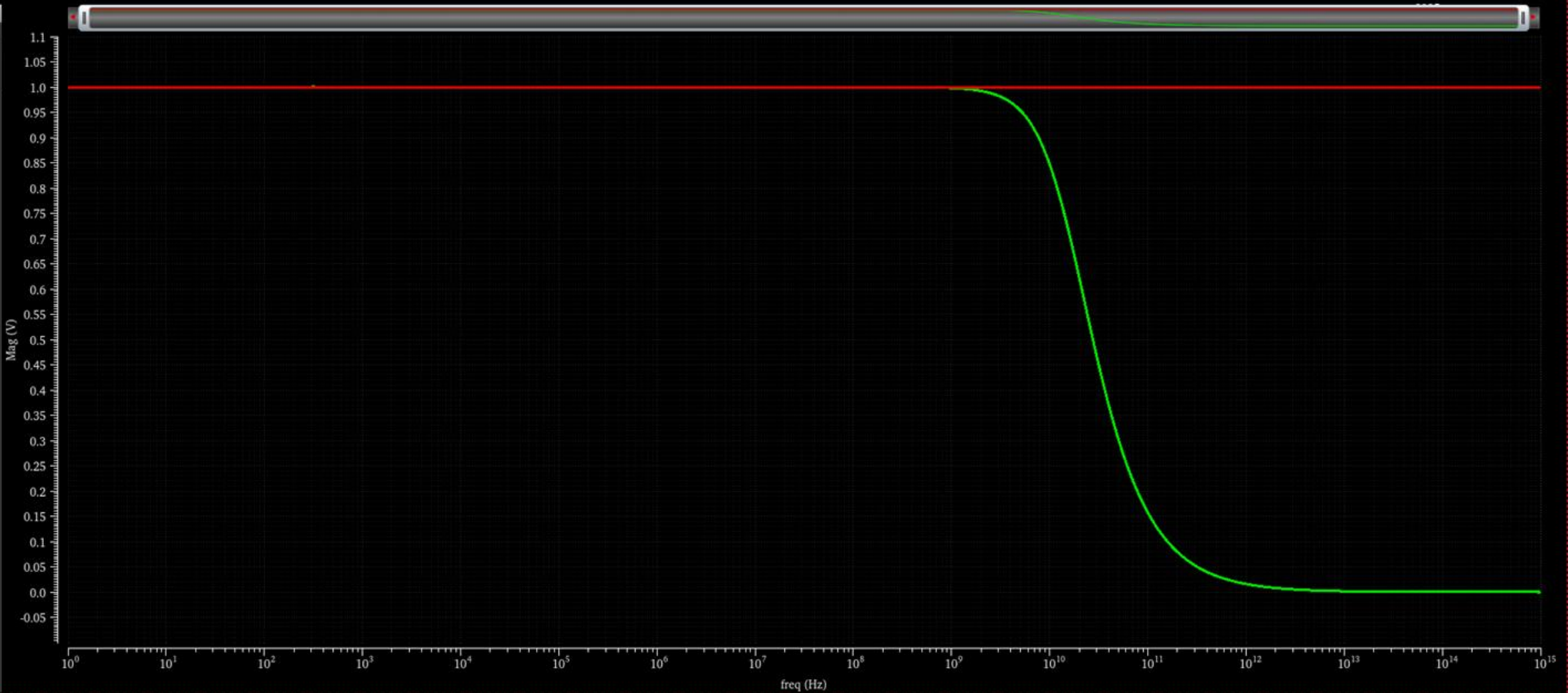


sailesh_rc_ckt_dc_1

AC Response

Name Vis

/vout	↕
/vin	↕

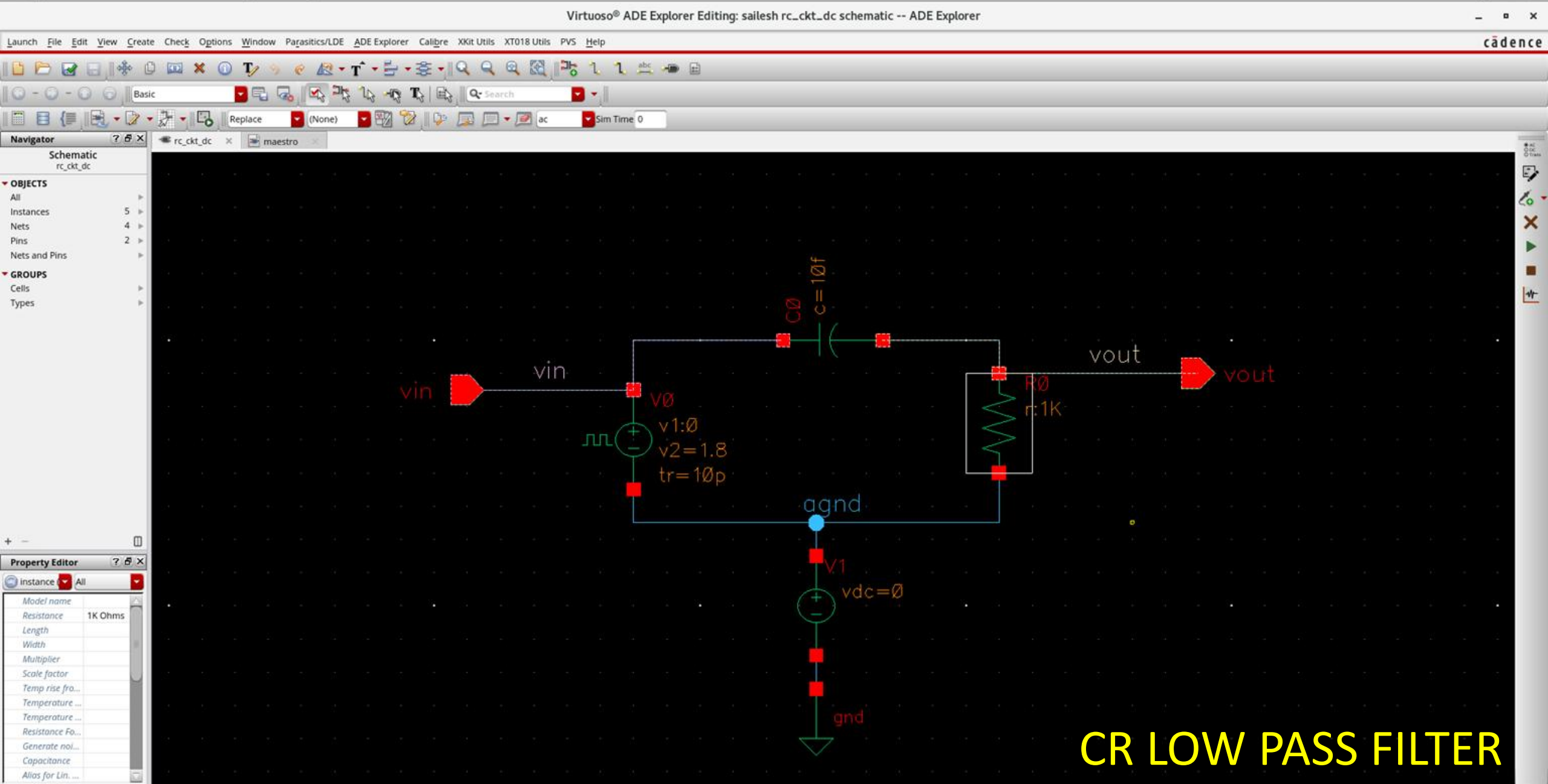


mouse L:

M:

R:

16(50) no error



Setup

Name	Value
Filter	Filter
sailesh_rc_ckt_dc_1	
Simulator	spectre
Analyses	
ac	1 1T 100 Logarithmic Points Per D...
Design Variables	
Parameters	
Corners	
Reliability Analyses	
Monte Carlo Sampling	
Checks/Asserts	

Name	Type	Details	Value	Plot	Save	Spec
	signal	/vout		☒	☒	
	signal	/vin		☒	☒	

Choosing Analyses -- ADE Explorer

Analysis

☐ tran ☐ dc ☒ ac ☐ noise

☐ xf ☐ sens ☐ dcmatch ☐ acmatch

☐ stb ☐ pz ☐ lf ☐ sp

☐ envlp ☐ pss ☐ pac ☐ pstb

☐ pnoise ☐ pxf ☐ psp ☐ qpss

☐ qpac ☐ qpnoise ☐ qpxf ☐ qpssp

☐ hb ☐ hbac ☐ hbstb ☐ hbnoise

☐ hbasp ☐ hbxf

AC Analysis

Sweep Variable

☒ Frequency

☐ Design Variable

☐ Temperature

☐ Component Parameter

☐ Model Parameter

☐ None

Sweep Range

☒ Start-Stop Start 1 Stop 1T

☐ Center-Span

Sweep Type

Logarithmic ☒ Points Per Decade 100

☐ Number of Steps

Add Specific Points ☐

Add Points By File ☐

Specialized Analyses

None

Enabled ☒

Options...

OK Cancel Defaults Apply Help



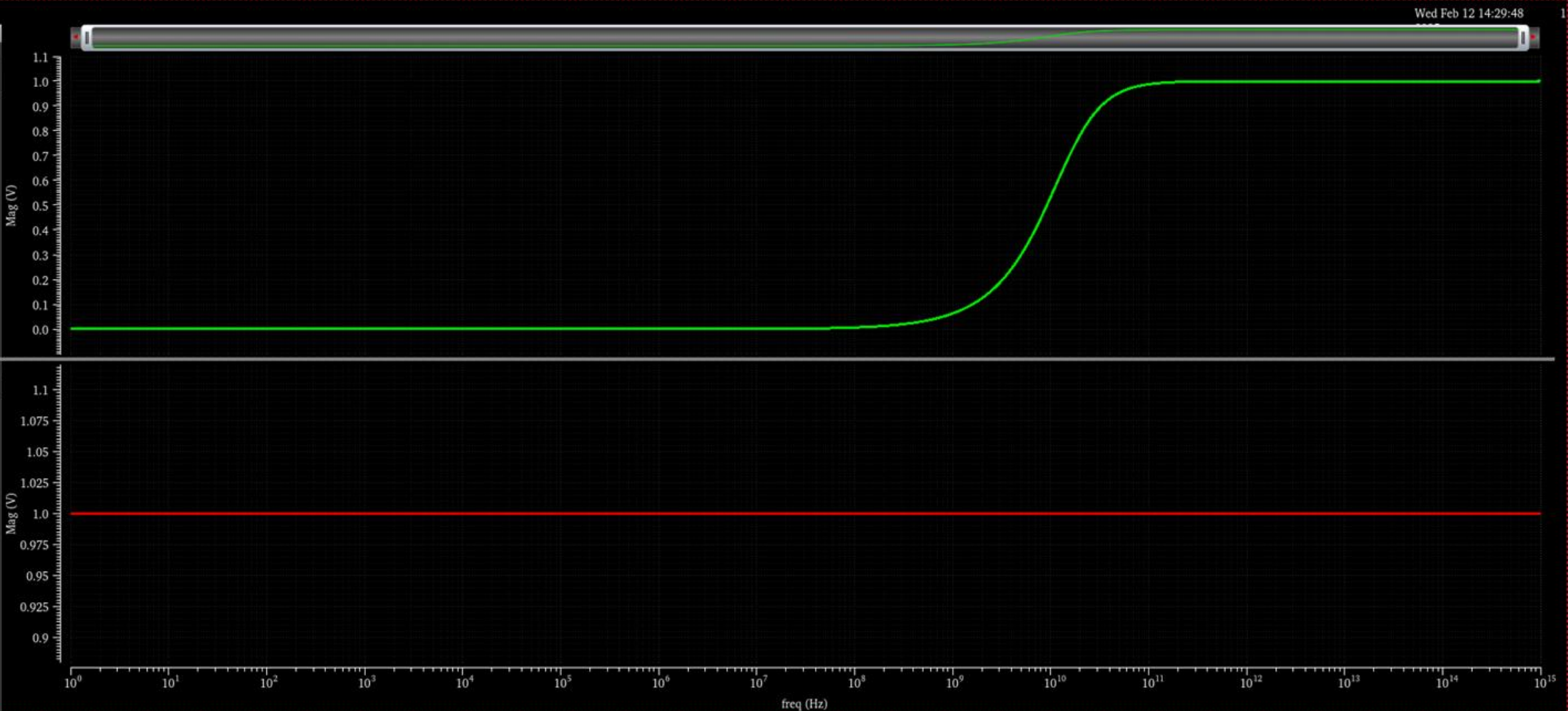
sailesh_rc_ckt_dc_1

AC Response

Name Vis

/vout

/vin



mouse L:

M:

R:

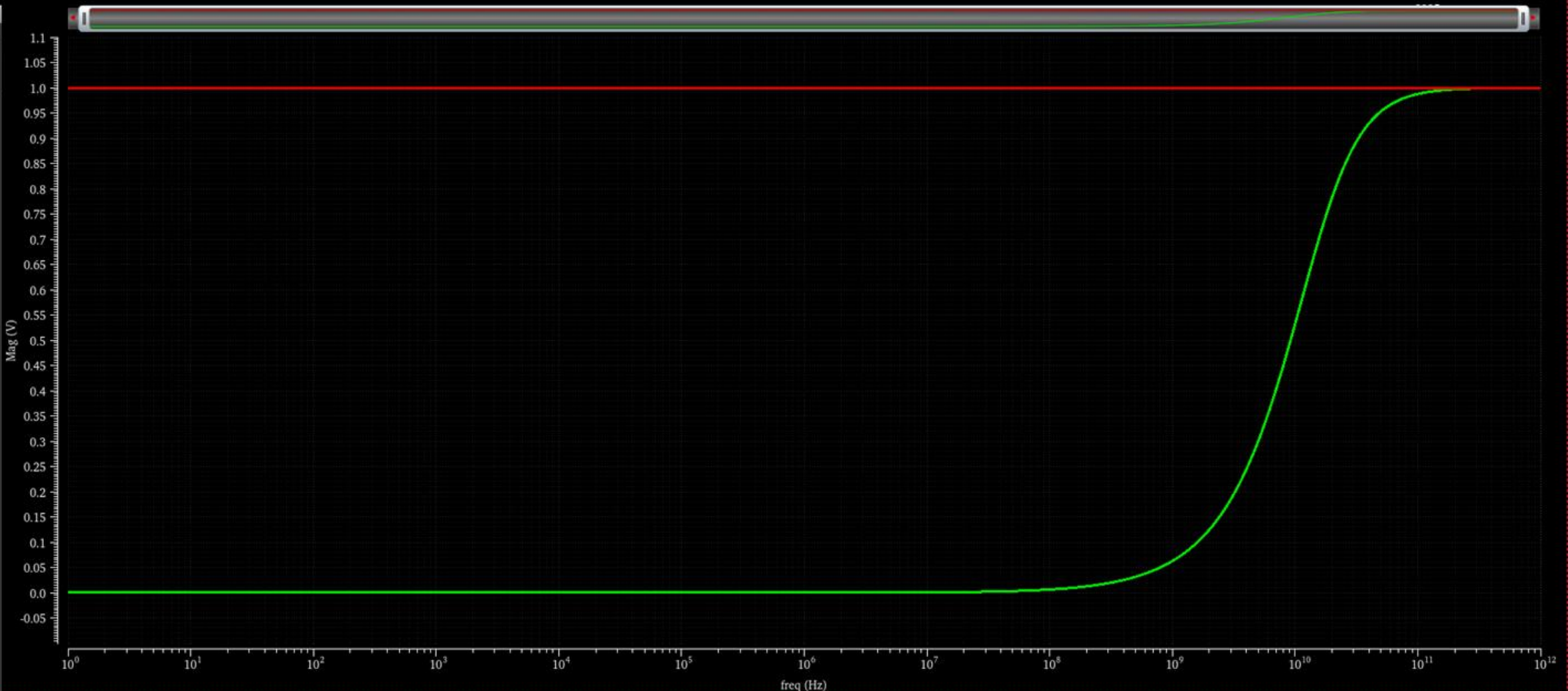
17(53) Trace: /vout; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom



sailesh_rc_ckt_dc_1

AC Response

Name	Vis
/vout	
/vin	



mouse L:

M:

R:

13(41) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom

File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help

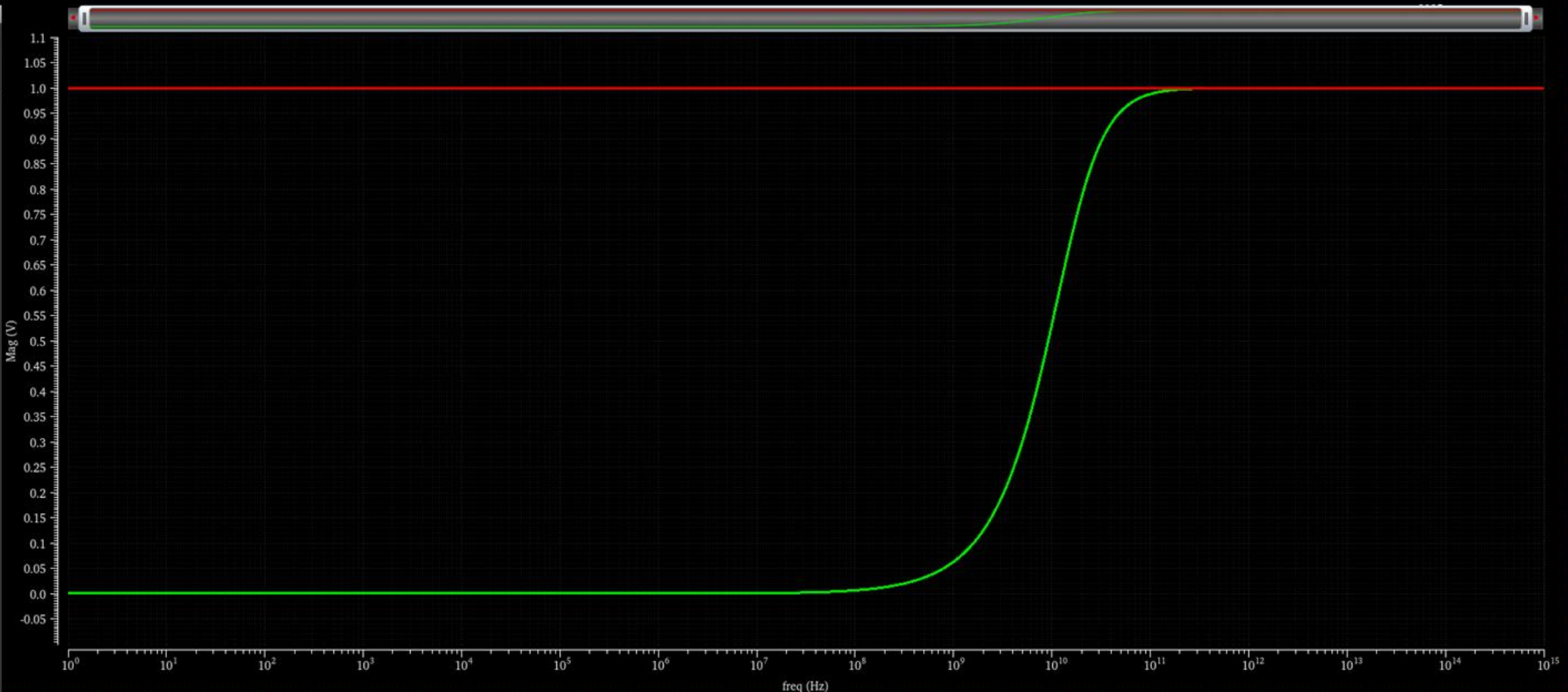


sailesh_rc_ckt_dc_1

AC Response

Name Vis

/vout
/vin



mouse L:

M:

R:

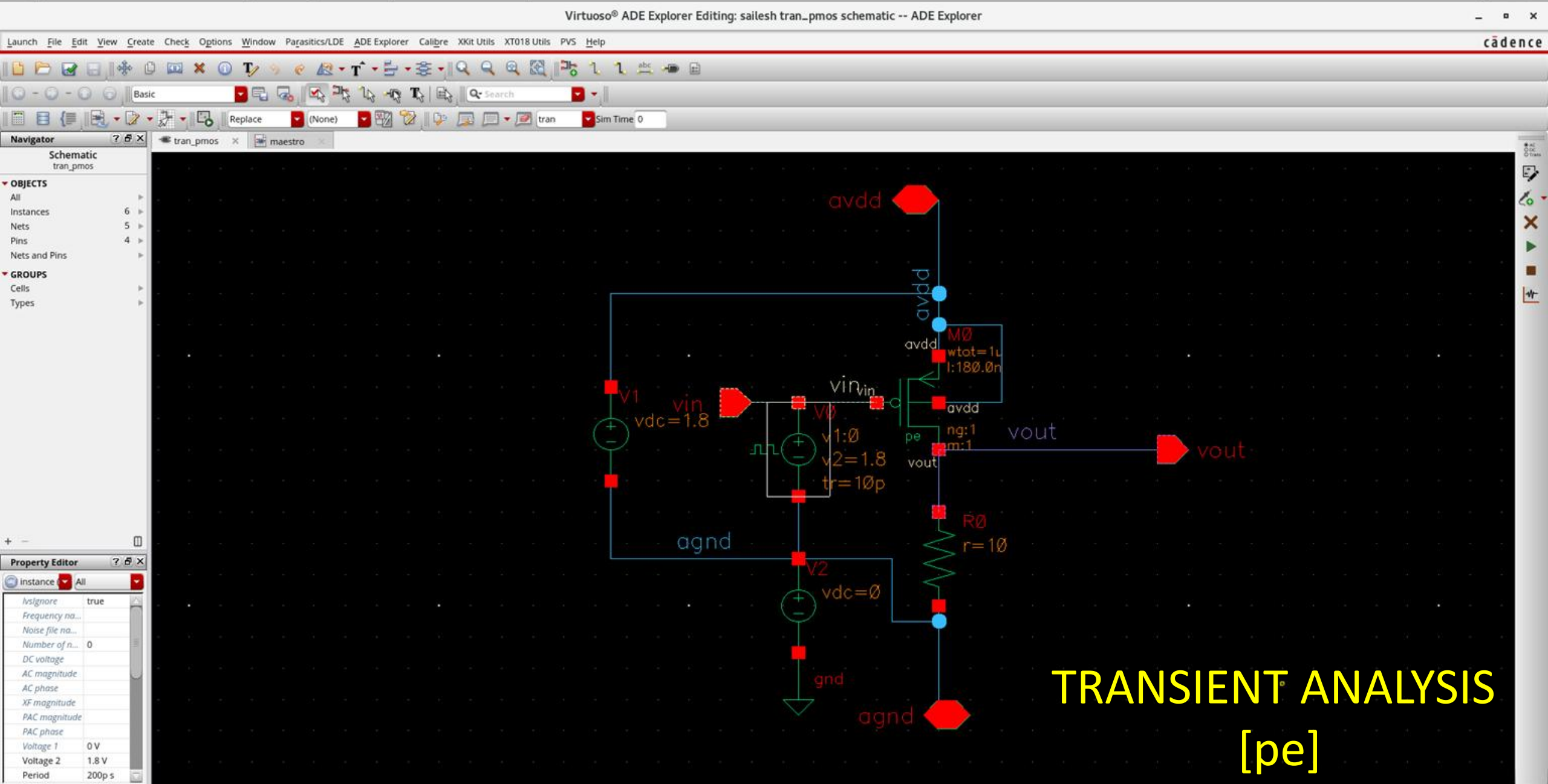
15(47) no error

MOSFET BIASING

Er. Sailesh Mishra

B.Tech (ECE | Silicon University)

Specialization in Microelectronics & VLSI



TRANSIENT ANALYSIS
[pe]

Launch Session Setup Analyses Variables Outputs Simulation Results Tools EAD Parasitics/LDE Window Help

Replace (None)

Setup

Name	Value
Filter	Filter
sailesh_tran_pmos_1	
Simulator spectre	
Analyses	
tran 0 800p moderate	
Click to add analysis	
Design Variables	
Click to add variable	
Parameters	
Corners	
Reliability Analyses	
Monte Carlo Sampling	
Checks/Asserts	

Name	Type	Details	Value	Plot	Save	Spec
	signal	/vout		☒	☒	
	signal	/vin		☒	☒	

Choosing Analyses -- ADE Explorer

- Analysis
- ☒ tran ☐ dc ☐ ac ☐ noise
- ☐ xf ☐ sens ☐ dcmatch ☐ acmatch
- ☐ stb ☐ pz ☐ lf ☐ sp
- ☐ envlp ☐ pss ☐ pac ☐ pstb
- ☐ pnoise ☐ pxf ☐ psp ☐ qpss
- ☐ qpac ☐ qpnoise ☐ qpxf ☐ qpdp
- ☐ hb ☐ hbac ☐ hbstb ☐ hbnoise
- ☐ hbasp ☐ hbxf

Transient Analysis

Stop Time 800p

Accuracy Defaults (errpreset)

☐ conservative ☒ moderate ☐ liberal☐ Transient NoiseDynamic Parameter ☐Enabled ☒

Options...

OK

Cancel

Defaults

Apply

Help



sailesh_tran_pmos_1 ×

Transient Response

Name Vis

/vin

/vout

V (V)

V (mV)

time (ps)

5(15) Trace: Avin; History: ExplorerRun.0; Test: sailesh_tran_pmos_1; Design Point: 1; Corner: nom

[Terminal]

[Virtuoso® 6.1.8-64b - Log: /home/...]

[Library Manager: Directory ...ainings...]

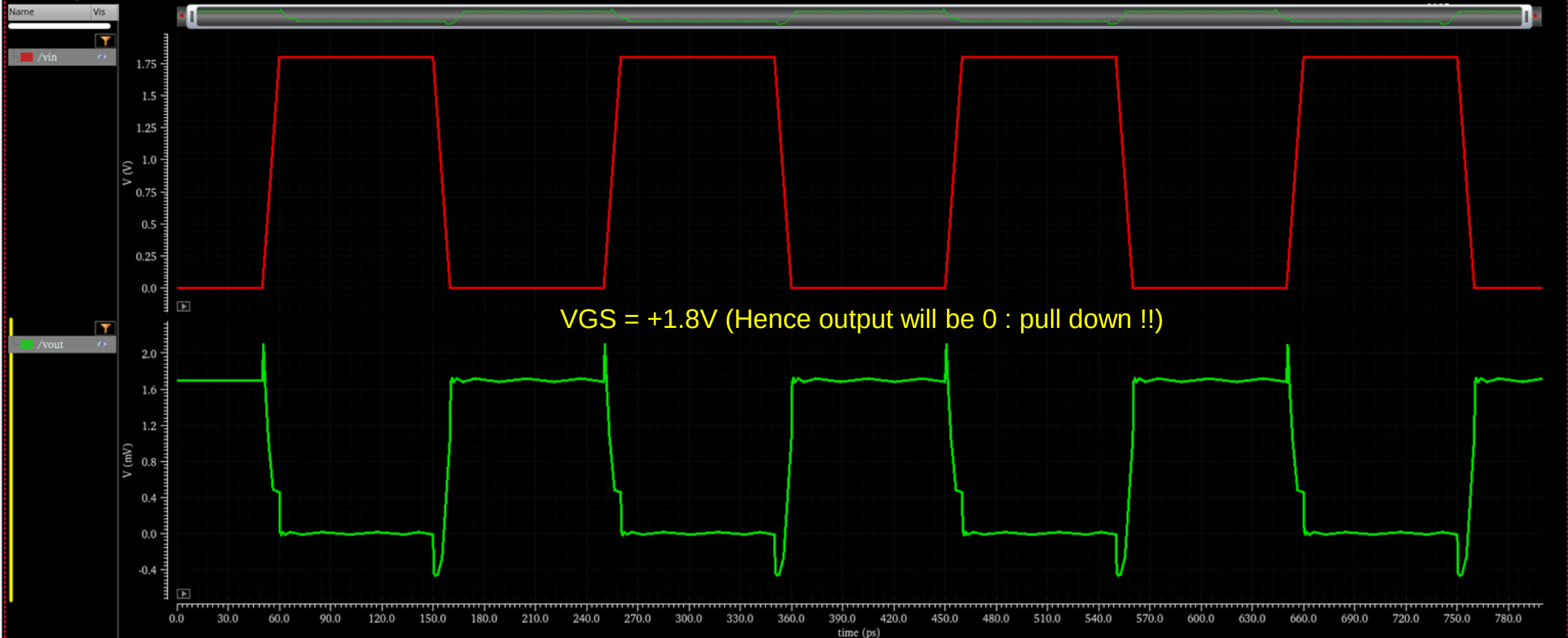
[Virtuoso® ADE Explorer Editing: sai...]

[/home/local/simulation/vlab14/saile...]

[Virtuoso (R) Visualization & Analysis...]

Wed Feb 12 15:28:43

1



mouse L:

M:

R:

5(15) Trace: Avin; History: ExplorerRun.0; Test: sailesh_tran_pmos_1; Design Point: 1; Corner: nom

[Terminal]

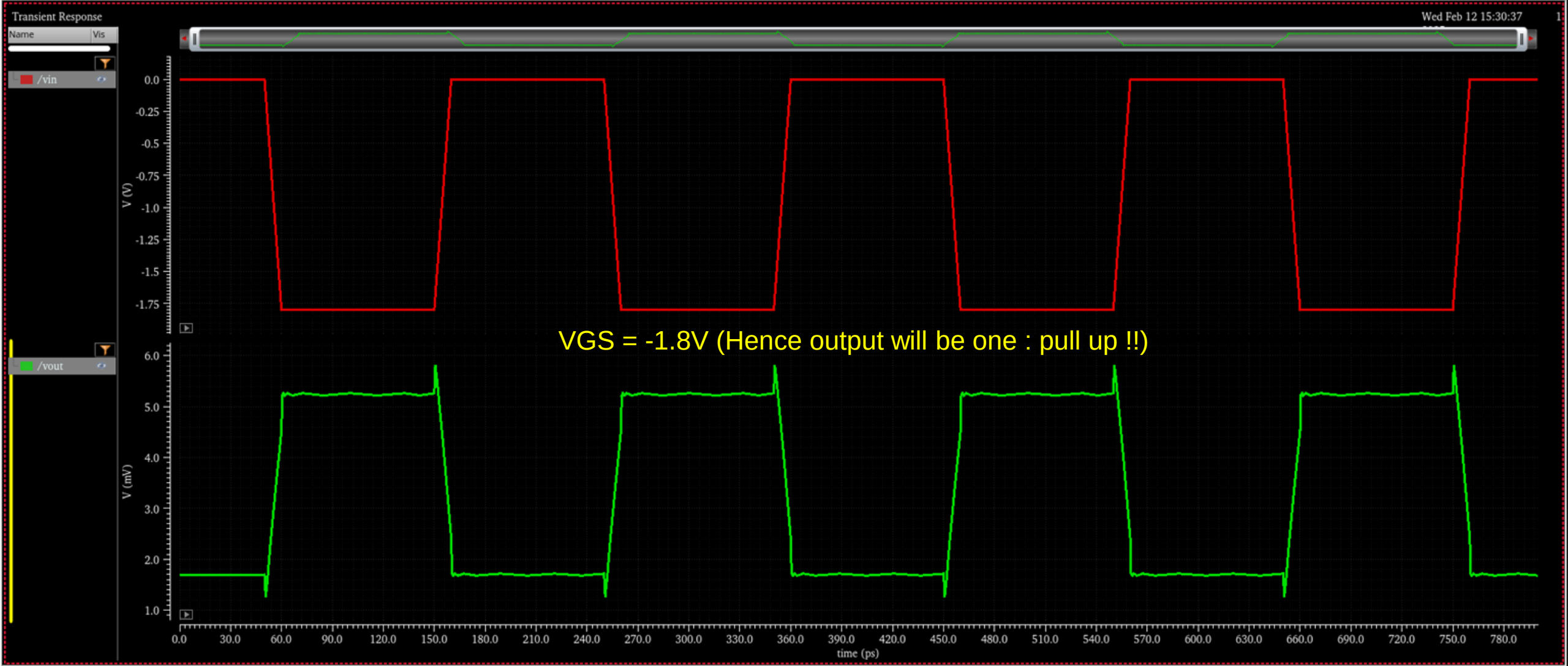
[Virtuoso® 6.1.8-64b - Log: /home/...]

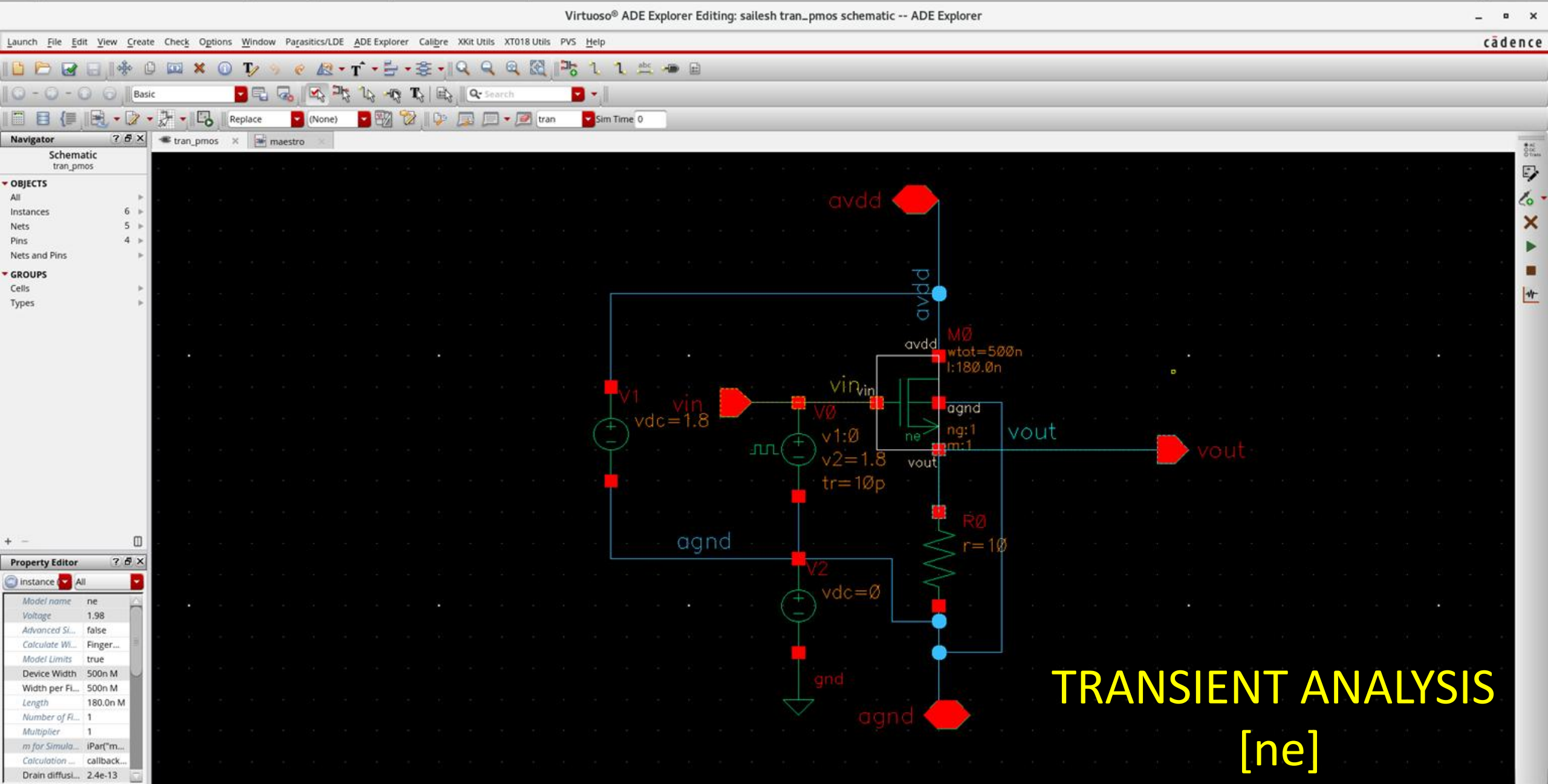
[Library Manager: Directory ...ainings...]

[Virtuoso® ADE Explorer Editing: sai...]

[/home/local/simulation/vlab14/saile...]

[Virtuoso (R) Visualization & Analysis...]

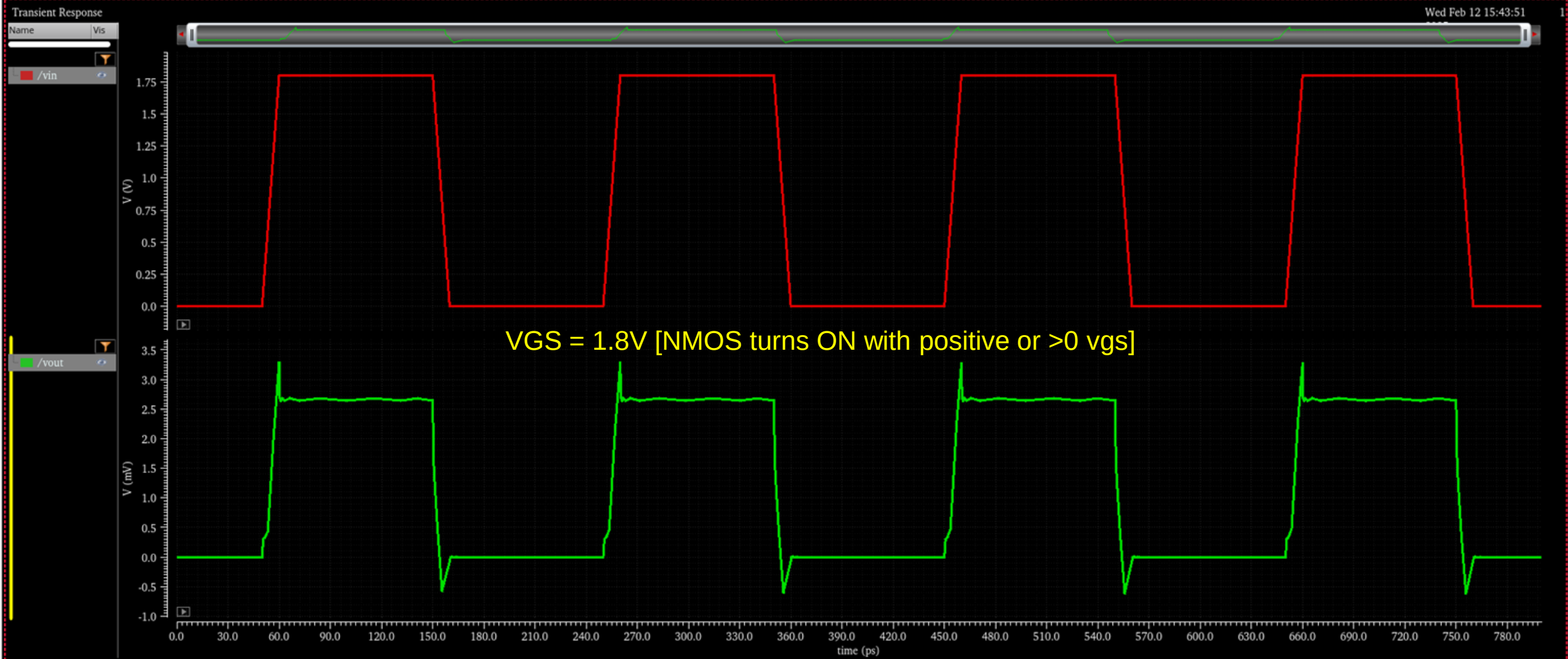




TRANSIENT ANALYSIS
[ne]



sailesh_tran_pmos_1



mouse L: M: R:

7(21) Trace: /vout; History: ExplorerRun.0; Test: sailesh_tran_pmos_1; Design Point: 1; Corner: nom



sailesh_tran_pmos_1

Transient Response

Name Vis

/vin

/vout

V (V)

V (uV)

time (ps)

0.0

-0.25

-0.5

-0.75

-1.0

-1.25

-1.5

-1.75

-2.0

-2.25

-2.5

-2.75

-3.0

-3.25

-3.5

-3.75

-4.0

-4.25

-4.5

-4.75

-5.0

-5.25

-5.5

-5.75

-6.0

-6.25

-6.5

-6.75

-7.0

-7.25

-7.5

-7.75

-8.0

-8.25

-8.5

-8.75

-9.0

-9.25

-9.5

-9.75

-10.0

-10.25

-10.5

-10.75

-11.0

-11.25

-11.5

-11.75

-12.0

-12.25

-12.5

-12.75

-13.0

-13.25

-13.5

-13.75

-14.0

-14.25

-14.5

-14.75

-15.0

-15.25

-15.5

-15.75

-16.0

-16.25

-16.5

-16.75

-17.0

-17.25

-17.5

-17.75

-18.0

-18.25

-18.5

-18.75

-19.0

-19.25

-19.5

-19.75

-20.0

-20.25

-20.5

-20.75

-21.0

-21.25

-21.5

-21.75

-22.0

-22.25

-22.5

-22.75

-23.0

-23.25

-23.5

-23.75

-24.0

-24.25

-24.5

-24.75

-25.0

-25.25

-25.5

-25.75

-26.0

-26.25

-26.5

-26.75

-27.0

-27.25

-27.5

-27.75

-28.0

-28.25

-28.5

-28.75

-29.0

-29.25

-29.5

-29.75

-30.0

-30.25

-30.5

-30.75

-31.0

-31.25

-31.5

-31.75

-32.0

-32.25

-32.5

-32.75

-33.0

-33.25

-33.5

-33.75

-34.0

-34.25

-34.5

-34.75

-35.0

-35.25

-35.5

-35.75

-36.0

-36.25

-36.5

-36.75

-37.0

-37.25

-37.5

-37.75

-38.0

-38.25

-38.5

-38.75

-39.0

-39.25

-39.5

-39.75

-40.0

-40.25

-40.5

-40.75

-41.0

-41.25

-41.5

-41.75

-42.0

-42.25

-42.5

-42.75

-43.0

-43.25

-43.5

-43.75

-44.0

-44.25

-44.5

-44.75

-45.0

-45.25

-45.5

-45.75

-46.0

-46.25

-46.5

-46.75

-47.0

-47.25

-47.5

-47.75

-48.0

-48.25

-48.5

-48.75

-49.0

-49.25

-49.5

-49.75

-50.0

-50.25

-50.5

-50.75

-51.0

-51.25

-51.5

-51.75

-52.0

-52.25

-52.5

-52.75

-53.0

-53.25

-53.5

-53.75

-54.0

-54.25

-54.5

-54.75

-55.0

-55.25

-55.5

-55.75

-56.0

-56.25

-56.5

-56.75

-57.0

-57.25

-57.5

-57.75

-58.0

-58.25

-58.5

-58.75

-59.0

-59.25

-59.5

-59.75

-60.0

-60.25

-60.5

-60.75

-61.0

-61.25

-61.5

-61.75

-62.0

-62.25

-62.5

-62.75

-63.0

-63.25

-63.5

-63.75

-64.0

-64.25

-64.5

-64.75

-65.0

-65.25

-65.5

-65.75

-66.0

-66.25

-66.5

-66.75

-67.0

-67.25

-67.5

-67.75

-68.0

-68.25

-68.5

-68.75

-69.0

-69.25

-69.5

-69.75

-70.0

-70.25

-70.5

-70.75

-71.0

-71.25

-71.5

-71.75

-72.0

-72.25

-72.5

-72.75

-73.0

-73.25

-73.5

-73.75

-74.0

-74.25

-74.5

-74.75

-75.0

-75.25

-75.5

-75.75

-76.0

-76.25

-76.5

-76.75

-77.0

-77.25

-77.5

-77.75

-78.0

-78.25

-78.5

-78.75

-79.0

-79.25

-79.5

-79.75

-80.0

-80.25

-80.5

-80.75

-81.0

-81.25

-81.5

-81.75

-82.0

-82.25

-82.5

-82.75

-83.0

-83.25

-83.5

-83.75

-84.0

-84.25

-84.5

-84.75

-85.0

-85.25

-85.5

-85.75

-86.0

-86.25

-86.5

-86.75

-87.0

-87.25

-87.5

-87.75

-88.0

-88.25

-88.5

-88.75

-89.0

TIMING PARAMETERS IN VLSI

Er. Sailesh Mishra

B.Tech (ECE | Silicon University)

Specialization in Microelectronics & VLSI

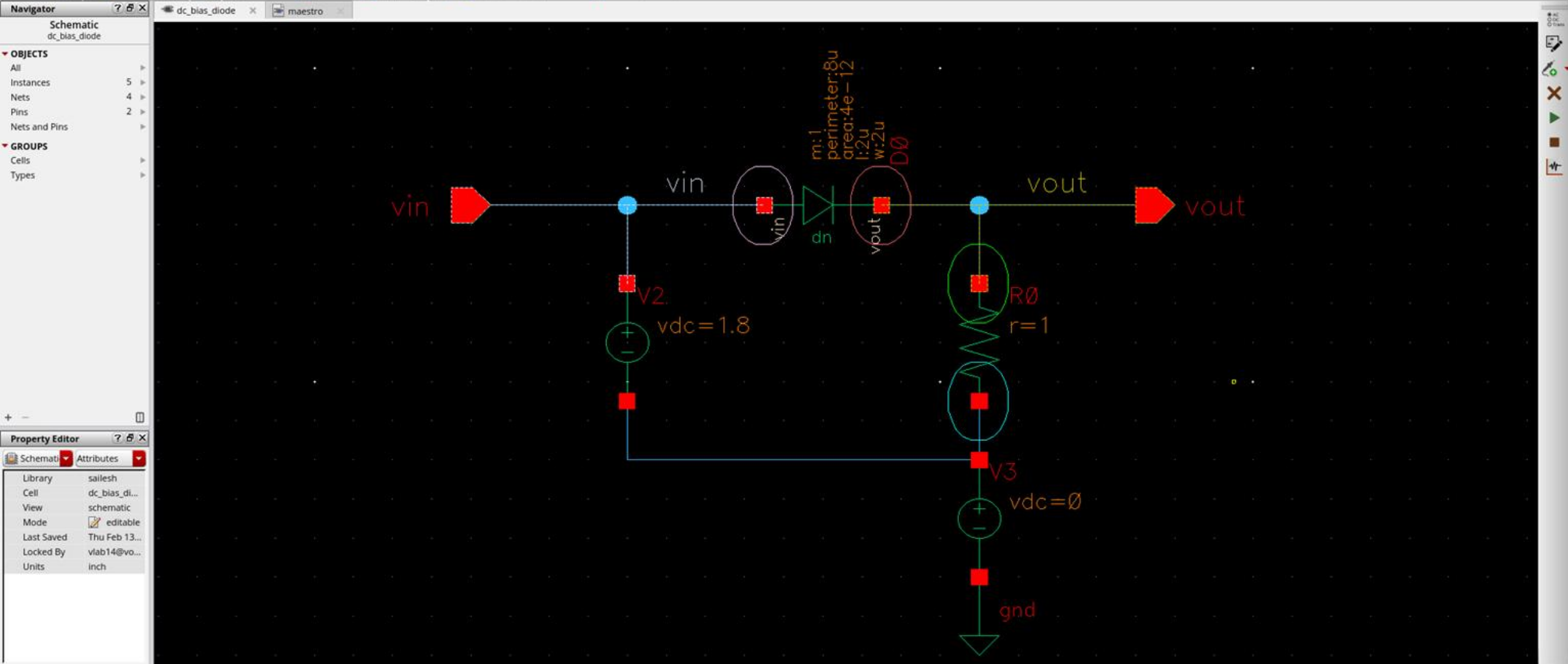
1. Propagation Delay
2. Switching Speed
3. T_{rise}
4. T_{fall}
5. Setup Time
6. Hold Time

DIODE CHARACTERISTICS

Er. Sailesh Mishra

B.Tech (ECE | Silicon University)

Specialization in Microelectronics & VLSI



Filter

sailesh_dc_bias_diode_1

Simulator spectre

Analyses

dc t 0 1.8 Automatic Start-Stop /V2

Click to add analysis

Design Variables

Click to add variable

Parameters

Corners

Reliability Analyses

Monte Carlo Sampling

Checks/Asserts

Name	Type	Details	Value	Plot	Save	Spec
signal (I)	/D0/PLUS					
signal (I)	/D0/MINUS					
signal (I)	/R0/PLUS					
signal (I)	/R0/MINUS					
signal	/vout					
signal	/vin					

Choosing Analyses -- ADE Explorer

Analysis

☐ tran ☒ dc ☐ ac ☐ noise

☐ xf ☐ sens ☐ dcmatch ☐ acmatch

☐ stb ☐ pz ☐ lf ☐ sp

☐ envlp ☐ pss ☐ pac ☐ pstb

☐ pnoise ☐ pxf ☐ psp ☐ qpss

☐ qpac ☐ qpnoise ☐ qpxf ☐ qpssp

☐ hb ☐ hbac ☐ hbstb ☐ hbnoise

☐ hbsp ☐ hbxf

DC Analysis

Save DC Operating Point ☒

Hysteresis Sweep ☐

Sweep Variable

☐ Temperature

☐ Design Variable

☒ Component Parameter

☐ Model Parameter

Component Name /V2

Select Component

Parameter Name dc

Sweep Range

☒ Start-Stop

☐ Center-Span

Start 0 Stop 1.8

Sweep Type

Automatic

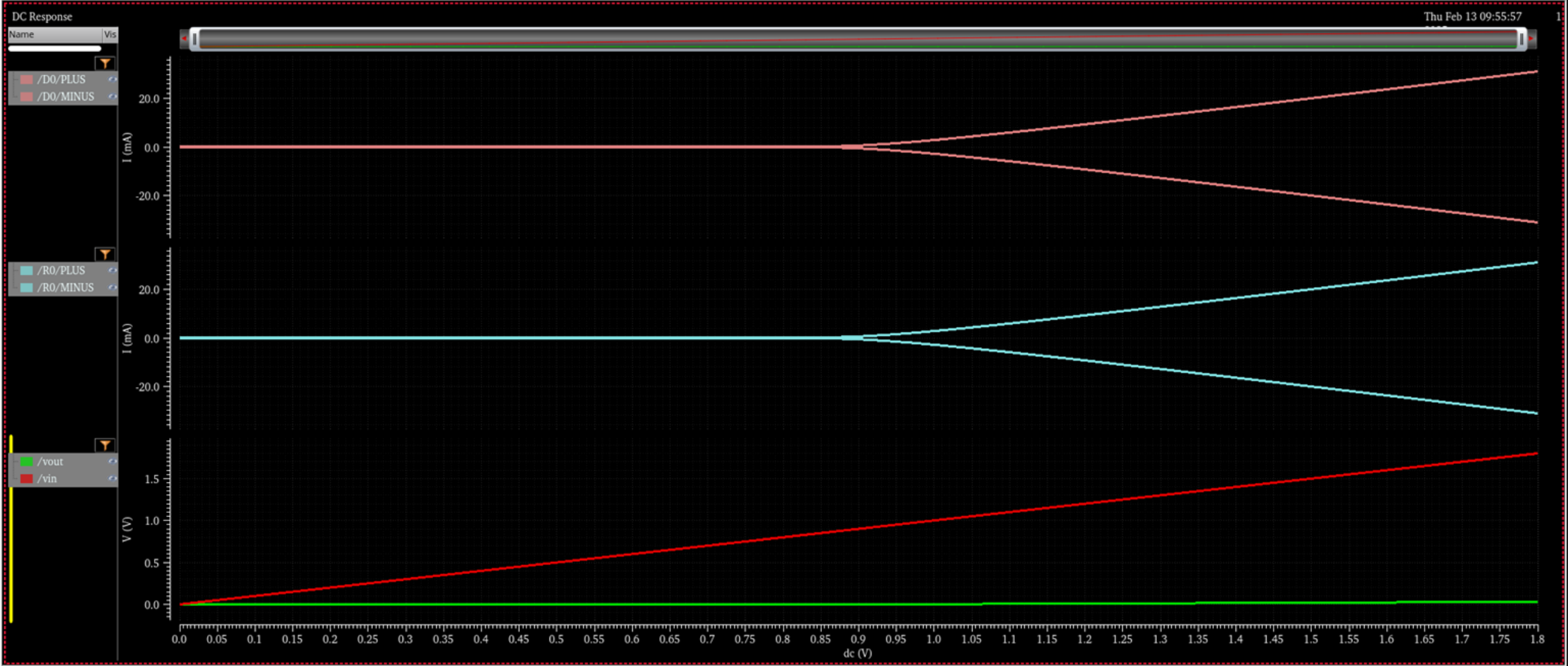
Add Specific Points ☐

Add Points By File ☐

Enabled ☒

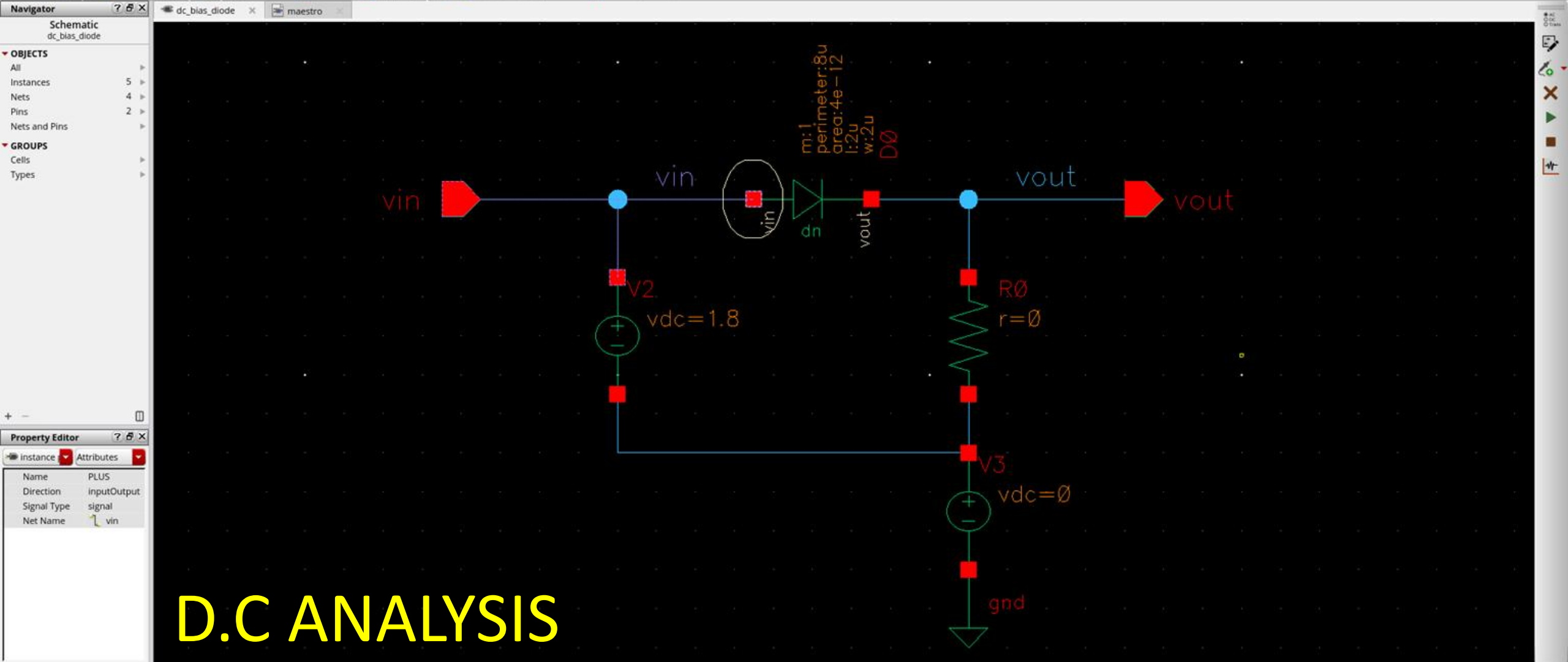
Options...

OK Cancel Defaults Apply Help



Launch File Edit View Create Check Options Window Parasitics/LDE ADE Explorer Calibre XKit Utils XT018 Utils PVS Help

Basic Search Replace (None) dc Sim Time 0



Name	Value
Filter	Filter
sailesh_dc_bias_diode_1	
Simulator spectre	
Analyses	
dc t 0 1.8 Automatic Start-Stop /V2	
Click to add analysis	
Design Variables	
Click to add variable	
Parameters	
Corners	
Reliability Analyses	
Monte Carlo Sampling	
Checks/Asserts	

Name	Type	Details	Value	Plot	Save	Spec
signal	/vin			☒	☐	
signal (I)	/D0/PLUS			☒	☐	

Choosing Analyses -- ADE Explorer

Analysis

☐ tran

☒ dc

☐ ac

☐ noise

☐ xf

☐ sens

☐ dcmatch

☐ acmatch

☐ stb

☐ pz

☐ lf

☐ sp

☐ envlp

☐ pss

☐ pac

☐ pstb

☐ pnoise

☐ pxf

☐ psp

☐ qpss

☐ qpac

☐ qpnoise

☐ qpxf

☐ qpdp

☐ hb

☐ hbac

☐ hbstb

☐ hbnoise

☐ hbsp

☐ hbxf

DC Analysis

Save DC Operating Point

☒

Hysteresis Sweep

☐

Sweep Variable

☐ Temperature

☐ Design Variable

☒ Component Parameter

☐ Model Parameter

Component Name

/V2

Select Component

Parameter Name

dc

Sweep Range

☒ Start-Stop

☐ Center-Span

Start

0

Stop

1.8

Sweep Type

Automatic

Add Specific Points

☐

Add Points By File

☐

Enabled

☒

Options...

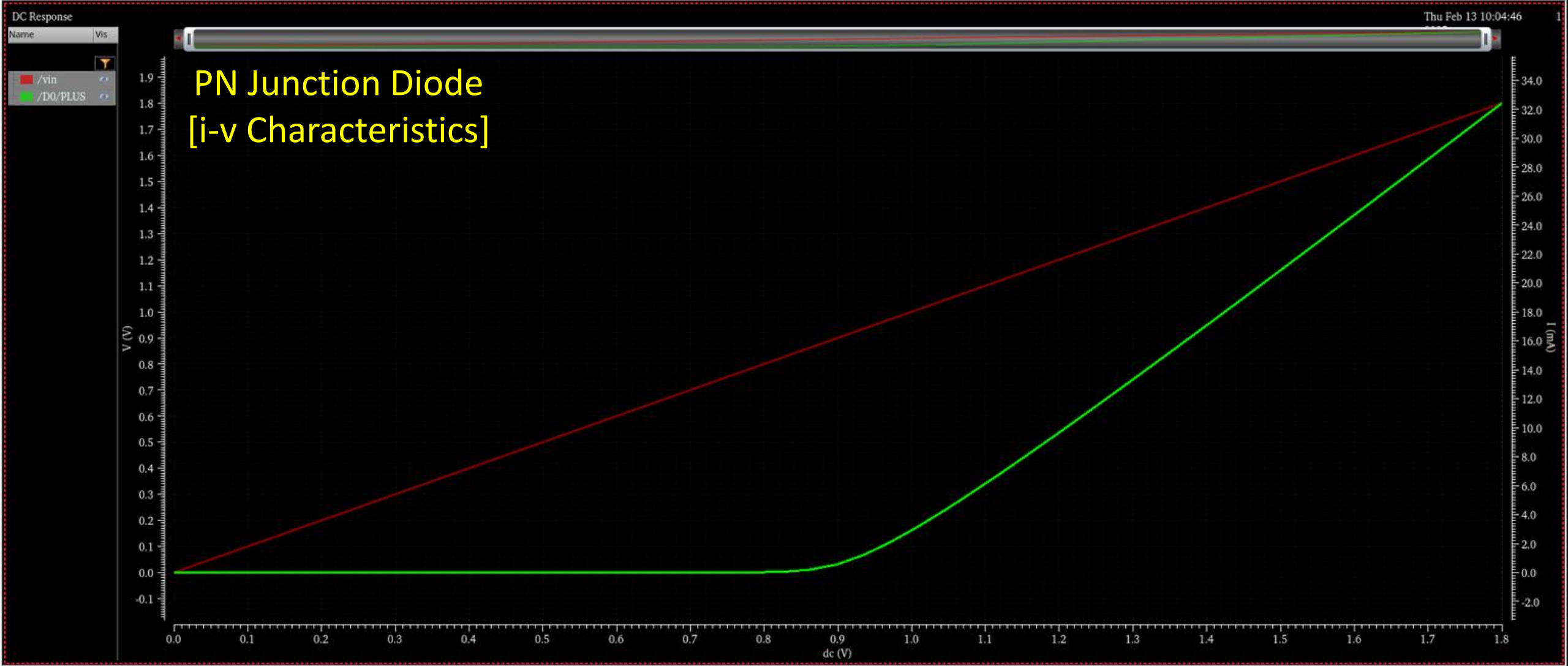
OK

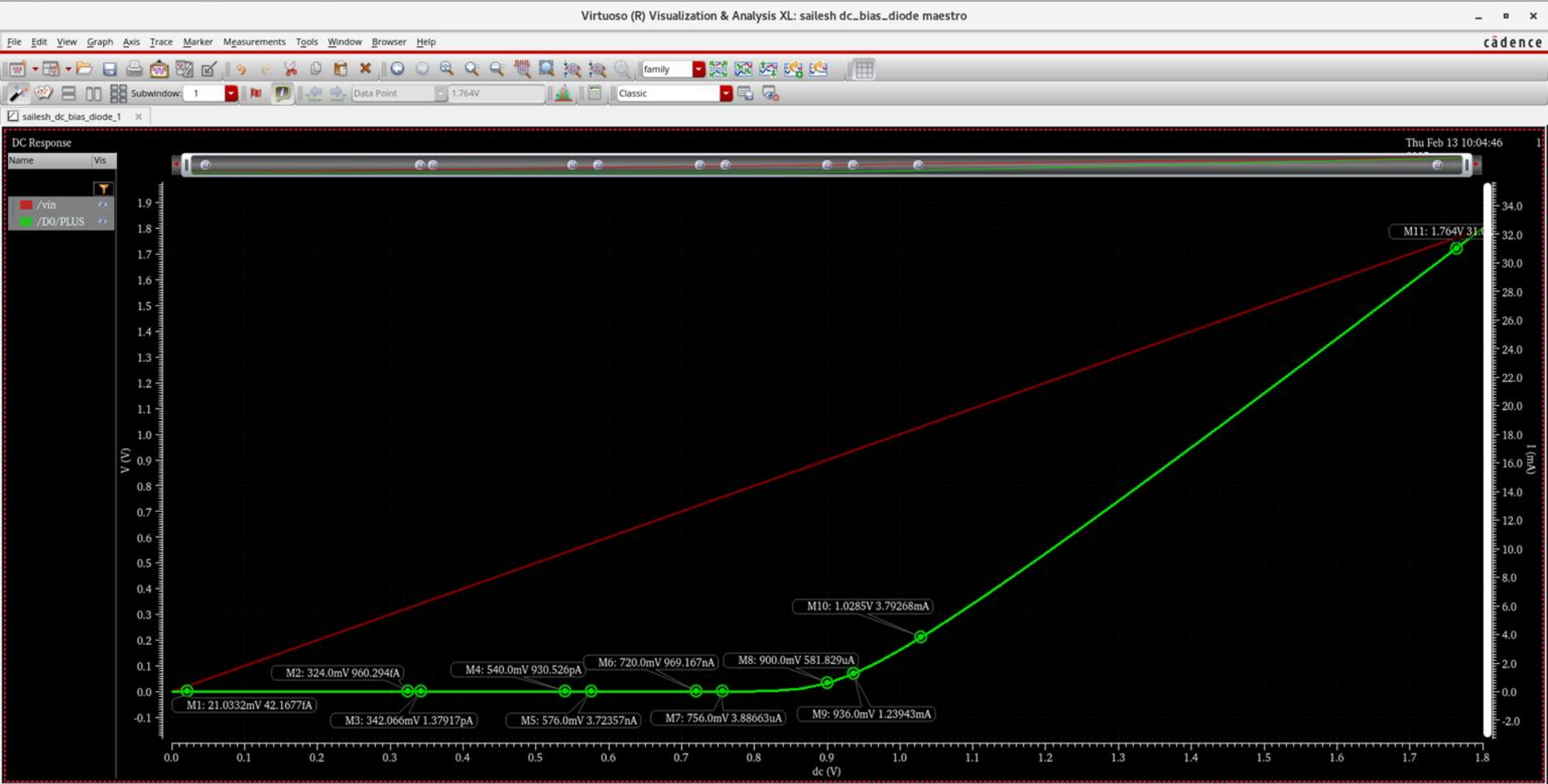
Cancel

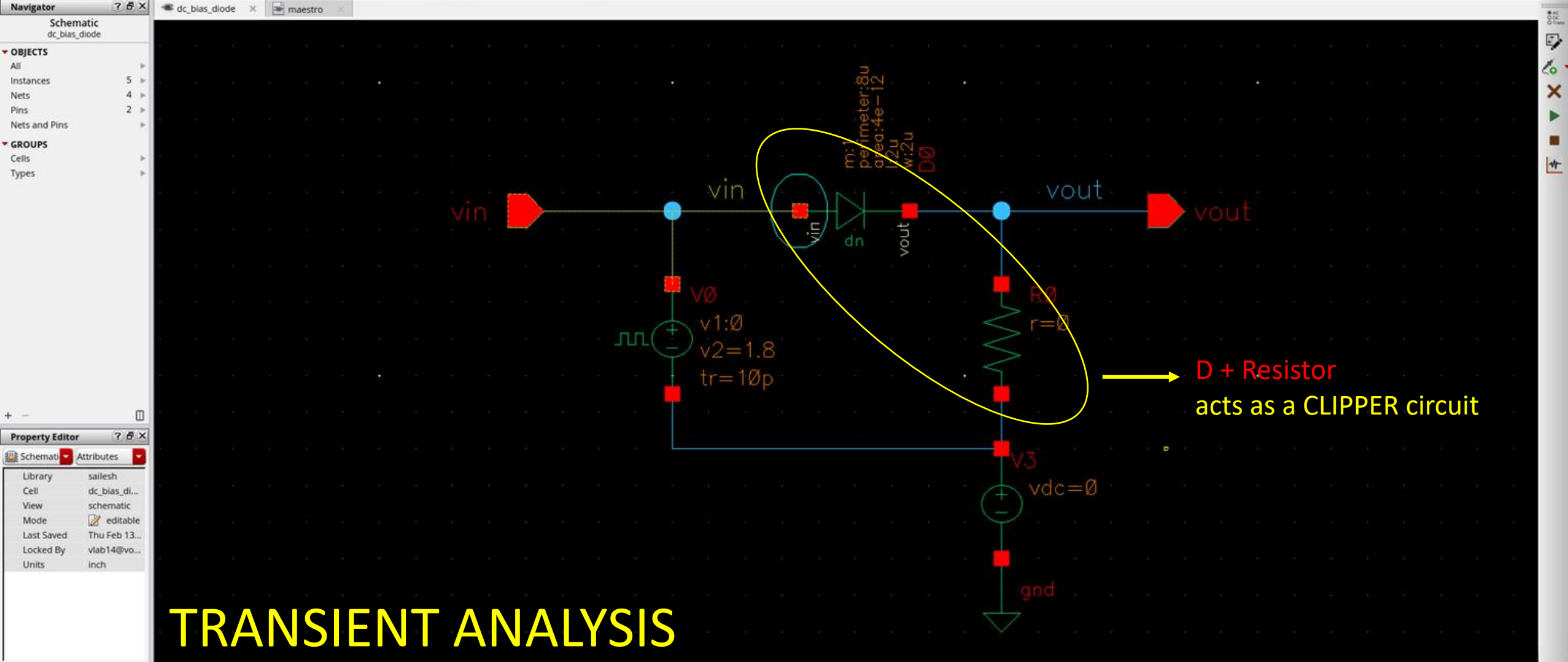
Defaults

Apply

Help







Setup

Name	Value
Filter	Filter
sailesh_dc_bias_diode_1	
Simulator	spectre
Analyses	
tran	0 800p moderate
Click to add analysis	
Design Variables	
Click to add variable	
Parameters	
Corners	
Reliability Analyses	
Monte Carlo Sampling	
Checks/Asserts	

Name	Type	Details	Value	Plot	Save	Spec
signal (I)	/D0/PLUS			☒	☒	
signal	/vin			☒	☒	

Choosing Analyses -- ADE Explorer

Analysis

☒ tran ☐ dc ☐ ac ☐ noise

☐ xf ☐ sens ☐ dcmatch ☐ acmatch

☐ stb ☐ pz ☐ lf ☐ sp

☐ envfp ☐ pss ☐ pac ☐ pstb

☐ pnoise ☐ pxf ☐ psp ☐ qpss

☐ qpac ☐ qpnoise ☐ qpxf ☐ qpdp

☐ hb ☐ hbac ☐ hbstb ☐ hbnoise

☐ hbsp ☐ hbxf

Transient Analysis

Stop Time 800p

Accuracy Defaults (errpreset)

☐ conservative ☒ moderate ☐ liberal

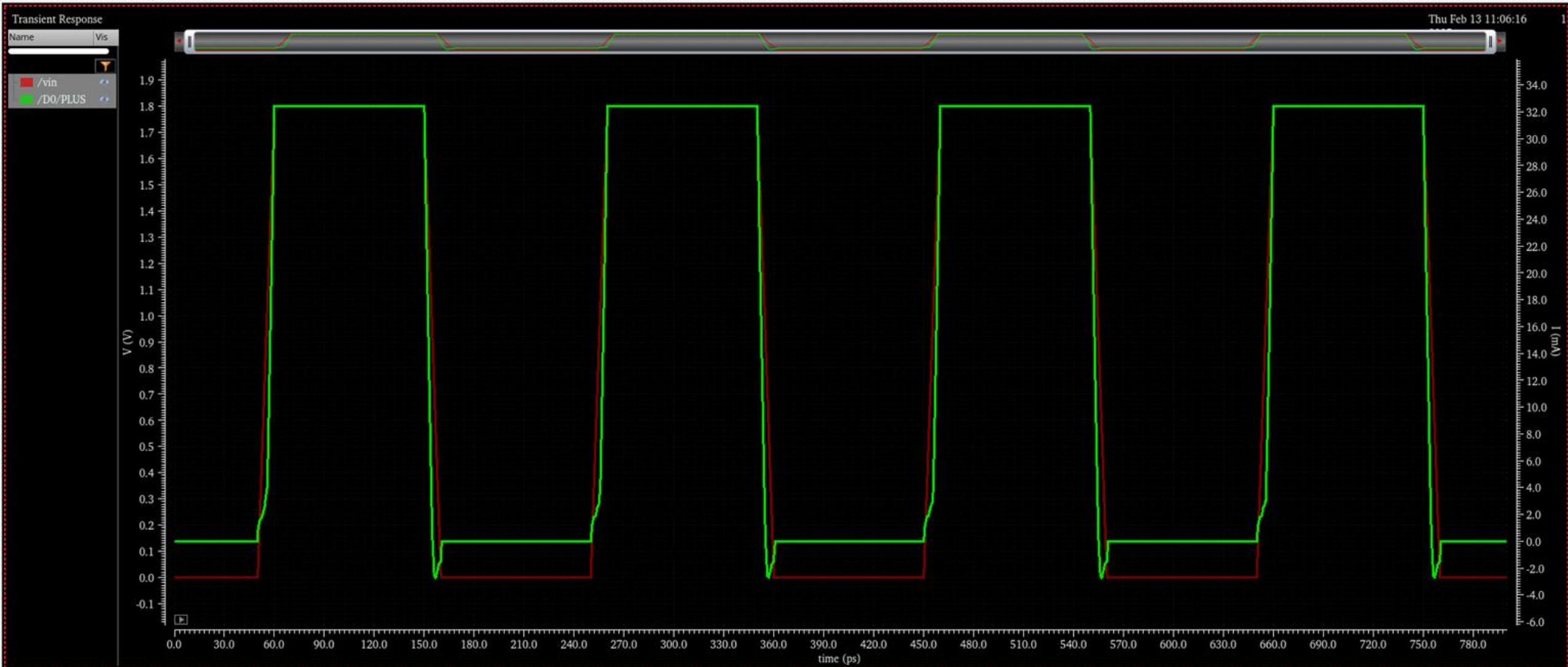
☐ Transient Noise

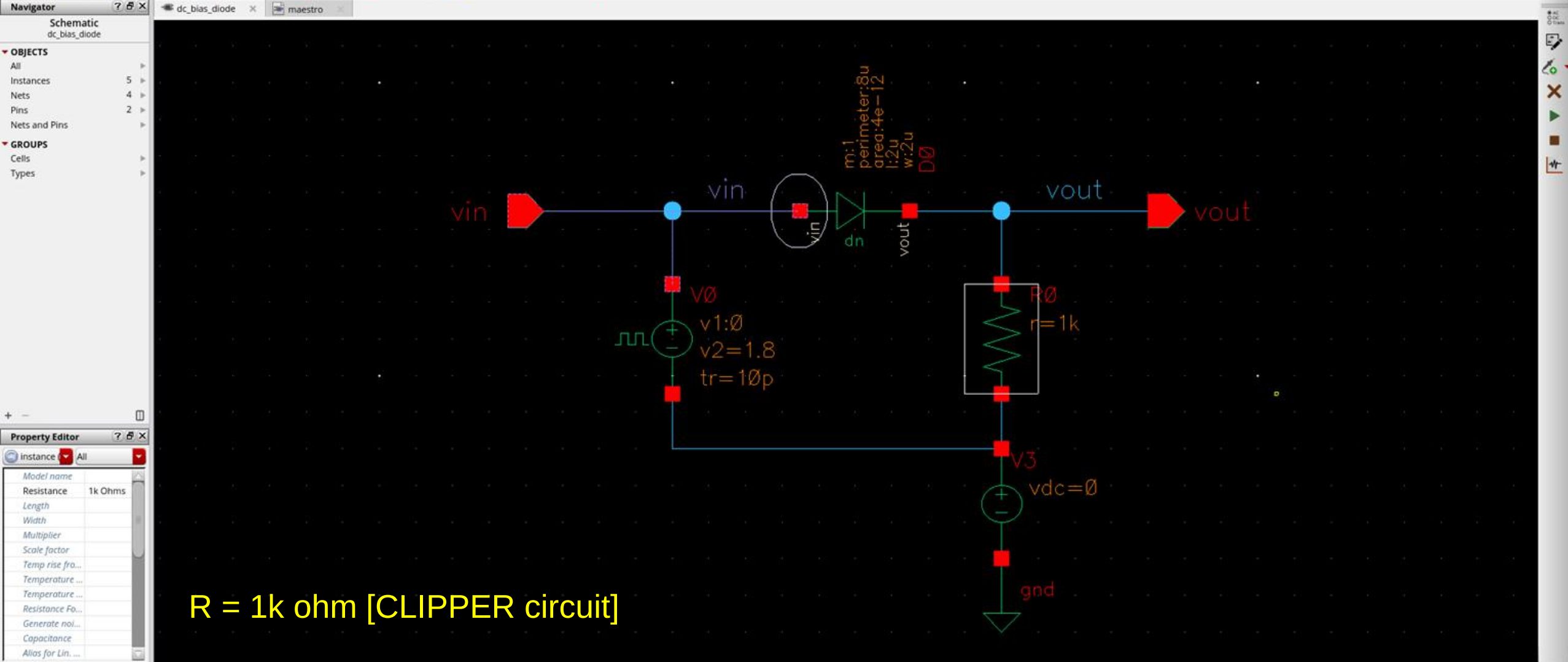
Dynamic Parameter

Enabled ☒

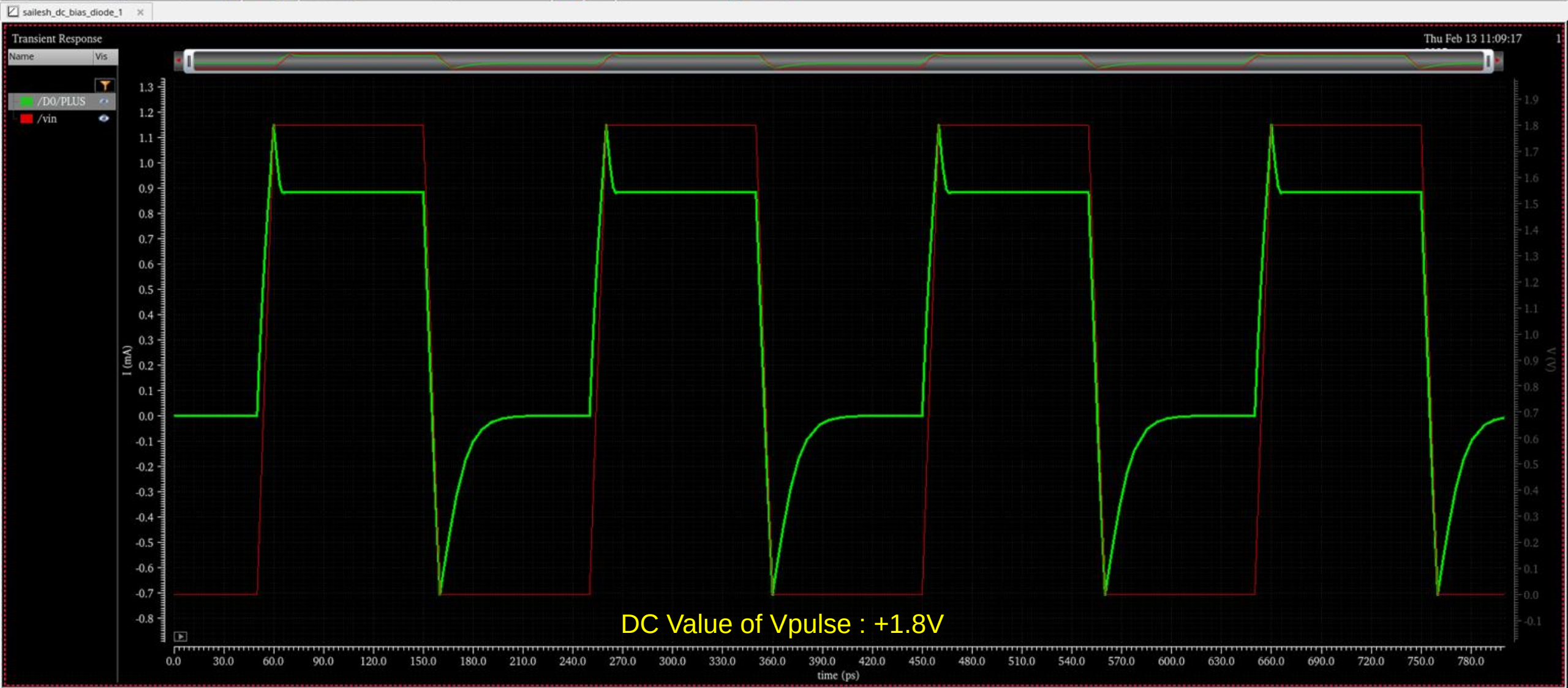
Options...

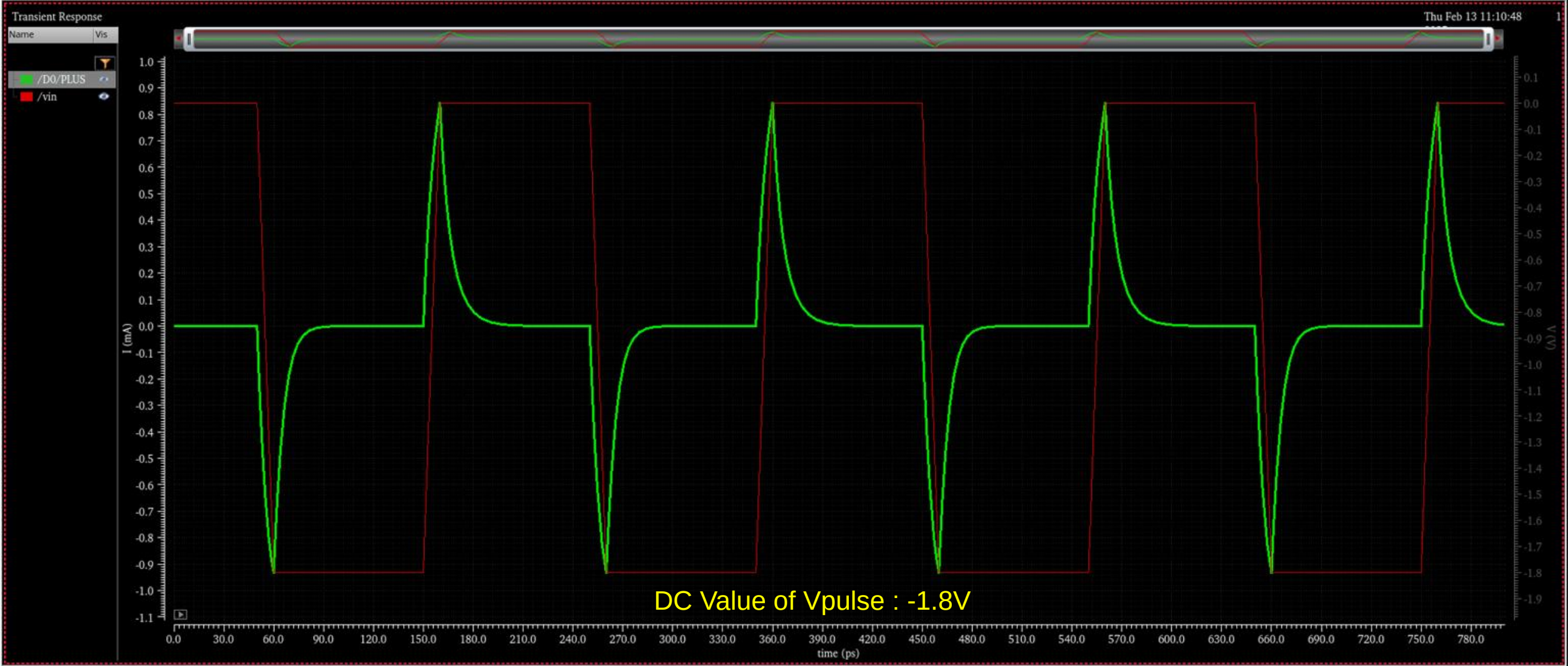
OK Cancel Defaults Apply Help





R = 1k ohm [CLIPPER circuit]



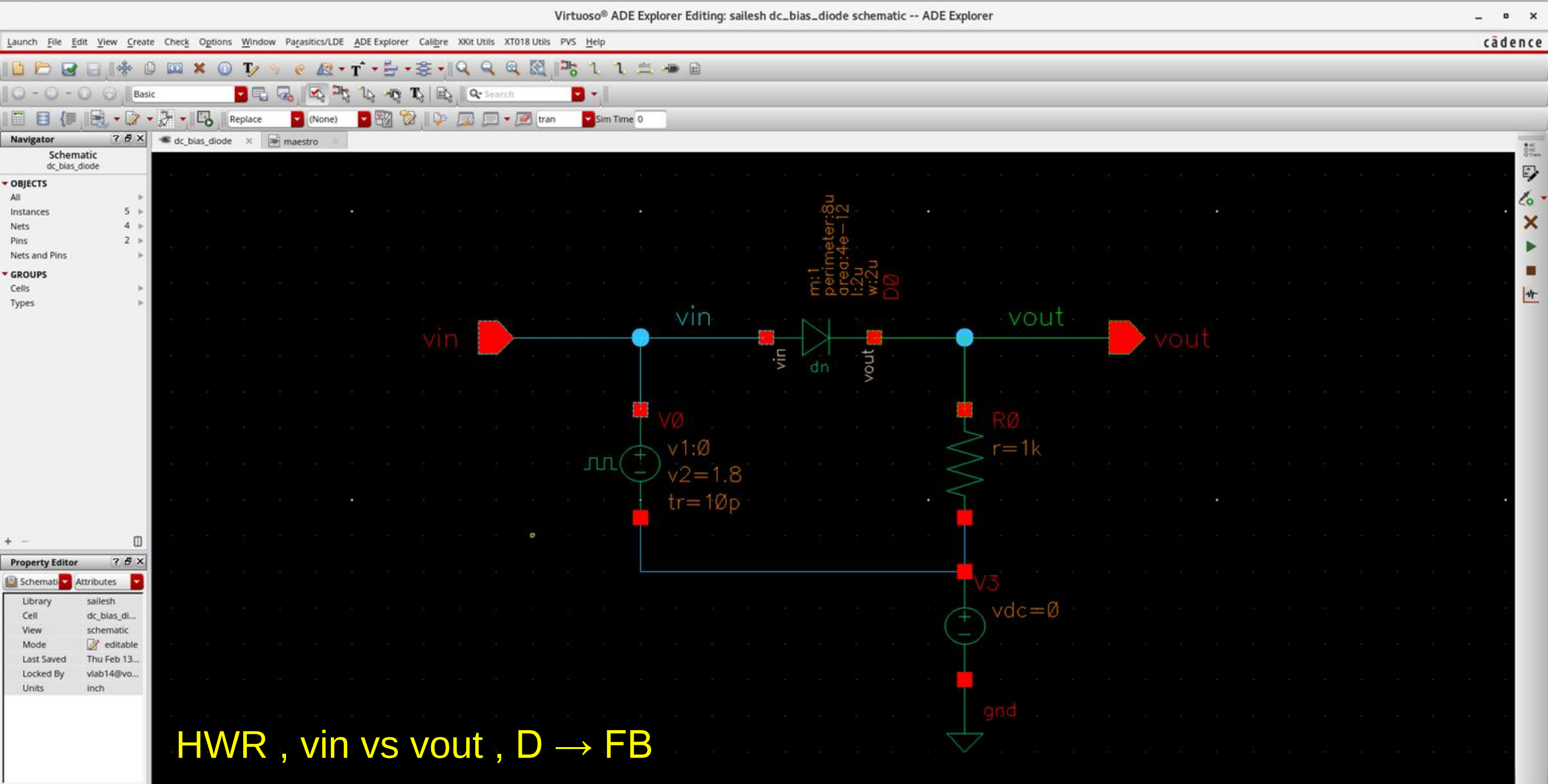


APPLICATION OF DIODE

Er. Sailesh Mishra

B.Tech (ECE | Silicon University)

Specialization in Microelectronics & VLSI



Name	Value
Filter	Filter
sailesh_dc_bias_diode_1	
Simulator spectre	
Analyses	
tran 0 800p moderate	
Click to add analysis	
Design Variables	
Click to add variable	
Parameters	
Corners	
Reliability Analyses	
Monte Carlo Sampling	
Checks/Asserts	

Name	Type	Details	Value	Plot	Save	Spec
	signal	/vout		☒	☐	
	signal	/vin		☒	☐	

Choosing Analyses -- ADE Explorer

Analysis

☒ tran

☐ dc

☐ ac

☐ noise

☐ xf

☐ sens

☐ dcmatch

☐ acmatch

☐ stb

☐ pz

☐ lf

☐ sp

☐ envlp

☐ pss

☐ pac

☐ pstb

☐ pnoise

☐ pxf

☐ psp

☐ qpss

☐ qpac

☐ qnoise

☐ qpxf

☐ qpdp

☐ hb

☐ hbac

☐ hbstb

☐ hbnoise

☐ hbsp

☐ hbxf

Transient Analysis

Stop Time 800p

Accuracy Defaults (errpreset)

☐ conservative ☒ moderate ☐ liberal

☐ Transient Noise

Dynamic Parameter

Enabled ☒

Options...

OK Cancel Defaults Apply Help

Virtuoso (R) Visualization & Analysis XL: sailesh dc_bias_diode maestro

File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help

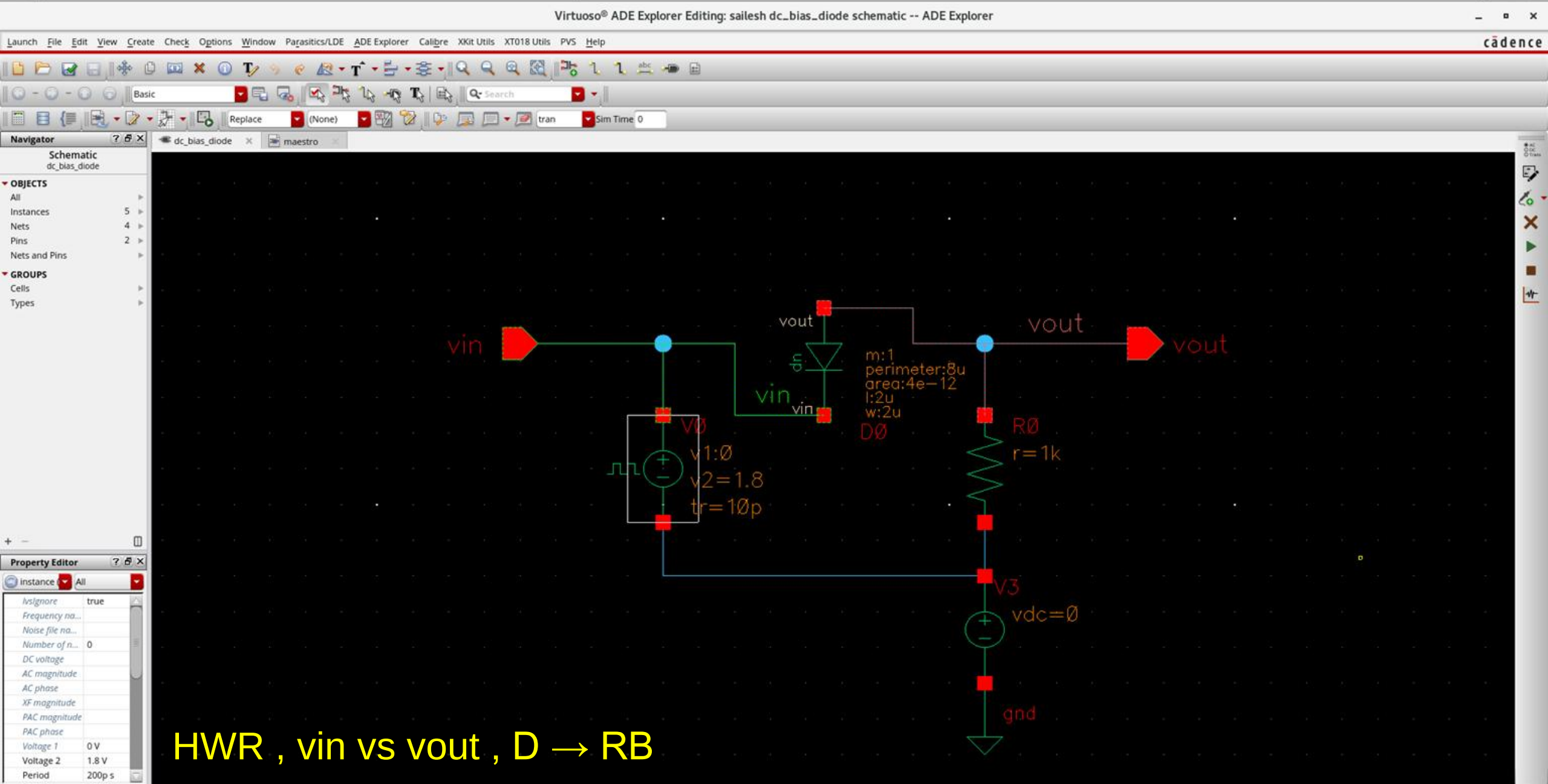
family

Subwindow: 1 Data Point 160.0ps Classic

sailesh_dc_bias_diode_1



mouse L: M: R:



Name	Value
Filter	Filter
sailesh_dc_bias_diode_1	
Simulator spectre	
Analyses	
tran 0 800p moderate	
Click to add analysis	
Design Variables	
Click to add variable	
Parameters	
Corners	
Reliability Analyses	
Monte Carlo Sampling	
Checks/Asserts	

Name	Type	Details	Value	Plot	Save	Spec
	signal	/vout		☒	☐	
	signal	/vin		☒	☐	

Choosing Analyses -- ADE Explorer

Analysis

☒ tran

☐ dc

☐ ac

☐ noise

☐ xf

☐ sens

☐ dcmatch

☐ acmatch

☐ stb

☐ pz

☐ lf

☐ sp

☐ envlp

☐ pss

☐ pac

☐ pstb

☐ pnoise

☐ pxf

☐ psp

☐ qpss

☐ qpac

☐ qnoise

☐ qpxf

☐ qpdp

☐ hb

☐ hbac

☐ hbstb

☐ hbnoise

☐ hbsp

☐ hbxf

Transient Analysis

Stop Time 800p

Accuracy Defaults (errpreset)

☐ conservative ☒ moderate ☐ liberal

☐ Transient Noise

Dynamic Parameter

Enabled ☒

Options...

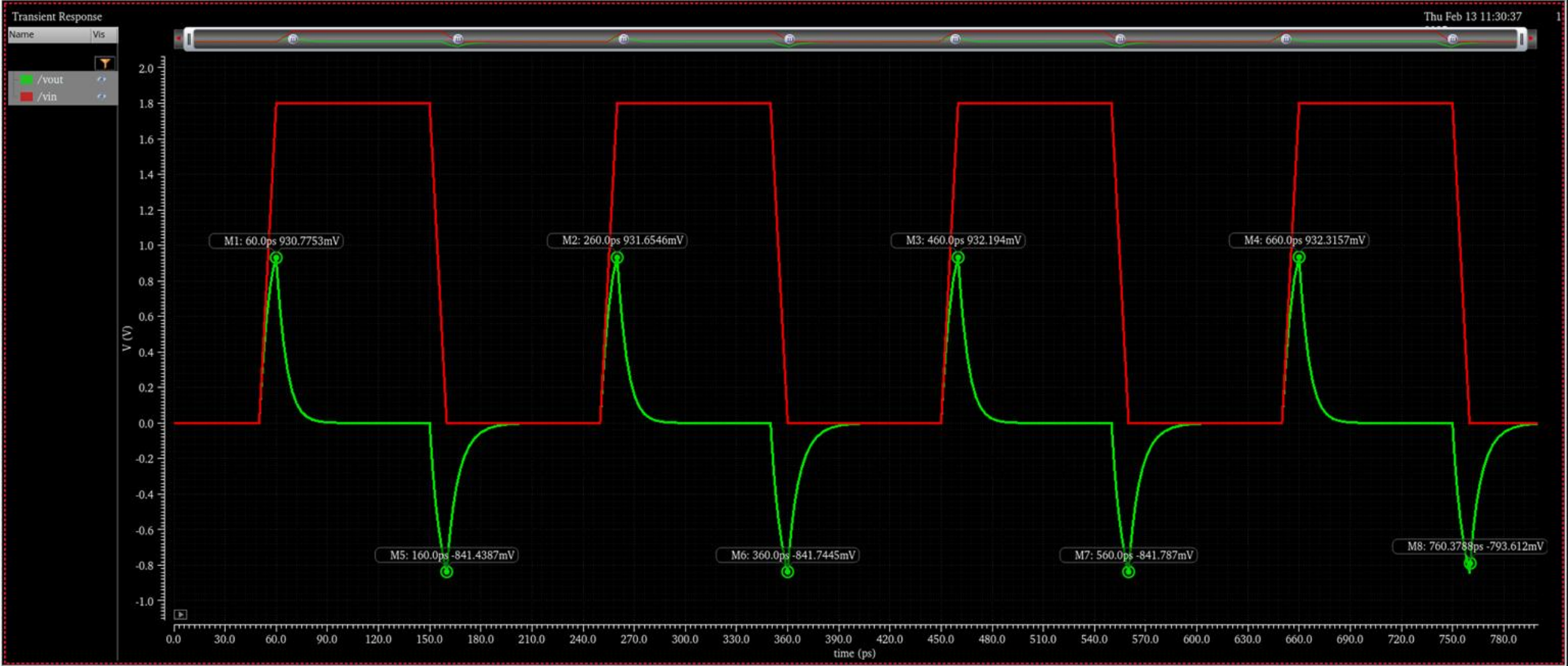
OK Cancel Defaults Apply Help

Virtuoso (R) Visualization & Analysis XL: sailesh dc_bias_diode maestro

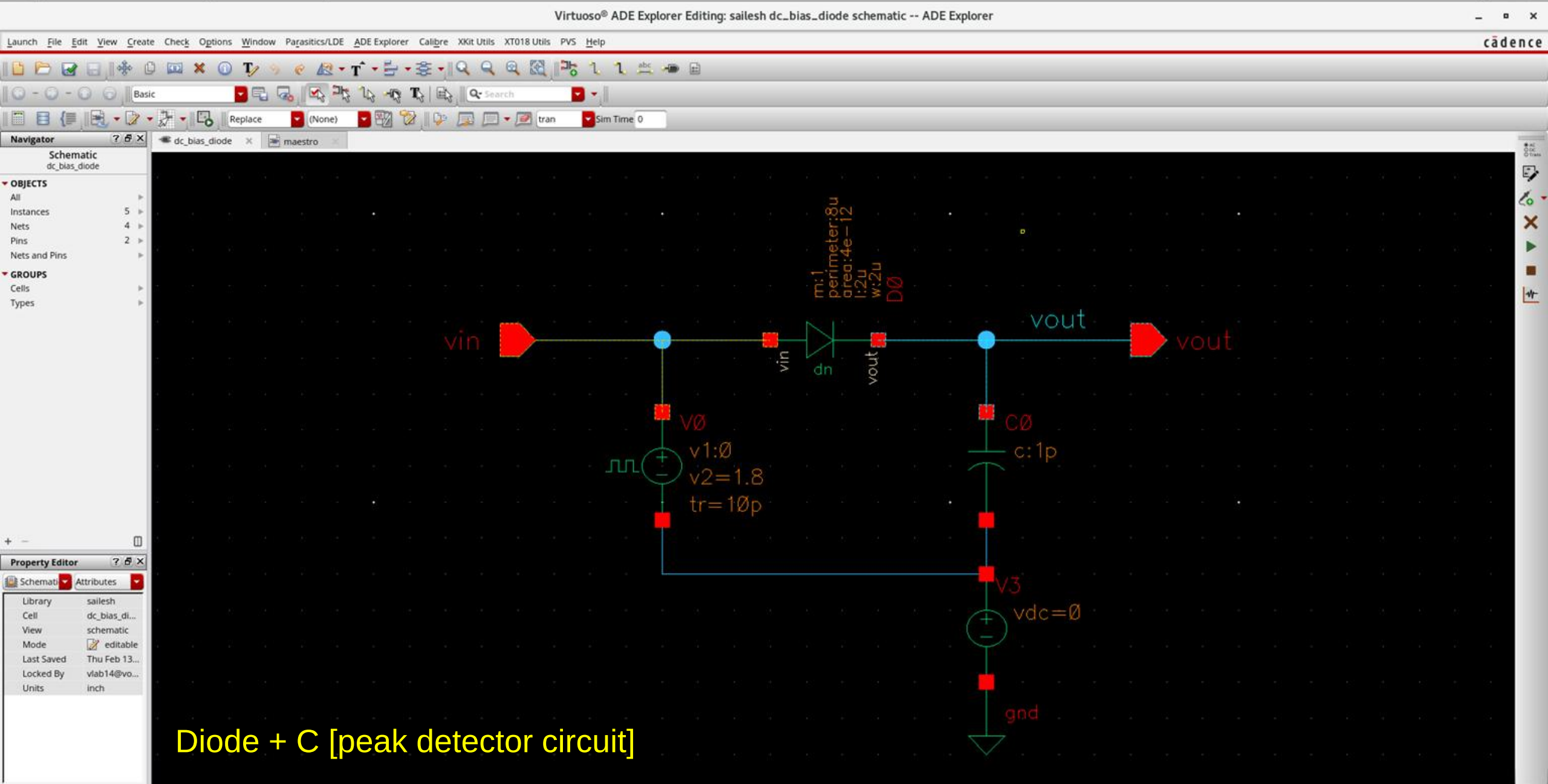
File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help

Subwindow: 1 Data Point: 760.378827838ps Classic

sailesh_dc_bias_diode_1



11(37) Trace: Vout; History: ExplorerRun.0; Test: sailesh_dc_bias_diode_1; Design Point: 1; Corner: nom



Name	Value
Filter	Filter
sailesh_dc_bias_diode_1	
Simulator spectre	
Analyses	
tran 0 800p moderate	
Click to add analysis	
Design Variables	
Click to add variable	
Parameters	
Corners	
Reliability Analyses	
Monte Carlo Sampling	
Checks/Asserts	

Name	Type	Details	Value	Plot	Save	Spec
signal	/vout			☒	☐	
signal	/vin			☒	☐	

Choosing Analyses -- ADE Explorer

Analysis

☒ tran

☐ dc

☐ ac

☐ noise

☐ xf

☐ sens

☐ dcmatch

☐ acmatch

☐ stb

☐ pz

☐ lf

☐ sp

☐ envip

☐ pss

☐ pac

☐ pstb

☐ pnoise

☐ pxf

☐ psp

☐ qpss

☐ qpac

☐ qpnoise

☐ qpxf

☐ qpdp

☐ hb

☐ hbac

☐ hbstb

☐ hbnoise

☐ hbsp

☐ hbxf

Transient Analysis

Stop Time800p

Accuracy Defaults (errpreset)

☐ conservative☒ moderate☐ liberal

☐ Transient Noise

Dynamic Parameter

Enabled☒

Options...

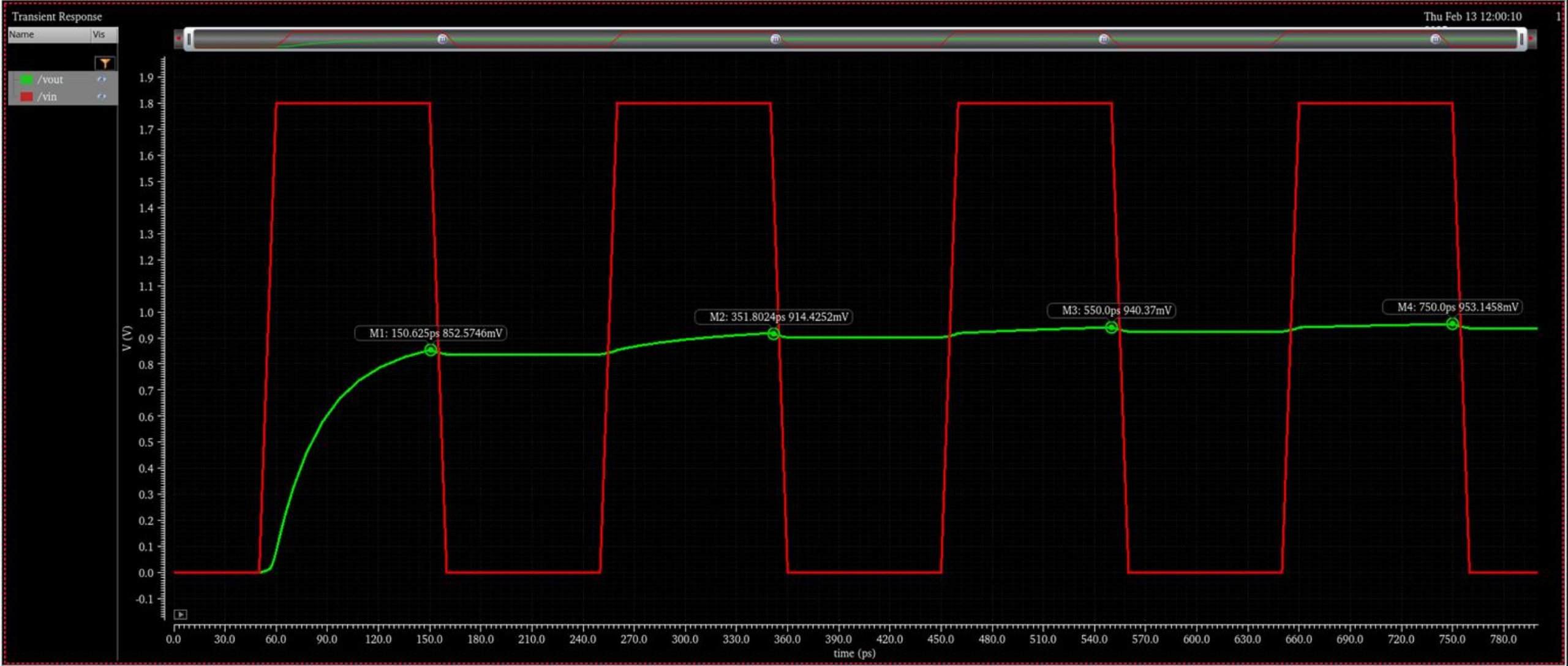
OK

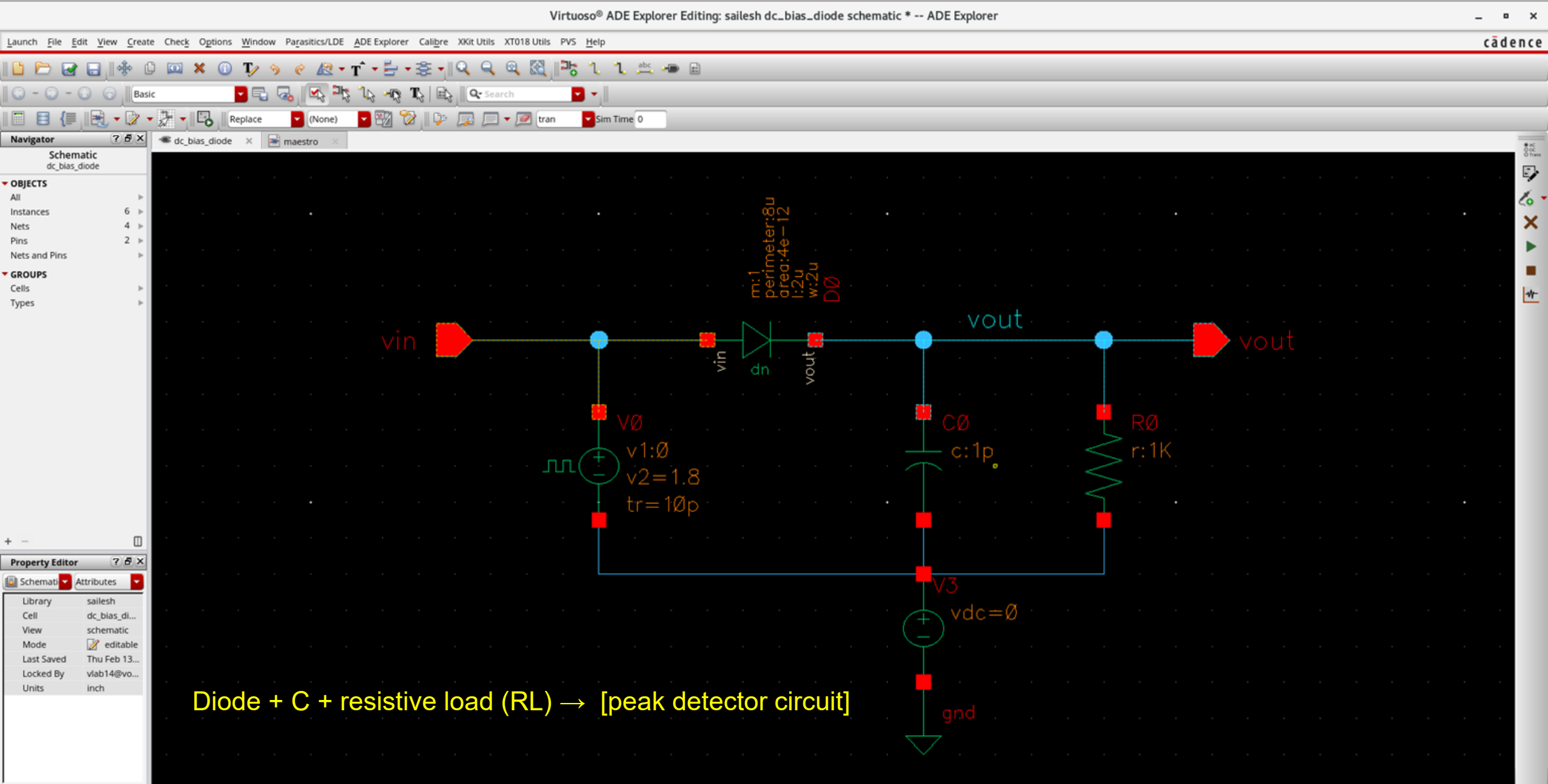
Cancel

Defaults

Apply

Help





Diode + C + resistive load (RL) → [peak detector circuit]

Setup

Name	Value
Filter	Filter
sailesh_dc_bias_diode_1	
Simulator spectre	
Analyses	
tran 0 800p moderate	
Click to add analysis	
Design Variables	
Click to add variable	
Parameters	
Corners	
Reliability Analyses	
Monte Carlo Sampling	
Checks/Asserts	

Name	Type	Details	Value	Plot	Save	Spec
	signal	/vout		☒	☐	
	signal	/vin		☒	☐	

Choosing Analyses -- ADE Explorer

Analysis

☒ tran	☐ dc	☐ ac	☐ noise
☐ xf	☐ sens	☐ dcmatch	☐ acmatch
☐ stb	☐ pz	☐ lf	☐ sp
☐ envlp	☐ pss	☐ pac	☐ pstb
☐ pnoise	☐ pxf	☐ psp	☐ qpss
☐ qpac	☐ qpnoise	☐ qpxf	☐ qpssp
☐ hb	☐ hbac	☐ hbstb	☐ hbnoise
☐ hbsp	☐ hbxf		

Transient Analysis

Stop Time

Accuracy Defaults (errpreset)

☐ conservative ☒ moderate ☐ liberal

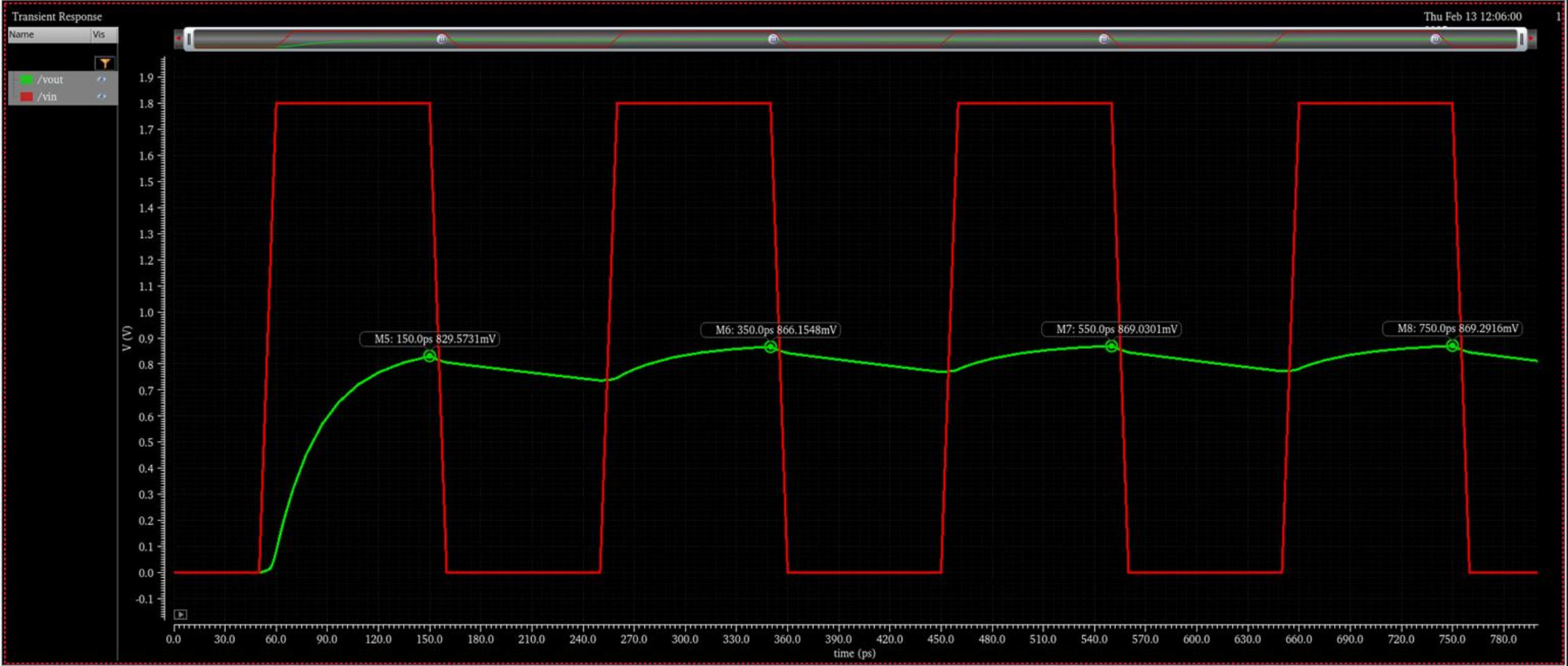
☐ Transient Noise

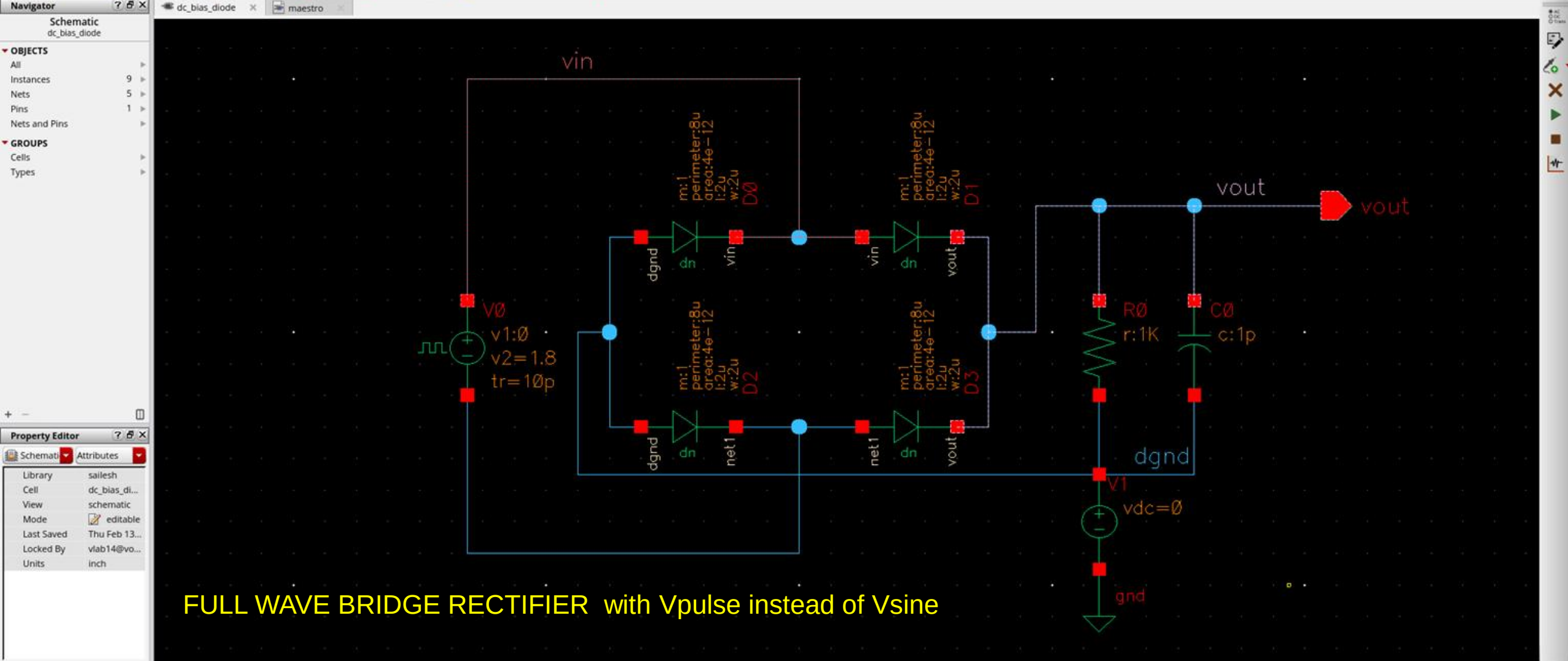
Dynamic Parameter ☐

Enabled ☒

Options...

OK Cancel Defaults Apply Help





FULL WAVE BRIDGE RECTIFIER with Vpulse instead of Vsine

Setup

Name	Value
Filter	Filter
sailesh_dc_bias_diode_1	
Simulator	spectre
Analyses	
tran	0 1000p moderate
Click to add analysis	
Design Variables	
Click to add variable	
Parameters	
Corners	
Reliability Analyses	
Monte Carlo Sampling	
Checks/Asserts	

Name	Type	Details	Value	Plot	Save	Spec
	signal	/vout		☒	☐	
	signal	/vin		☒	☐	

Choosing Analyses -- ADE Explorer

Analysis

☒ tran	☐ dc	☐ ac	☐ noise
☐ xf	☐ sens	☐ dcmatch	☐ acmatch
☐ stb	☐ pz	☐ lf	☐ sp
☐ envip	☐ pss	☐ pac	☐ pstb
☐ pnoise	☐ pxf	☐ psp	☐ qpss
☐ qpac	☐ qpnoise	☐ qpxf	☐ qpss
☐ hb	☐ hbac	☐ hbstb	☐ hbnoise
☐ hbsp	☐ hbxf		

Transient Analysis

Stop Time: 1000p

Accuracy Defaults (errpreset)

☐ conservative ☒ moderate ☐ liberal

☐ Transient Noise

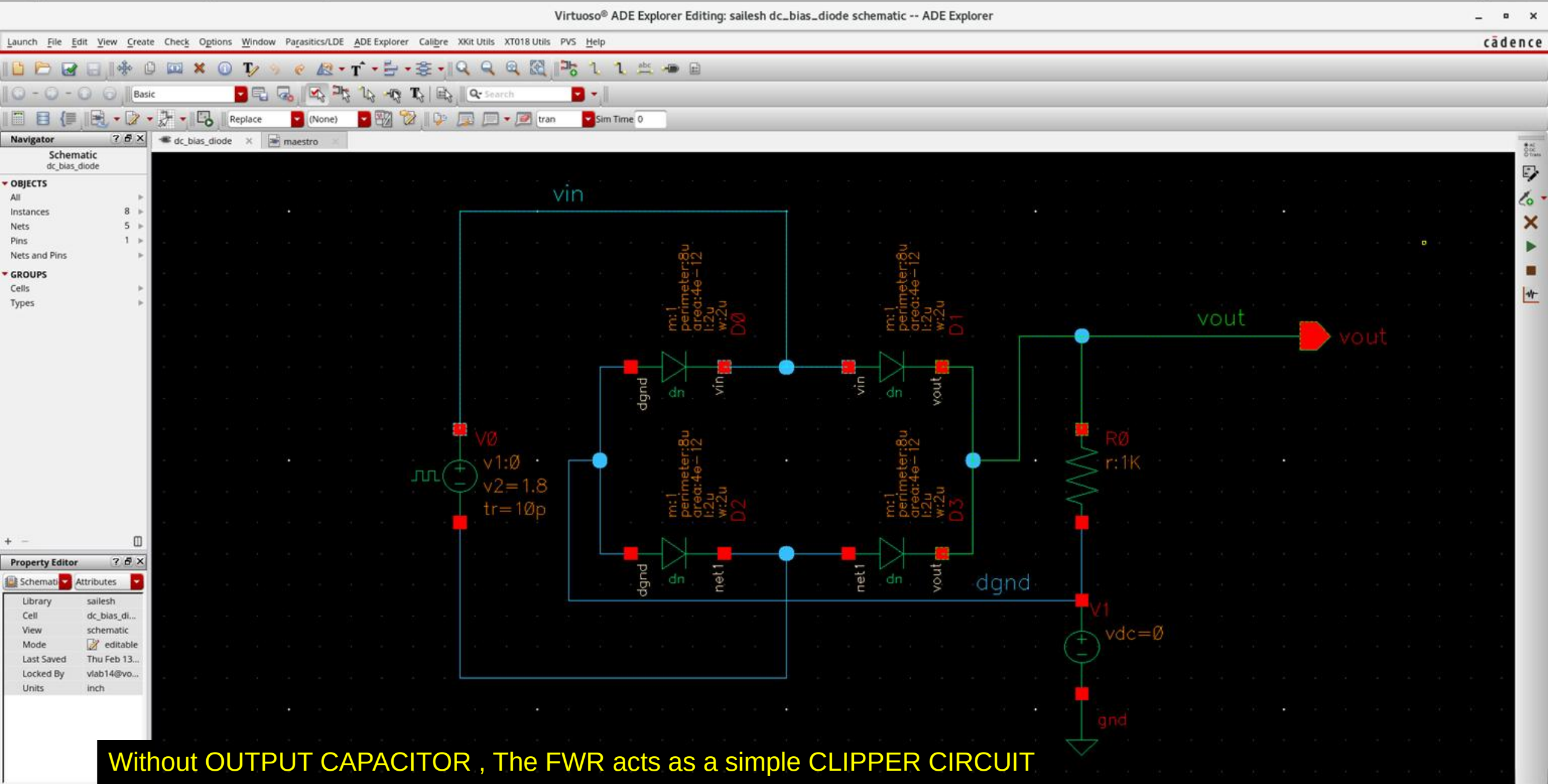
Dynamic Parameter ☐

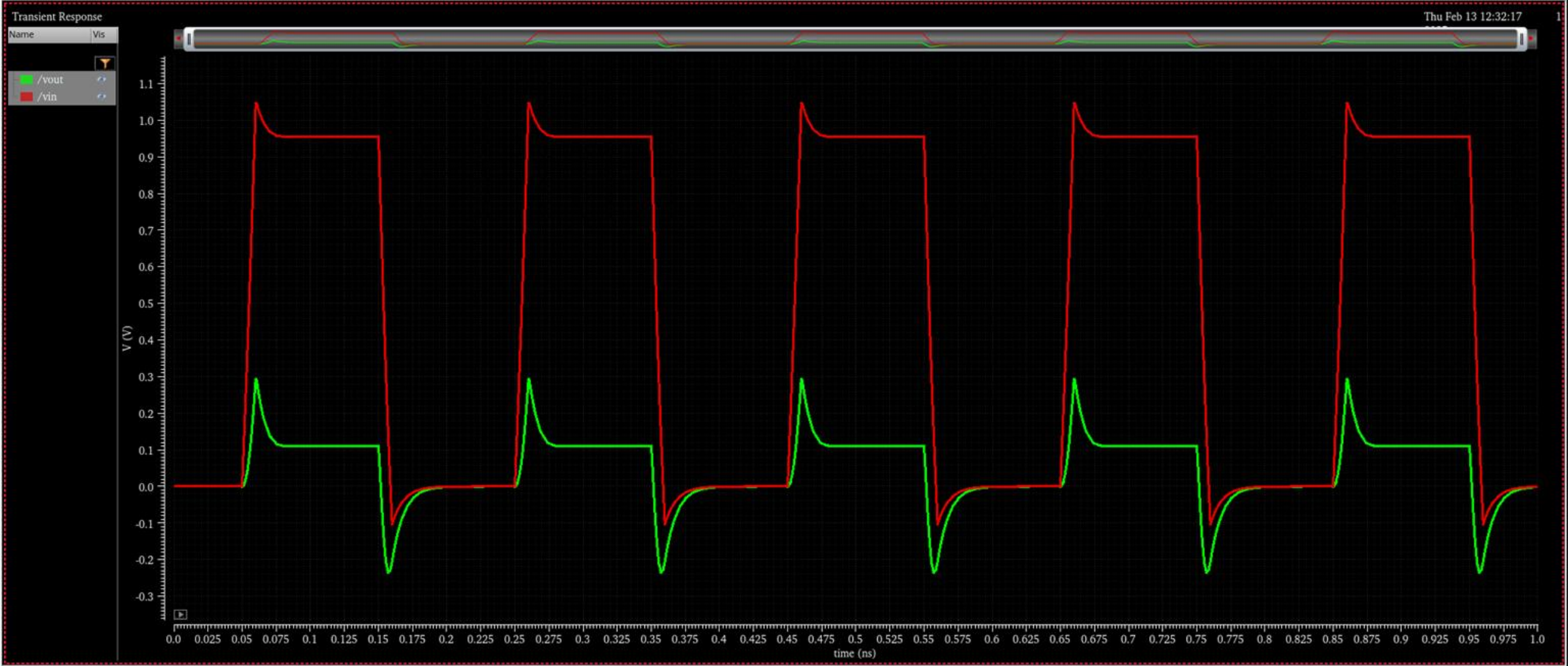
Enabled ☒

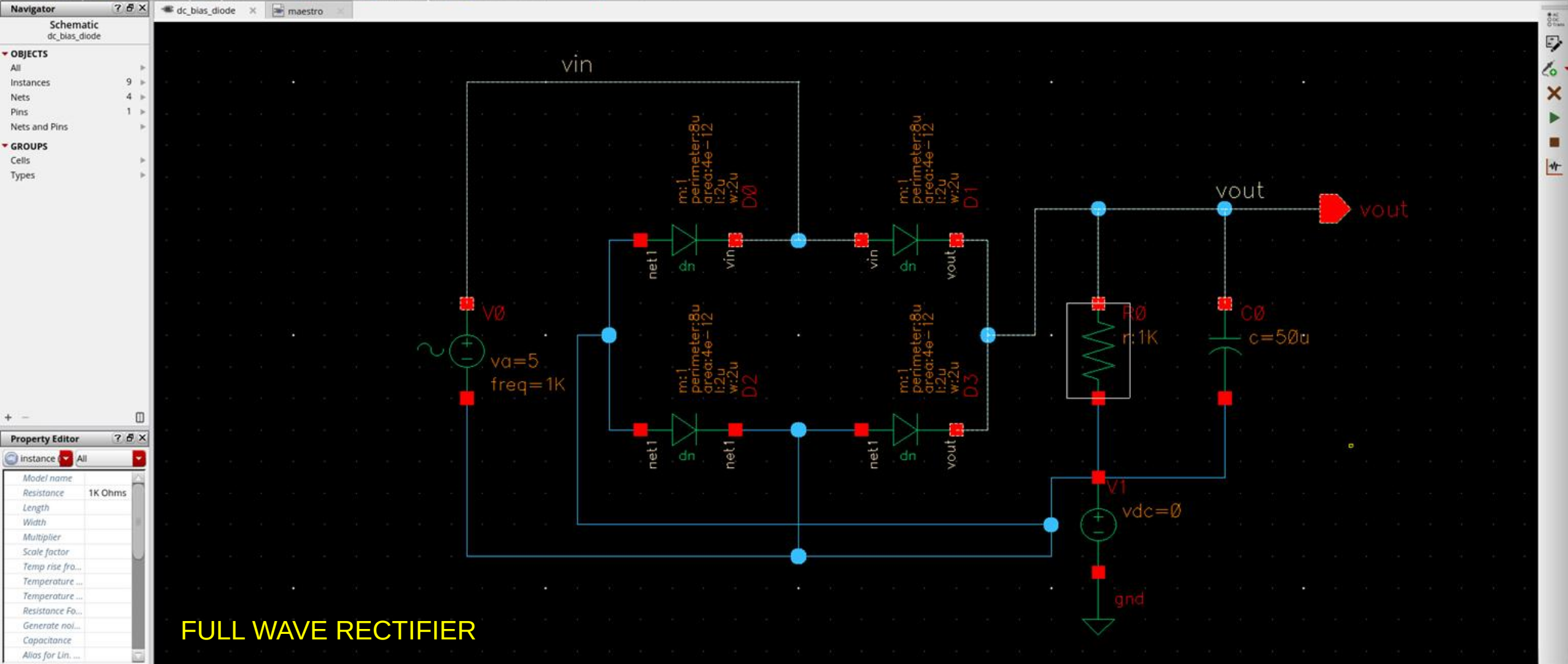
Options...

OK Cancel Defaults Apply Help









Name	Value
Filter	Filter
sailesh_dc_bias_diode_1	
Simulator spectre	
Analyses	
tran 0.25m moderate	
Click to add analysis	
Design Variables	
Click to add variable	
Parameters	
Corners	
Reliability Analyses	
Monte Carlo Sampling	
Checks/Asserts	

Name	Type	Details	Value	Plot	Save	Spec
signal	/vout			☒	☐	
signal	/vin			☒	☐	

Choosing Analyses -- ADE Explorer

Analysis

☒ tran

☐ dc

☐ ac

☐ noise

☐ xf

☐ sens

☐ dcmatch

☐ acmatch

☐ stb

☐ pz

☐ lf

☐ sp

☐ envip

☐ pss

☐ pac

☐ pstb

☐ pnoise

☐ pxf

☐ psp

☐ qpss

☐ qpac

☐ qpnoise

☐ qpxf

☐ qpss

☐ hb

☐ hbac

☐ hbstb

☐ hbnoise

☐ hbsp

☐ hbxf

Transient Analysis

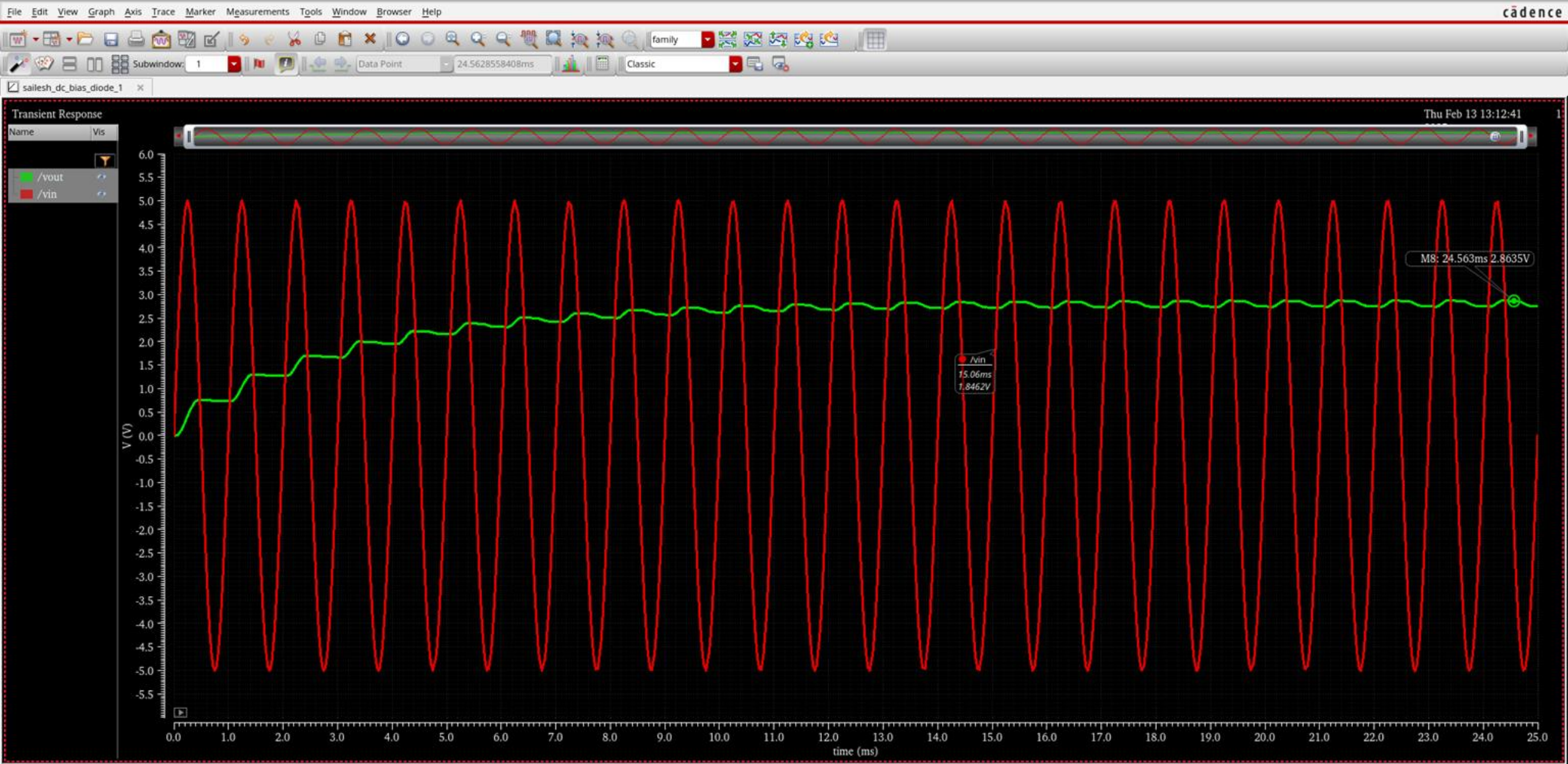
Stop Time 25m

Accuracy Defaults (errpreset)
☐ conservative ☒ moderate ☐ liberal☐ Transient NoiseDynamic Parameter ☐Enabled ☒

Options...

OK Cancel Defaults Apply Help

Virtuoso (R) Visualization & Analysis XL: sailesh dc_bias_diode maestro



BJT DC ANALYSIS